

# Tối ưu hóa Danh mục Cổ phiếu Ngân hàng Việt Nam

Ứng dụng Mô hình Black-Litterman kết hợp Phân cụm K-means và Monte Carlo

Universe: 25 cổ phiếu ngân hàng niêm yết tại HOSE / HNX

Dữ liệu: Giá đóng cửa tháng & Vol hóa trị trung bình | 2014-06 → 2026-05

Cửa sổ huấn luyện: 36 tháng (rolling) | OOS: 106 tháng (2017-08 → 2026-05)

Lãi suất phi rủi ro: VN10Y / 12 (time-varying)

Danh mục: MKT · TAN · MV · BL · EW (1/N) · RP (Risk Parity)

| Danh mục | Ann. Return | Ann. Vol | Sharpe | Sortino | MDD     |
|----------|-------------|----------|--------|---------|---------|
| MKT      | +23.40%     | 27.05%   | 0.729  | 0.763   | -32.51% |
| TAN      | +19.20%     | 27.81%   | 0.558  | 0.549   | -51.54% |
| MV       | +16.24%     | 25.40%   | 0.494  | 0.539   | -38.44% |
| BL       | +25.29%     | 29.77%   | 0.726  | 0.715   | -51.13% |
| EW       | +26.62%     | 27.38%   | 0.837  | 0.904   | -36.55% |
| RP       | +26.62%     | 27.38%   | 0.837  | 0.904   | -36.55% |

## 1. Hiệu suất Out-of-Sample (Tóm tắt)

Lãi nhuận netm hoá, biến động, Sharpe, Sortino, MDD, Calmar và tail risk của 6 danh mục trên 106 tháng OOS (08/2017 – 05/2026). Lãi suất phi rủi ro sử dụng VN10Y / 12 theo tháng netm.

| Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino  | MDD       | Calmar   | Win_Rate |
|-----------|------------|----------|----------|----------|-----------|----------|----------|
| MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| TAN       | 0.191990   | 0.278101 | 0.557740 | 0.872662 | -0.515448 | 0.372472 | 0.622642 |
| MV        | 0.162408   | 0.253993 | 0.494209 | 0.801307 | -0.384445 | 0.422448 | 0.575472 |
| BL        | 0.301188   | 0.281283 | 0.939644 | 1.6281   | -0.306594 | 0.982370 | 0.613208 |
| EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |

## 2. Kiểm định Thống kê

### 2a. Paired t-test ( $H_0: E[r_{BL}] = E[r_{other}]$ )

Kiểm định t ghép cặp hai phía so sánh lãi nhuận tháng OOS của BL với từng danh mục còn lại.

| Pair      | t_statistic | p_value  | is_significant_5pct |
|-----------|-------------|----------|---------------------|
| BL vs MKT | 1.1420      | 0.256054 | False               |
| BL vs TAN | 2.0206      | 0.045872 | True                |
| BL vs MV  | 2.7305      | 0.007418 | True                |
| BL vs EW  | 0.774217    | 0.440543 | False               |
| BL vs RP  | 0.992437    | 0.323266 | False               |

### 2b. Jobson-Korkie Test ( $H_0: SR_{BL} = SR_{other}$ )

Kiểm định Jobson & Korkie (1981) so sánh trực tiếp hai hệ số Sharpe (dùng thống kê Sharpe theo tháng, không annualise). Thống kê z kiểm định phân phối chuẩn  $N(0,1)$ .

| Pair      | z_statistic | p_value  | is_significant_5pct |
|-----------|-------------|----------|---------------------|
| BL vs MKT | 0.965171    | 0.334459 | False               |
| BL vs TAN | 1.9395      | 0.052441 | False               |
| BL vs MV  | 2.3255      | 0.020046 | True                |
| BL vs EW  | 0.606804    | 0.543981 | False               |
| BL vs RP  | 0.775014    | 0.438331 | False               |

### 3. Kiểm định Phân phối Lợi nhuận

#### 3a. Jarque-Bera (H0: phân phối chuẩn)

Kiểm tra liệu lợi nhuận tháng OOS có phân phối chuẩn không. Bác bỏ H0 ( $p < 0.05$ ) nghĩa là lợi nhuận có thể lệch hoặc đuôi dày.

| Portfolio | JB_stat | JB_pval  | Skewness  | Excess_Kurtosis | is_normal_5pct |
|-----------|---------|----------|-----------|-----------------|----------------|
| MKT       | 10.1416 | 0.006277 | -0.175466 | 1.6061          | False          |
| TAN       | 16.2824 | 0.000291 | -0.212219 | 2.0242          | False          |
| MV        | 83.7728 | 0.000000 | 0.143990  | 4.6169          | False          |
| BL        | 16.2666 | 0.000294 | -0.013640 | 2.0714          | False          |
| EW        | 7.2799  | 0.026253 | 0.198406  | 1.3413          | False          |
| RP        | 10.7793 | 0.004564 | 0.150547  | 1.6675          | False          |

#### 3b. Ljung-Box (H0: không có tự tương quan)

Kiểm tra tự tương quan chuỗi lợi nhuận tháng tại các trễ 5, 10, 20.  $p > 0.05$  nghĩa là không có bằng chứng tự tương quan.

| Portfolio | LB5_stat | LB5_pval | LB5_no_autocorr | LB10_stat | LB10_pval | LB10_no_autocorr | LB20_stat | LB20_pval | LB20_no_autocorr |
|-----------|----------|----------|-----------------|-----------|-----------|------------------|-----------|-----------|------------------|
| MKT       | 8.8677   | 0.114457 | True            | 19.7177   | 0.032039  | False            | 34.2652   | 0.024385  | False            |
| TAN       | 4.5857   | 0.468501 | True            | 11.5802   | 0.314135  | True             | 25.5470   | 0.181299  | True             |
| MV        | 2.0715   | 0.839169 | True            | 16.2845   | 0.091772  | True             | 25.7963   | 0.172645  | True             |
| BL        | 5.9811   | 0.308063 | True            | 12.4184   | 0.258027  | True             | 22.7773   | 0.299862  | True             |
| EW        | 3.0475   | 0.692660 | True            | 12.8746   | 0.230769  | True             | 23.9440   | 0.244848  | True             |
| RP        | 2.6747   | 0.749978 | True            | 13.3277   | 0.205920  | True             | 24.2006   | 0.233738  | True             |

## 4. Phân tích Drawdown

### 4a. Tóm tắt Drawdown

Số lần sụt giảm, độ sâu trung bình, thời gian phục hồi trung bình, số tháng danh mục vượt ngưỡng -20%.

| Portfolio | N_Drawdowns | Max_DD_pct | Avg_DD_Depth_pct | Avg_Recovery_Months | Months_Below_Neg20 |
|-----------|-------------|------------|------------------|---------------------|--------------------|
| MKT       | 9           | -32.5100   | -15.8300         | 6.9000              | 19                 |
| TAN       | 5           | -51.5400   | -24.9000         | 9.6000              | 51                 |
| MV        | 5           | -38.4400   | -21.3200         | 11.8000             | 26                 |
| BL        | 12          | -30.6600   | -9.4800          | 3.8000              | 21                 |
| EW        | 7           | -36.5500   | -15.3500         | 9.0000              | 34                 |
| RP        | 7           | -35.7200   | -15.6100         | 6.5000              | 34                 |

### 4b. Chi tiết từng đợt Drawdown

Ngày bắt đầu, ngày đáy, ngày phục hồi, độ sâu (%), số tháng liên tiếp và số tháng phục hồi cho từng đợt sụt giảm.

| Portfolio | DD_Start   | DD_Trough  | DD_Recovery         | Depth_pct | Months_to_Trough | Recovery_Months | Total_Months |
|-----------|------------|------------|---------------------|-----------|------------------|-----------------|--------------|
| MKT       | 2018-04-30 | 2018-06-30 | 2019-12-31 00:00:00 | -26.9400  | 2                | 18.0000         | 20           |
| MKT       | 2020-02-29 | 2020-03-31 | 2020-11-30 00:00:00 | -31.0700  | 1                | 8.0000          | 9            |
| MKT       | 2021-01-31 | 2021-01-31 | 2021-02-28 00:00:00 | -7.8800   | 0                | 1.0000          | 1            |
| MKT       | 2021-07-31 | 2021-08-31 | 2022-01-31 00:00:00 | -14.3800  | 1                | 5.0000          | 6            |
| MKT       | 2022-02-28 | 2022-10-31 | 2024-02-29 00:00:00 | -32.5100  | 8                | 16.0000         | 24           |
| MKT       | 2024-04-30 | 2024-06-30 | 2024-08-31 00:00:00 | -6.9900   | 2                | 2.0000          | 4            |
| MKT       | 2024-10-31 | 2024-11-30 | 2025-01-31 00:00:00 | -2.8100   | 1                | 2.0000          | 3            |
| MKT       | 2025-04-30 | 2025-04-30 | 2025-07-31 00:00:00 | -7.6900   | 0                | 3.0000          | 3            |
| MKT       | 2025-09-30 | 2026-03-31 | Not recovered       | -12.2300  | 6                | nan             | 8            |

| Portfolio | DD_Start   | DD_Trough  | DD_Recovery         | Depth_pct | Months_to_Trough | Recovery_Months | Total_Months |
|-----------|------------|------------|---------------------|-----------|------------------|-----------------|--------------|
| TAN       | 2018-04-30 | 2020-03-31 | 2020-12-31 00:00:00 | -38.0500  | 23               | 9.0000          | 32           |
| TAN       | 2021-01-31 | 2021-01-31 | 2021-02-28 00:00:00 | -8.0800   | 0                | 1.0000          | 1            |
| TAN       | 2021-07-31 | 2021-08-31 | 2021-12-31 00:00:00 | -15.6200  | 1                | 4.0000          | 5            |
| TAN       | 2022-02-28 | 2022-12-31 | 2025-08-31 00:00:00 | -51.5400  | 10               | 32.0000         | 42           |
| TAN       | 2025-09-30 | 2026-03-31 | 2026-05-31 00:00:00 | -11.1900  | 6                | 2.0000          | 8            |
| MV        | 2018-04-30 | 2020-03-31 | 2020-12-31 00:00:00 | -38.4400  | 23               | 9.0000          | 32           |
| MV        | 2021-01-31 | 2021-01-31 | 2021-03-31 00:00:00 | -8.0800   | 0                | 2.0000          | 2            |
| MV        | 2021-06-30 | 2021-08-31 | 2022-01-31 00:00:00 | -13.0300  | 2                | 5.0000          | 7            |
| MV        | 2022-02-28 | 2022-12-31 | 2025-07-31 00:00:00 | -31.6700  | 10               | 31.0000         | 41           |
| MV        | 2025-09-30 | 2026-03-31 | Not recovered       | -15.3700  | 6                | nan             | 8            |
| BL        | 2017-10-31 | 2017-10-31 | 2017-11-30 00:00:00 | -2.0500   | 0                | 1.0000          | 1            |
| BL        | 2017-12-31 | 2017-12-31 | 2018-01-31 00:00:00 | -1.1400   | 0                | 1.0000          | 1            |
| BL        | 2018-04-30 | 2019-01-31 | 2020-01-31 00:00:00 | -30.4700  | 9                | 12.0000         | 21           |
| BL        | 2020-02-29 | 2020-03-31 | 2020-09-30 00:00:00 | -30.6600  | 1                | 6.0000          | 7            |
| BL        | 2021-01-31 | 2021-01-31 | 2021-02-28 00:00:00 | -3.2400   | 0                | 1.0000          | 1            |
| BL        | 2021-06-30 | 2022-10-31 | 2024-01-31 00:00:00 | -25.4300  | 16               | 15.0000         | 31           |
| BL        | 2024-06-30 | 2024-06-30 | 2024-07-31 00:00:00 | -3.9000   | 0                | 1.0000          | 1            |
| BL        | 2024-11-30 | 2024-11-30 | 2024-12-31 00:00:00 | -0.850000 | 0                | 1.0000          | 1            |
| BL        | 2025-03-31 | 2025-04-30 | 2025-06-30 00:00:00 | -5.9300   | 1                | 2.0000          | 3            |
| BL        | 2025-09-30 | 2025-09-30 | 2025-10-31 00:00:00 | -2.0600   | 0                | 1.0000          | 1            |
| BL        | 2025-11-30 | 2025-11-30 | 2025-12-31 00:00:00 | -1.4700   | 0                | 1.0000          | 1            |
| BL        | 2026-01-31 | 2026-03-31 | Not recovered       | -6.5600   | 2                | nan             | 4            |
| EW        | 2018-04-30 | 2018-12-31 | 2020-09-30 00:00:00 | -30.3600  | 8                | 21.0000         | 29           |
| EW        | 2021-01-31 | 2021-01-31 | 2021-02-28 00:00:00 | -6.2900   | 0                | 1.0000          | 1            |
| EW        | 2021-06-30 | 2021-08-31 | 2022-01-31 00:00:00 | -13.5800  | 2                | 5.0000          | 7            |

| Portfolio | DD_Start   | DD_Trough  | DD_Recovery         | Depth_pct | Months_to_Trough | Recovery_Months | Total_Months |
|-----------|------------|------------|---------------------|-----------|------------------|-----------------|--------------|
| EW        | 2022-02-28 | 2022-10-31 | 2024-09-30 00:00:00 | -36.5500  | 8                | 23.0000         | 31           |
| EW        | 2024-10-31 | 2024-11-30 | 2025-02-28 00:00:00 | -4.6700   | 1                | 3.0000          | 4            |
| EW        | 2025-04-30 | 2025-04-30 | 2025-05-31 00:00:00 | -4.8700   | 0                | 1.0000          | 1            |
| EW        | 2025-09-30 | 2026-03-31 | Not recovered       | -11.1000  | 6                | nan             | 8            |
| RP        | 2018-04-30 | 2020-03-31 | 2020-09-30 00:00:00 | -32.1400  | 23               | 6.0000          | 29           |
| RP        | 2021-01-31 | 2021-01-31 | 2021-02-28 00:00:00 | -6.0000   | 0                | 1.0000          | 1            |
| RP        | 2021-06-30 | 2021-08-31 | 2022-01-31 00:00:00 | -13.7700  | 2                | 5.0000          | 7            |
| RP        | 2022-02-28 | 2022-10-31 | 2024-09-30 00:00:00 | -35.7200  | 8                | 23.0000         | 31           |
| RP        | 2024-10-31 | 2024-11-30 | 2025-02-28 00:00:00 | -4.5500   | 1                | 3.0000          | 4            |
| RP        | 2025-04-30 | 2025-04-30 | 2025-05-31 00:00:00 | -5.3300   | 0                | 1.0000          | 1            |
| RP        | 2025-09-30 | 2026-03-31 | Not recovered       | -11.7400  | 6                | nan             | 8            |

## 5. Phân tích One-at-a-time

Mỗi tham số thay đổi một lần, các tham số còn lại giữ nguyên giá trị cũ (lookback=36, max\_weight=30%, k=3, rf=VN10Y). Chọn BL Sharpe để dùng để đánh giá mức hiệu quả.

Tham số: lookback

| Value | Is_Baseline | Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino  | MDD       | Calmar   | Win_Rate |
|-------|-------------|-----------|------------|----------|----------|----------|-----------|----------|----------|
| 24    | False       | MKT       | 0.235121   | 0.265019 | 0.739276 | 1.2286   | -0.326455 | 0.720227 | 0.593220 |
| 24    | False       | TAN       | 0.232108   | 0.274176 | 0.703592 | 1.2695   | -0.517867 | 0.448199 | 0.593220 |
| 24    | False       | MV        | 0.208152   | 0.240141 | 0.703555 | 1.2976   | -0.334275 | 0.622695 | 0.567797 |
| 24    | False       | BL        | 0.237854   | 0.282611 | 0.702925 | 1.2523   | -0.374048 | 0.635890 | 0.593220 |
| 24    | False       | EW        | 0.270876   | 0.263778 | 0.878302 | 1.5424   | -0.379934 | 0.712955 | 0.652542 |
| 24    | False       | RP        | 0.262675   | 0.254143 | 0.879330 | 1.5569   | -0.366140 | 0.717416 | 0.635593 |
| 36    | True        | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| 36    | True        | TAN       | 0.191990   | 0.278101 | 0.557740 | 0.872662 | -0.515448 | 0.372472 | 0.622642 |
| 36    | True        | MV        | 0.162408   | 0.253993 | 0.494210 | 0.801308 | -0.384445 | 0.422449 | 0.575472 |
| 36    | True        | BL        | 0.301188   | 0.281283 | 0.939644 | 1.6281   | -0.306594 | 0.982370 | 0.613208 |
| 36    | True        | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| 36    | True        | RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |
| 48    | False       | MKT       | 0.195444   | 0.251826 | 0.636748 | 0.985068 | -0.317482 | 0.615607 | 0.595745 |
| 48    | False       | TAN       | 0.228688   | 0.271121 | 0.714048 | 1.1190   | -0.481909 | 0.474545 | 0.617021 |
| 48    | False       | MV        | 0.141531   | 0.227842 | 0.467152 | 0.703178 | -0.316631 | 0.446992 | 0.563830 |
| 48    | False       | BL        | 0.268895   | 0.264060 | 0.885409 | 1.4524   | -0.309510 | 0.868776 | 0.617021 |
| 48    | False       | EW        | 0.241854   | 0.252678 | 0.818272 | 1.3455   | -0.362445 | 0.667283 | 0.606383 |
| 48    | False       | RP        | 0.229664   | 0.249331 | 0.780366 | 1.2617   | -0.352908 | 0.650777 | 0.617021 |



### Tham số: max\_weight

| Value | Is_Baseline | Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino  | MDD       | Calmar   | Win_Rate |
|-------|-------------|-----------|------------|----------|----------|----------|-----------|----------|----------|
| 20%   | False       | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| 20%   | False       | TAN       | 0.220025   | 0.281697 | 0.650139 | 1.0469   | -0.440590 | 0.499386 | 0.632075 |
| 20%   | False       | MV        | 0.179781   | 0.260903 | 0.547709 | 0.887567 | -0.405647 | 0.443195 | 0.594340 |
| 20%   | False       | BL        | 0.271627   | 0.269711 | 0.870359 | 1.4472   | -0.340115 | 0.798635 | 0.613208 |
| 20%   | False       | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| 20%   | False       | RP        | 0.257003   | 0.271531 | 0.810669 | 1.3922   | -0.357157 | 0.719581 | 0.622642 |
| 30%   | True        | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| 30%   | True        | TAN       | 0.191990   | 0.278101 | 0.557740 | 0.872662 | -0.515448 | 0.372472 | 0.622642 |
| 30%   | True        | MV        | 0.162408   | 0.253993 | 0.494210 | 0.801308 | -0.384445 | 0.422449 | 0.575472 |
| 30%   | True        | BL        | 0.301188   | 0.281283 | 0.939644 | 1.6281   | -0.306594 | 0.982370 | 0.613208 |
| 30%   | True        | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| 30%   | True        | RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |
| 40%   | False       | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| 40%   | False       | TAN       | 0.180892   | 0.281645 | 0.511316 | 0.799540 | -0.516288 | 0.350370 | 0.603774 |
| 40%   | False       | MV        | 0.155090   | 0.257685 | 0.458730 | 0.750771 | -0.385704 | 0.402096 | 0.566038 |
| 40%   | False       | BL        | 0.292378   | 0.283401 | 0.901537 | 1.5546   | -0.337344 | 0.866707 | 0.622642 |
| 40%   | False       | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| 40%   | False       | RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |

### Tham số: k\_clusters

| Value | Is_Baseline | Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino  | MDD       | Calmar   | Win_Rate |
|-------|-------------|-----------|------------|----------|----------|----------|-----------|----------|----------|
| 2     | False       | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| 2     | False       | TAN       | 0.191990   | 0.278101 | 0.557740 | 0.872662 | -0.515448 | 0.372472 | 0.622642 |
| 2     | False       | MV        | 0.162408   | 0.253993 | 0.494210 | 0.801308 | -0.384445 | 0.422449 | 0.575472 |

| Value | Is_Baseline | Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino  | MDD       | Calmar   | Win_Rate |
|-------|-------------|-----------|------------|----------|----------|----------|-----------|----------|----------|
| 2     | False       | BL        | 0.240639   | 0.281538 | 0.723727 | 1.2007   | -0.420598 | 0.572136 | 0.575472 |
| 2     | False       | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| 2     | False       | RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |
| 3     | False       | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| 3     | False       | TAN       | 0.191990   | 0.278101 | 0.557740 | 0.872662 | -0.515448 | 0.372472 | 0.622642 |
| 3     | False       | MV        | 0.162408   | 0.253993 | 0.494210 | 0.801308 | -0.384445 | 0.422449 | 0.575472 |
| 3     | False       | BL        | 0.274545   | 0.285040 | 0.833787 | 1.4258   | -0.328046 | 0.836910 | 0.603774 |
| 3     | False       | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| 3     | False       | RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |
| 4     | True        | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| 4     | True        | TAN       | 0.191990   | 0.278101 | 0.557740 | 0.872662 | -0.515448 | 0.372472 | 0.622642 |
| 4     | True        | MV        | 0.162408   | 0.253993 | 0.494210 | 0.801308 | -0.384445 | 0.422449 | 0.575472 |
| 4     | True        | BL        | 0.301188   | 0.281283 | 0.939644 | 1.6281   | -0.306594 | 0.982370 | 0.613208 |
| 4     | True        | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| 4     | True        | RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |
| 5     | False       | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| 5     | False       | TAN       | 0.191990   | 0.278101 | 0.557740 | 0.872662 | -0.515448 | 0.372472 | 0.622642 |
| 5     | False       | MV        | 0.162408   | 0.253993 | 0.494210 | 0.801308 | -0.384445 | 0.422449 | 0.575472 |
| 5     | False       | BL        | 0.303473   | 0.283770 | 0.939461 | 1.6614   | -0.308210 | 0.984630 | 0.613208 |
| 5     | False       | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| 5     | False       | RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |
| 6     | False       | MKT       | 0.199880   | 0.266787 | 0.626252 | 0.973019 | -0.325098 | 0.614829 | 0.586667 |
| 6     | False       | TAN       | 0.174373   | 0.293778 | 0.481892 | 0.729138 | -0.515448 | 0.338295 | 0.600000 |
| 6     | False       | MV        | 0.136926   | 0.252509 | 0.412351 | 0.624286 | -0.316737 | 0.432302 | 0.560000 |
| 6     | False       | BL        | 0.318746   | 0.274950 | 1.0400   | 1.8278   | -0.247620 | 1.2872   | 0.640000 |

| Value | Is_Baseline | Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino | MDD       | Calmar   | Win_Rate |
|-------|-------------|-----------|------------|----------|----------|---------|-----------|----------|----------|
| 6     | False       | EW        | 0.277651   | 0.273653 | 0.894735 | 1.5269  | -0.365473 | 0.759704 | 0.626667 |
| 6     | False       | RP        | 0.266168   | 0.272824 | 0.855365 | 1.4381  | -0.357157 | 0.745242 | 0.626667 |

Tham số: rf\_rate

| Value           | Is_Baseline | Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino  | MDD       | Calmar   | Win_Rate |
|-----------------|-------------|-----------|------------|----------|----------|----------|-----------|----------|----------|
| 4% fixed        | False       | MKT       | 0.234033   | 0.270493 | 0.717331 | 1.1624   | -0.325098 | 0.719883 | 0.613208 |
| 4% fixed        | False       | TAN       | 0.192536   | 0.278191 | 0.548314 | 0.856354 | -0.515448 | 0.373532 | 0.622642 |
| 4% fixed        | False       | MV        | 0.162408   | 0.253993 | 0.481935 | 0.779136 | -0.384445 | 0.422449 | 0.575472 |
| 4% fixed        | False       | BL        | 0.304884   | 0.282740 | 0.936849 | 1.6275   | -0.306046 | 0.996204 | 0.613208 |
| 4% fixed        | False       | EW        | 0.266157   | 0.273821 | 0.825931 | 1.4304   | -0.365473 | 0.728253 | 0.622642 |
| 4% fixed        | False       | RP        | 0.256658   | 0.271436 | 0.798190 | 1.3674   | -0.357157 | 0.718613 | 0.622642 |
| 5% fixed        | False       | MKT       | 0.234033   | 0.270493 | 0.680361 | 1.0937   | -0.325098 | 0.719883 | 0.613208 |
| 5% fixed        | False       | TAN       | 0.192949   | 0.277950 | 0.514298 | 0.796449 | -0.514923 | 0.374715 | 0.622642 |
| 5% fixed        | False       | MV        | 0.162408   | 0.253993 | 0.442564 | 0.709654 | -0.384445 | 0.422449 | 0.575472 |
| 5% fixed        | False       | BL        | 0.308893   | 0.284987 | 0.908438 | 1.5691   | -0.306493 | 1.0078   | 0.613208 |
| 5% fixed        | False       | EW        | 0.266157   | 0.273821 | 0.789410 | 1.3555   | -0.365473 | 0.728253 | 0.622642 |
| 5% fixed        | False       | RP        | 0.256658   | 0.271436 | 0.761349 | 1.2933   | -0.357157 | 0.718613 | 0.622642 |
| 6% fixed        | False       | MKT       | 0.234033   | 0.270493 | 0.643392 | 1.0260   | -0.325098 | 0.719883 | 0.613208 |
| 6% fixed        | False       | TAN       | 0.194646   | 0.277630 | 0.484982 | 0.745961 | -0.513095 | 0.379356 | 0.632075 |
| 6% fixed        | False       | MV        | 0.162408   | 0.253993 | 0.403192 | 0.641243 | -0.384445 | 0.422449 | 0.575472 |
| 6% fixed        | False       | BL        | 0.310200   | 0.286957 | 0.871907 | 1.4982   | -0.306994 | 1.0104   | 0.613208 |
| 6% fixed        | False       | EW        | 0.266157   | 0.273821 | 0.752890 | 1.2817   | -0.365473 | 0.728253 | 0.622642 |
| 6% fixed        | False       | RP        | 0.256658   | 0.271436 | 0.724508 | 1.2203   | -0.357157 | 0.718613 | 0.622642 |
| VN10Y (dynamic) | True        | MKT       | 0.234033   | 0.270493 | 0.728858 | 1.1842   | -0.325098 | 0.719883 | 0.613208 |
| VN10Y (dynamic) | True        | TAN       | 0.191990   | 0.278101 | 0.557740 | 0.872662 | -0.515448 | 0.372472 | 0.622642 |

| Value           | Is_Baseline | Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino  | MDD       | Calmar   | Win_Rate |
|-----------------|-------------|-----------|------------|----------|----------|----------|-----------|----------|----------|
| VN10Y (dynamic) | True        | MV        | 0.162408   | 0.253993 | 0.494210 | 0.801308 | -0.384445 | 0.422449 | 0.575472 |
| VN10Y (dynamic) | True        | BL        | 0.301188   | 0.281283 | 0.939644 | 1.6281   | -0.306594 | 0.982370 | 0.613208 |
| VN10Y (dynamic) | True        | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 | 0.622642 |
| VN10Y (dynamic) | True        | RP        | 0.256658   | 0.271436 | 0.809677 | 1.3900   | -0.357157 | 0.718613 | 0.622642 |

## 6. Monte Carlo – b n v ng Sharpe tr c sai s d báo

V i m i m c nhi u  $\delta \in \{-10\%, -5\%, 0\%, +5\%, +10\%\}$ , view vector  $q$  c nhi u hoá 2000 l n theo phân ph i chu n.  $\mu_{BL}$  c tính closed-form (tuy n tính trong  $q$ ), Sharpe c a  $w_{BL}$  c nh c ánh giá d i m i  $q$  nhi u. C t TAN là Sharpe c nh (không dùng views).

| Delta_Noise | Port_Type | Mean_Sharpe | Std_Sharpe | Min_Sharpe | Max_Sharpe |
|-------------|-----------|-------------|------------|------------|------------|
| -0.100000   | BL        | 0.828200    | 0.527900   | -0.173700  | 2.5327     |
| -0.100000   | TAN       | 1.0271      | 0.398100   | 0.319600   | 2.3520     |
| -0.050000   | BL        | 0.831400    | 0.530900   | -0.175700  | 2.5461     |
| -0.050000   | TAN       | 1.0271      | 0.398100   | 0.319600   | 2.3520     |
| 0.000000    | BL        | 0.831100    | 0.530600   | -0.175500  | 2.5447     |
| 0.000000    | TAN       | 1.0271      | 0.398100   | 0.319600   | 2.3520     |
| 0.050000    | BL        | 0.830700    | 0.530200   | -0.175200  | 2.5429     |
| 0.050000    | TAN       | 1.0271      | 0.398100   | 0.319600   | 2.3520     |
| 0.100000    | BL        | 0.830400    | 0.530000   | -0.175100  | 2.5418     |
| 0.100000    | TAN       | 1.0271      | 0.398100   | 0.319600   | 2.3520     |

Full Monte Carlo data (first 60 rows shown):

| Date       | Delta_Noise | Port_Type | Expected_Sharpe |
|------------|-------------|-----------|-----------------|
| 2017-08-31 | -0.100000   | BL        | 0.257974        |
| 2017-08-31 | -0.100000   | TAN       | 0.852097        |
| 2017-08-31 | -0.050000   | BL        | 0.259102        |
| 2017-08-31 | -0.050000   | TAN       | 0.852097        |
| 2017-08-31 | 0.000000    | BL        | 0.258983        |
| 2017-08-31 | 0.000000    | TAN       | 0.852097        |
| 2017-08-31 | 0.050000    | BL        | 0.258831        |
| 2017-08-31 | 0.050000    | TAN       | 0.852097        |

| Date       | Delta_Noise | Port_Type | Expected_Sharpe |
|------------|-------------|-----------|-----------------|
| 2017-08-31 | 0.100000    | BL        | 0.258744        |
| 2017-08-31 | 0.100000    | TAN       | 0.852097        |
| 2017-09-30 | -0.100000   | BL        | 0.350599        |
| 2017-09-30 | -0.100000   | TAN       | 0.917878        |
| 2017-09-30 | -0.050000   | BL        | 0.352073        |
| 2017-09-30 | -0.050000   | TAN       | 0.917878        |
| 2017-09-30 | 0.000000    | BL        | 0.351917        |
| 2017-09-30 | 0.000000    | TAN       | 0.917878        |
| 2017-09-30 | 0.050000    | BL        | 0.351720        |
| 2017-09-30 | 0.050000    | TAN       | 0.917878        |
| 2017-09-30 | 0.100000    | BL        | 0.351606        |
| 2017-09-30 | 0.100000    | TAN       | 0.917878        |
| 2017-10-31 | -0.100000   | BL        | 0.380144        |
| 2017-10-31 | -0.100000   | TAN       | 0.879333        |
| 2017-10-31 | -0.050000   | BL        | 0.382273        |
| 2017-10-31 | -0.050000   | TAN       | 0.879333        |
| 2017-10-31 | 0.000000    | BL        | 0.382048        |
| 2017-10-31 | 0.000000    | TAN       | 0.879333        |
| 2017-10-31 | 0.050000    | BL        | 0.381763        |
| 2017-10-31 | 0.050000    | TAN       | 0.879333        |
| 2017-10-31 | 0.100000    | BL        | 0.381599        |
| 2017-10-31 | 0.100000    | TAN       | 0.879333        |
| 2017-11-30 | -0.100000   | BL        | 0.446482        |
| 2017-11-30 | -0.100000   | TAN       | 0.975314        |
| 2017-11-30 | -0.050000   | BL        | 0.448579        |

| Date       | Delta_Noise | Port_Type | Expected_Sharpe |
|------------|-------------|-----------|-----------------|
| 2017-11-30 | -0.050000   | TAN       | 0.975314        |
| 2017-11-30 | 0.000000    | BL        | 0.448358        |
| 2017-11-30 | 0.000000    | TAN       | 0.975314        |
| 2017-11-30 | 0.050000    | BL        | 0.448077        |
| 2017-11-30 | 0.050000    | TAN       | 0.975314        |
| 2017-11-30 | 0.100000    | BL        | 0.447915        |
| 2017-11-30 | 0.100000    | TAN       | 0.975314        |
| 2017-12-31 | -0.100000   | BL        | 0.363535        |
| 2017-12-31 | -0.100000   | TAN       | 1.0086          |
| 2017-12-31 | -0.050000   | BL        | 0.364918        |
| 2017-12-31 | -0.050000   | TAN       | 1.0086          |
| 2017-12-31 | 0.000000    | BL        | 0.364772        |
| 2017-12-31 | 0.000000    | TAN       | 1.0086          |
| 2017-12-31 | 0.050000    | BL        | 0.364587        |
| 2017-12-31 | 0.050000    | TAN       | 1.0086          |
| 2017-12-31 | 0.100000    | BL        | 0.364480        |
| 2017-12-31 | 0.100000    | TAN       | 1.0086          |
| 2018-01-31 | -0.100000   | BL        | 0.110884        |
| 2018-01-31 | -0.100000   | TAN       | 1.2179          |
| 2018-01-31 | -0.050000   | BL        | 0.108213        |
| 2018-01-31 | -0.050000   | TAN       | 1.2179          |
| 2018-01-31 | 0.000000    | BL        | 0.108495        |
| 2018-01-31 | 0.000000    | TAN       | 1.2179          |
| 2018-01-31 | 0.050000    | BL        | 0.108853        |
| 2018-01-31 | 0.050000    | TAN       | 1.2179          |

| Date       | Delta_Noise | Port_Type | Expected_Sharpe |
|------------|-------------|-----------|-----------------|
| 2018-01-31 | 0.100000    | BL        | 0.109059        |
| 2018-01-31 | 0.100000    | TAN       | 1.2179          |



## 7. Chuỗi Lãi nhuận Out-of-Sample (106 tháng)

Lãi nhuận tháng (simple return) của 6 danh mục và lãi suất phi rủi ro cho từng tháng OOS.

| Date       | RF_monthly | MKT       | TAN       | MV        | BL        | EW        | RP        |
|------------|------------|-----------|-----------|-----------|-----------|-----------|-----------|
| 2017-08-31 | 0.004833   | -0.026592 | 0.021803  | 0.000532  | -0.050765 | -0.024700 | -0.020210 |
| 2017-09-30 | 0.004833   | 0.008600  | 0.017614  | 0.012287  | 0.007288  | 0.014080  | 0.012381  |
| 2017-10-31 | 0.004833   | 0.059921  | 0.050255  | 0.029226  | -0.020522 | 0.017326  | 0.018877  |
| 2017-11-30 | 0.004833   | 0.184409  | 0.152730  | 0.163976  | 0.175780  | 0.187071  | 0.182169  |
| 2017-12-31 | 0.004833   | 0.048090  | 0.050632  | 0.034358  | -0.011364 | 0.018884  | 0.020052  |
| 2018-01-31 | 0.003817   | 0.243752  | 0.230535  | 0.294043  | 0.267976  | 0.267098  | 0.266837  |
| 2018-02-28 | 0.003817   | 0.120337  | 0.099277  | 0.077586  | 0.134487  | 0.098473  | 0.096365  |
| 2018-03-31 | 0.003817   | 0.016710  | 0.017842  | 0.015931  | 0.041644  | 0.020543  | 0.021760  |
| 2018-04-30 | 0.003817   | -0.146733 | -0.126264 | -0.123802 | -0.138059 | -0.131849 | -0.131825 |
| 2018-05-31 | 0.003817   | -0.114886 | -0.073255 | -0.076191 | -0.114640 | -0.111764 | -0.103784 |
| 2018-06-30 | 0.003817   | -0.032586 | -0.053069 | -0.064188 | -0.063714 | -0.069962 | -0.069599 |
| 2018-07-31 | 0.003817   | 0.007452  | 0.022622  | 0.022947  | -0.005468 | 0.006366  | 0.009336  |
| 2018-08-31 | 0.003817   | 0.123037  | 0.060966  | 0.055417  | 0.172304  | 0.095768  | 0.090940  |
| 2018-09-30 | 0.003817   | 0.018549  | 0.002697  | 0.004246  | 0.012967  | 0.040664  | 0.033908  |
| 2018-10-31 | 0.003817   | -0.121796 | -0.095982 | -0.095374 | -0.126462 | -0.115741 | -0.111643 |
| 2018-11-30 | 0.003817   | 0.007542  | -0.013142 | -0.013195 | -0.004620 | -0.005717 | -0.007262 |
| 2018-12-31 | 0.003817   | -0.018578 | -0.045817 | -0.044849 | -0.051438 | -0.037550 | -0.042356 |
| 2019-01-31 | 0.003817   | 0.012980  | 0.048089  | 0.056572  | -0.000935 | 0.021086  | 0.028444  |
| 2019-02-28 | 0.003817   | 0.041454  | 0.019705  | 0.016662  | 0.034640  | 0.024012  | 0.023248  |
| 2019-03-31 | 0.003817   | 0.087474  | 0.063028  | 0.052068  | 0.074795  | 0.052151  | 0.052914  |
| 2019-04-30 | 0.003817   | -0.009915 | -0.002335 | -0.001570 | -0.007965 | -0.014232 | -0.012657 |

| Date       | RF_monthly | MKT       | TAN       | MV        | BL        | EW        | RP        |
|------------|------------|-----------|-----------|-----------|-----------|-----------|-----------|
| 2019-05-31 | 0.003817   | -0.040294 | -0.036642 | -0.028837 | -0.033866 | -0.045280 | -0.042339 |
| 2019-06-30 | 0.003817   | 0.018816  | 0.015486  | 0.011442  | 0.004369  | -0.006345 | -0.003536 |
| 2019-07-31 | 0.003817   | 0.103573  | 0.079358  | 0.061666  | 0.099161  | 0.045768  | 0.049044  |
| 2019-08-31 | 0.003817   | 0.003178  | 0.004126  | -0.007868 | 0.014018  | -0.003207 | -0.004538 |
| 2019-09-30 | 0.003817   | 0.044382  | 0.054395  | 0.054182  | 0.053843  | 0.042613  | 0.044720  |
| 2019-10-31 | 0.003817   | 0.047669  | 0.035522  | 0.034822  | 0.044811  | 0.028980  | 0.030364  |
| 2019-11-30 | 0.003817   | -0.019920 | -0.028705 | -0.038766 | -0.023996 | -0.042676 | -0.042495 |
| 2019-12-31 | 0.003817   | 0.067450  | 0.011788  | -0.001566 | 0.045882  | 0.032970  | 0.026890  |
| 2020-01-31 | 0.002500   | 0.040052  | 0.009322  | -0.000004 | 0.081208  | 0.064014  | 0.053729  |
| 2020-02-29 | 0.002500   | -0.047792 | -0.041175 | 0.029620  | -0.026162 | 0.089194  | 0.078052  |
| 2020-03-31 | 0.002500   | -0.276107 | -0.300868 | -0.301494 | -0.287965 | -0.240255 | -0.256221 |
| 2020-04-30 | 0.002500   | 0.137555  | 0.150589  | 0.155972  | 0.134515  | 0.178490  | 0.171579  |
| 2020-05-31 | 0.002500   | 0.168756  | 0.089504  | 0.089163  | 0.068182  | 0.104038  | 0.112812  |
| 2020-06-30 | 0.002500   | -0.046926 | -0.055452 | -0.035664 | -0.015210 | -0.018955 | -0.017534 |
| 2020-07-31 | 0.002500   | -0.040333 | -0.053315 | -0.048023 | -0.051459 | -0.035201 | -0.033178 |
| 2020-08-31 | 0.002500   | 0.121801  | 0.114106  | 0.125604  | 0.146489  | 0.146584  | 0.145434  |
| 2020-09-30 | 0.002500   | 0.034012  | 0.054371  | 0.061032  | 0.130943  | 0.097577  | 0.096664  |
| 2020-10-31 | 0.002500   | -0.000661 | 0.013751  | 0.034318  | 0.017359  | 0.034211  | 0.037935  |
| 2020-11-30 | 0.002500   | 0.114471  | 0.091426  | 0.111259  | 0.109581  | 0.107254  | 0.107803  |
| 2020-12-31 | 0.002500   | 0.100638  | 0.157318  | 0.165052  | 0.128925  | 0.139943  | 0.145595  |
| 2021-01-31 | 0.002108   | -0.078755 | -0.080811 | -0.080838 | -0.032355 | -0.062929 | -0.059956 |
| 2021-02-28 | 0.002108   | 0.128424  | 0.104445  | 0.056616  | 0.132551  | 0.144913  | 0.127255  |
| 2021-03-31 | 0.002108   | 0.043929  | 0.195558  | 0.224080  | 0.238249  | 0.147558  | 0.164876  |
| 2021-04-30 | 0.002108   | 0.069686  | 0.087668  | 0.039030  | 0.155218  | 0.089302  | 0.062330  |
| 2021-05-31 | 0.002108   | 0.154716  | 0.220957  | 0.144040  | 0.201190  | 0.233709  | 0.228602  |

| Date       | RF_monthly | MKT       | TAN       | MV        | BL        | EW        | RP        |
|------------|------------|-----------|-----------|-----------|-----------|-----------|-----------|
| 2021-06-30 | 0.002108   | 0.044947  | 0.004665  | -0.018138 | -0.068264 | -0.004647 | -0.006477 |
| 2021-07-31 | 0.002108   | -0.113116 | -0.123328 | -0.111084 | -0.094478 | -0.091124 | -0.092756 |
| 2021-08-31 | 0.002108   | -0.034625 | -0.037530 | -0.003530 | -0.038708 | -0.044764 | -0.043336 |
| 2021-09-30 | 0.002108   | 0.000996  | 0.011400  | 0.025518  | -0.028558 | 0.004185  | 0.007465  |
| 2021-10-31 | 0.002108   | 0.030575  | 0.031964  | 0.028788  | 0.032399  | 0.040120  | 0.038113  |
| 2021-11-30 | 0.002108   | 0.032911  | 0.057504  | 0.030239  | 0.059137  | 0.052475  | 0.051667  |
| 2021-12-31 | 0.002108   | 0.020766  | 0.083291  | 0.030334  | 0.031570  | 0.032872  | 0.034591  |
| 2022-01-31 | 0.003333   | 0.104257  | 0.031341  | 0.043366  | 0.035040  | 0.076226  | 0.074845  |
| 2022-02-28 | 0.003333   | -0.043408 | -0.019388 | -0.027255 | -0.020250 | -0.038160 | -0.037485 |
| 2022-03-31 | 0.003333   | -0.020615 | -0.022193 | -0.016968 | 0.063683  | 0.004651  | 0.002219  |
| 2022-04-30 | 0.003333   | -0.082787 | -0.147377 | -0.099271 | -0.107490 | -0.109177 | -0.105956 |
| 2022-05-31 | 0.003333   | -0.072449 | -0.103551 | -0.044184 | -0.041574 | -0.081414 | -0.075345 |
| 2022-06-30 | 0.003333   | -0.068432 | -0.136007 | -0.070779 | -0.082785 | -0.087762 | -0.087492 |
| 2022-07-31 | 0.003333   | 0.044292  | 0.090316  | 0.018105  | 0.065715  | 0.059262  | 0.055490  |
| 2022-08-31 | 0.003333   | 0.074039  | 0.013024  | 0.046948  | 0.088153  | 0.050773  | 0.050525  |
| 2022-09-30 | 0.003333   | -0.143466 | -0.127780 | -0.109730 | -0.138716 | -0.136735 | -0.133303 |
| 2022-10-31 | 0.003333   | -0.053868 | -0.057153 | -0.033293 | -0.007417 | -0.084503 | -0.080754 |
| 2022-11-30 | 0.003333   | 0.083200  | -0.084619 | 0.002064  | 0.029794  | 0.028596  | 0.027381  |
| 2022-12-31 | 0.003333   | -0.006743 | -0.079643 | -0.028363 | 0.009177  | -0.010396 | -0.011879 |
| 2023-01-31 | 0.002917   | 0.148118  | 0.157561  | 0.143657  | 0.149694  | 0.152493  | 0.153030  |
| 2023-02-28 | 0.002917   | -0.047656 | -0.080255 | 0.005233  | -0.052600 | -0.061735 | -0.054593 |
| 2023-03-31 | 0.002917   | 0.047184  | 0.070172  | -0.007017 | 0.050817  | 0.052515  | 0.047303  |
| 2023-04-30 | 0.002917   | -0.020110 | -0.037522 | -0.006488 | -0.021901 | -0.006776 | -0.007503 |
| 2023-05-31 | 0.002917   | 0.021645  | 0.031663  | 0.037142  | 0.019517  | 0.023716  | 0.024116  |
| 2023-06-30 | 0.002917   | 0.046573  | 0.048782  | 0.045352  | 0.045279  | 0.056366  | 0.052440  |

| Date       | RF_monthly | MKT       | TAN       | MV        | BL        | EW        | RP        |
|------------|------------|-----------|-----------|-----------|-----------|-----------|-----------|
| 2023-07-31 | 0.002917   | 0.072933  | 0.058160  | 0.042184  | 0.044361  | 0.064792  | 0.063650  |
| 2023-08-31 | 0.002917   | -0.002617 | 0.011854  | -0.017438 | 0.029900  | 0.016439  | 0.012578  |
| 2023-09-30 | 0.002917   | -0.031677 | -0.030950 | -0.019040 | -0.039045 | -0.045869 | -0.041493 |
| 2023-10-31 | 0.002917   | -0.059835 | -0.003531 | -0.033969 | -0.054354 | -0.075824 | -0.070516 |
| 2023-11-30 | 0.002917   | 0.024465  | 0.009890  | 0.007350  | 0.018884  | 0.044460  | 0.040951  |
| 2023-12-31 | 0.002917   | 0.023527  | 0.008199  | 0.017215  | 0.029754  | 0.036214  | 0.037548  |
| 2024-01-31 | 0.002550   | 0.091943  | 0.075801  | 0.078301  | 0.099873  | 0.072711  | 0.076635  |
| 2024-02-29 | 0.002550   | 0.095842  | 0.043038  | 0.063421  | 0.051506  | 0.065798  | 0.070645  |
| 2024-03-31 | 0.002550   | 0.003849  | 0.003879  | -0.000978 | 0.006890  | 0.004252  | 0.004793  |
| 2024-04-30 | 0.002550   | -0.048663 | -0.013038 | -0.042120 | 0.017273  | -0.029587 | -0.033917 |
| 2024-05-31 | 0.002550   | -0.003662 | 0.002642  | 0.010818  | 0.010148  | 0.022505  | 0.020859  |
| 2024-06-30 | 0.002550   | -0.018775 | -0.022561 | -0.020244 | -0.039007 | -0.012736 | -0.012871 |
| 2024-07-31 | 0.002550   | 0.048634  | 0.052790  | 0.019355  | 0.047735  | 0.037203  | 0.038247  |
| 2024-08-31 | 0.002550   | 0.025287  | 0.051975  | 0.001160  | 0.033135  | 0.018478  | 0.016333  |
| 2024-09-30 | 0.002550   | 0.026921  | 0.012138  | -0.015353 | 0.036651  | 0.039055  | 0.034229  |
| 2024-10-31 | 0.002550   | -0.008231 | -0.015224 | -0.011221 | 0.010314  | -0.024802 | -0.022269 |
| 2024-11-30 | 0.002550   | -0.020013 | -0.021911 | -0.013213 | -0.008478 | -0.022484 | -0.023717 |
| 2024-12-31 | 0.002550   | 0.018309  | 0.093132  | 0.011599  | 0.040576  | 0.030247  | 0.029882  |
| 2025-01-31 | 0.002583   | 0.018841  | 0.049506  | 0.037886  | 0.063690  | 0.011546  | 0.010500  |
| 2025-02-28 | 0.002583   | 0.028168  | 0.014633  | 0.032052  | 0.039368  | 0.045352  | 0.044752  |
| 2025-03-31 | 0.002583   | 0.000756  | -0.039288 | 0.006031  | -0.024999 | 0.000248  | 0.001343  |
| 2025-04-30 | 0.002583   | -0.076861 | -0.054662 | -0.063103 | -0.035208 | -0.048695 | -0.053265 |
| 2025-05-31 | 0.002583   | 0.042136  | 0.026155  | 0.015781  | 0.040975  | 0.077463  | 0.068187  |
| 2025-06-30 | 0.002583   | 0.038606  | 0.018894  | 0.012406  | 0.063788  | 0.029487  | 0.029292  |
| 2025-07-31 | 0.002583   | 0.101024  | 0.086195  | 0.093231  | 0.115972  | 0.156467  | 0.153496  |

| Date       | RF_monthly | MKT       | TAN       | MV        | BL        | EW        | RP        |
|------------|------------|-----------|-----------|-----------|-----------|-----------|-----------|
| 2025-08-31 | 0.002583   | 0.191455  | 0.174109  | 0.093140  | 0.092556  | 0.155768  | 0.152875  |
| 2025-09-30 | 0.002583   | -0.061845 | -0.026854 | -0.084397 | -0.020579 | -0.059003 | -0.065239 |
| 2025-10-31 | 0.002583   | -0.042226 | -0.006632 | -0.027556 | 0.024278  | -0.019916 | -0.022683 |
| 2025-11-30 | 0.002583   | -0.019884 | -0.018711 | -0.037211 | -0.014664 | -0.016465 | -0.017814 |
| 2025-12-31 | 0.002583   | 0.025236  | -0.006659 | -0.001797 | 0.053382  | 0.016349  | 0.014466  |
| 2026-01-31 | 0.003625   | 0.098177  | 0.000622  | 0.047368  | -0.015152 | 0.034101  | 0.042349  |
| 2026-02-28 | 0.003625   | -0.023014 | 0.007064  | -0.013919 | 0.013093  | -0.004318 | -0.007500 |
| 2026-03-31 | 0.003625   | -0.093967 | -0.064750 | -0.042369 | -0.063529 | -0.063432 | -0.062731 |
| 2026-04-30 | 0.003625   | 0.029591  | 0.091541  | 0.029153  | 0.033291  | 0.036212  | 0.033699  |
| 2026-05-31 | 0.003625   | 0.029712  | 0.079321  | 0.011041  | 0.010291  | 0.011828  | 0.009975  |

## 8. Tín hiệu Phân cụm K-means

Nhấn cụm và vị thế view (Long / Short / Neutral) của từng cổ phiếu từ mồi khởi tái cân bằng. Phân cụm dựa trên idiosyncratic momentum 6-1 và low-volatility feature được chuẩn hoá trước khi đưa vào K-means (k=4). Tín hiệu từng hập theo Jegadeesh & Titman (1993) và Baker, Bradley & Wurgler (2011).

Tổng số quan sát: 1,561 dòng

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2017-08-31 | VCB    | 0             | Short                  |
| 2017-08-31 | BID    | 2             | Neutral                |
| 2017-08-31 | CTG    | 2             | Neutral                |
| 2017-08-31 | MBB    | 3             | Long                   |
| 2017-08-31 | ACB    | 1             | Neutral                |
| 2017-08-31 | STB    | 1             | Neutral                |
| 2017-08-31 | SHB    | 1             | Neutral                |
| 2017-09-30 | VCB    | 2             | Short                  |
| 2017-09-30 | BID    | 0             | Neutral                |
| 2017-09-30 | CTG    | 0             | Neutral                |
| 2017-09-30 | MBB    | 1             | Long                   |
| 2017-09-30 | ACB    | 3             | Neutral                |
| 2017-09-30 | STB    | 0             | Neutral                |
| 2017-09-30 | SHB    | 3             | Neutral                |
| 2017-10-31 | VCB    | 2             | Short                  |
| 2017-10-31 | BID    | 0             | Neutral                |
| 2017-10-31 | CTG    | 0             | Neutral                |
| 2017-10-31 | MBB    | 3             | Long                   |
| 2017-10-31 | ACB    | 2             | Short                  |
| 2017-10-31 | STB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2017-10-31 | SHB    | 1             | Neutral                |
| 2017-11-30 | VCB    | 0             | Short                  |
| 2017-11-30 | BID    | 2             | Neutral                |
| 2017-11-30 | CTG    | 2             | Neutral                |
| 2017-11-30 | MBB    | 3             | Long                   |
| 2017-11-30 | ACB    | 1             | Neutral                |
| 2017-11-30 | STB    | 2             | Neutral                |
| 2017-11-30 | SHB    | 1             | Neutral                |
| 2017-12-31 | VCB    | 2             | Short                  |
| 2017-12-31 | BID    | 0             | Neutral                |
| 2017-12-31 | CTG    | 0             | Neutral                |
| 2017-12-31 | MBB    | 3             | Long                   |
| 2017-12-31 | ACB    | 1             | Neutral                |
| 2017-12-31 | STB    | 0             | Neutral                |
| 2017-12-31 | SHB    | 1             | Neutral                |
| 2018-01-31 | VCB    | 0             | Neutral                |
| 2018-01-31 | BID    | 3             | Neutral                |
| 2018-01-31 | CTG    | 1             | Short                  |
| 2018-01-31 | MBB    | 2             | Long                   |
| 2018-01-31 | ACB    | 0             | Neutral                |
| 2018-01-31 | STB    | 1             | Short                  |
| 2018-01-31 | SHB    | 1             | Short                  |
| 2018-02-28 | VCB    | 1             | Neutral                |
| 2018-02-28 | BID    | 0             | Neutral                |
| 2018-02-28 | CTG    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2018-02-28 | MBB    | 2             | Long                   |
| 2018-02-28 | ACB    | 1             | Neutral                |
| 2018-02-28 | STB    | 3             | Short                  |
| 2018-02-28 | SHB    | 1             | Neutral                |
| 2018-03-31 | VCB    | 2             | Long                   |
| 2018-03-31 | BID    | 0             | Neutral                |
| 2018-03-31 | CTG    | 0             | Neutral                |
| 2018-03-31 | MBB    | 3             | Neutral                |
| 2018-03-31 | ACB    | 2             | Long                   |
| 2018-03-31 | STB    | 1             | Short                  |
| 2018-03-31 | SHB    | 0             | Neutral                |
| 2018-04-30 | VCB    | 0             | Long                   |
| 2018-04-30 | BID    | 2             | Neutral                |
| 2018-04-30 | CTG    | 1             | Short                  |
| 2018-04-30 | MBB    | 3             | Neutral                |
| 2018-04-30 | ACB    | 0             | Long                   |
| 2018-04-30 | STB    | 1             | Short                  |
| 2018-04-30 | SHB    | 2             | Neutral                |
| 2018-05-31 | VCB    | 2             | Long                   |
| 2018-05-31 | BID    | 0             | Neutral                |
| 2018-05-31 | CTG    | 0             | Neutral                |
| 2018-05-31 | MBB    | 3             | Neutral                |
| 2018-05-31 | ACB    | 3             | Neutral                |
| 2018-05-31 | STB    | 1             | Short                  |
| 2018-05-31 | SHB    | 0             | Neutral                |



| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2018-06-30 | VCB    | 0             | Neutral                |
| 2018-06-30 | BID    | 3             | Long                   |
| 2018-06-30 | CTG    | 1             | Neutral                |
| 2018-06-30 | MBB    | 0             | Neutral                |
| 2018-06-30 | ACB    | 0             | Neutral                |
| 2018-06-30 | STB    | 2             | Short                  |
| 2018-06-30 | SHB    | 1             | Neutral                |
| 2018-07-31 | VCB    | 0             | Neutral                |
| 2018-07-31 | BID    | 1             | Long                   |
| 2018-07-31 | CTG    | 1             | Long                   |
| 2018-07-31 | MBB    | 2             | Neutral                |
| 2018-07-31 | ACB    | 0             | Neutral                |
| 2018-07-31 | STB    | 3             | Short                  |
| 2018-07-31 | SHB    | 1             | Long                   |
| 2018-08-31 | VCB    | 2             | Neutral                |
| 2018-08-31 | BID    | 0             | Neutral                |
| 2018-08-31 | CTG    | 0             | Neutral                |
| 2018-08-31 | MBB    | 2             | Neutral                |
| 2018-08-31 | ACB    | 3             | Long                   |
| 2018-08-31 | STB    | 1             | Short                  |
| 2018-08-31 | SHB    | 0             | Neutral                |
| 2018-09-30 | VCB    | 2             | Long                   |
| 2018-09-30 | BID    | 0             | Neutral                |
| 2018-09-30 | CTG    | 0             | Neutral                |
| 2018-09-30 | MBB    | 2             | Long                   |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2018-09-30 | ACB    | 2             | Long                   |
| 2018-09-30 | STB    | 3             | Short                  |
| 2018-09-30 | SHB    | 1             | Neutral                |
| 2018-10-31 | VCB    | 2             | Long                   |
| 2018-10-31 | BID    | 3             | Neutral                |
| 2018-10-31 | CTG    | 0             | Neutral                |
| 2018-10-31 | MBB    | 2             | Long                   |
| 2018-10-31 | ACB    | 2             | Long                   |
| 2018-10-31 | STB    | 3             | Neutral                |
| 2018-10-31 | SHB    | 1             | Short                  |
| 2018-11-30 | VCB    | 2             | Long                   |
| 2018-11-30 | BID    | 3             | Neutral                |
| 2018-11-30 | CTG    | 0             | Neutral                |
| 2018-11-30 | MBB    | 2             | Long                   |
| 2018-11-30 | ACB    | 2             | Long                   |
| 2018-11-30 | STB    | 1             | Short                  |
| 2018-11-30 | SHB    | 1             | Short                  |
| 2018-12-31 | VCB    | 2             | Long                   |
| 2018-12-31 | BID    | 3             | Neutral                |
| 2018-12-31 | CTG    | 0             | Neutral                |
| 2018-12-31 | MBB    | 0             | Neutral                |
| 2018-12-31 | ACB    | 2             | Long                   |
| 2018-12-31 | STB    | 3             | Neutral                |
| 2018-12-31 | SHB    | 1             | Short                  |
| 2019-01-31 | VCB    | 0             | Long                   |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2019-01-31 | BID    | 3             | Neutral                |
| 2019-01-31 | CTG    | 2             | Neutral                |
| 2019-01-31 | MBB    | 0             | Long                   |
| 2019-01-31 | ACB    | 2             | Neutral                |
| 2019-01-31 | STB    | 3             | Neutral                |
| 2019-01-31 | SHB    | 1             | Short                  |
| 2019-02-28 | VCB    | 2             | Long                   |
| 2019-02-28 | BID    | 3             | Neutral                |
| 2019-02-28 | CTG    | 0             | Short                  |
| 2019-02-28 | MBB    | 1             | Neutral                |
| 2019-02-28 | ACB    | 1             | Neutral                |
| 2019-02-28 | STB    | 3             | Neutral                |
| 2019-02-28 | SHB    | 0             | Short                  |
| 2019-03-31 | VCB    | 0             | Neutral                |
| 2019-03-31 | BID    | 2             | Long                   |
| 2019-03-31 | CTG    | 1             | Short                  |
| 2019-03-31 | MBB    | 0             | Neutral                |
| 2019-03-31 | ACB    | 0             | Neutral                |
| 2019-03-31 | STB    | 3             | Neutral                |
| 2019-03-31 | SHB    | 1             | Short                  |
| 2019-04-30 | VCB    | 0             | Neutral                |
| 2019-04-30 | BID    | 2             | Long                   |
| 2019-04-30 | CTG    | 1             | Short                  |
| 2019-04-30 | MBB    | 0             | Neutral                |
| 2019-04-30 | ACB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2019-04-30 | STB    | 3             | Neutral                |
| 2019-04-30 | SHB    | 1             | Short                  |
| 2019-05-31 | VCB    | 2             | Neutral                |
| 2019-05-31 | BID    | 3             | Neutral                |
| 2019-05-31 | CTG    | 0             | Short                  |
| 2019-05-31 | MBB    | 2             | Neutral                |
| 2019-05-31 | ACB    | 2             | Neutral                |
| 2019-05-31 | STB    | 1             | Long                   |
| 2019-05-31 | SHB    | 3             | Neutral                |
| 2019-06-30 | VCB    | 0             | Long                   |
| 2019-06-30 | BID    | 2             | Neutral                |
| 2019-06-30 | CTG    | 1             | Short                  |
| 2019-06-30 | MBB    | 3             | Neutral                |
| 2019-06-30 | ACB    | 3             | Neutral                |
| 2019-06-30 | STB    | 1             | Short                  |
| 2019-06-30 | SHB    | 1             | Short                  |
| 2019-07-31 | VCB    | 2             | Long                   |
| 2019-07-31 | BID    | 3             | Neutral                |
| 2019-07-31 | CTG    | 0             | Short                  |
| 2019-07-31 | MBB    | 1             | Neutral                |
| 2019-07-31 | ACB    | 1             | Neutral                |
| 2019-07-31 | STB    | 0             | Short                  |
| 2019-07-31 | SHB    | 0             | Short                  |
| 2019-08-31 | VCB    | 2             | Long                   |
| 2019-08-31 | BID    | 3             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2019-08-31 | CTG    | 0             | Short                  |
| 2019-08-31 | MBB    | 1             | Neutral                |
| 2019-08-31 | ACB    | 1             | Neutral                |
| 2019-08-31 | STB    | 0             | Short                  |
| 2019-08-31 | SHB    | 3             | Neutral                |
| 2019-09-30 | VCB    | 2             | Long                   |
| 2019-09-30 | BID    | 3             | Short                  |
| 2019-09-30 | CTG    | 0             | Neutral                |
| 2019-09-30 | MBB    | 1             | Neutral                |
| 2019-09-30 | ACB    | 1             | Neutral                |
| 2019-09-30 | STB    | 0             | Neutral                |
| 2019-09-30 | SHB    | 3             | Short                  |
| 2019-10-31 | VCB    | 2             | Long                   |
| 2019-10-31 | BID    | 0             | Neutral                |
| 2019-10-31 | CTG    | 0             | Neutral                |
| 2019-10-31 | MBB    | 1             | Neutral                |
| 2019-10-31 | ACB    | 1             | Neutral                |
| 2019-10-31 | STB    | 3             | Short                  |
| 2019-10-31 | SHB    | 3             | Short                  |
| 2019-11-30 | VCB    | 0             | Long                   |
| 2019-11-30 | BID    | 2             | Neutral                |
| 2019-11-30 | CTG    | 1             | Short                  |
| 2019-11-30 | MBB    | 3             | Neutral                |
| 2019-11-30 | ACB    | 3             | Neutral                |
| 2019-11-30 | STB    | 1             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2019-11-30 | SHB    | 1             | Short                  |
| 2019-12-31 | VCB    | 0             | Long                   |
| 2019-12-31 | BID    | 2             | Neutral                |
| 2019-12-31 | CTG    | 1             | Short                  |
| 2019-12-31 | MBB    | 0             | Long                   |
| 2019-12-31 | ACB    | 3             | Neutral                |
| 2019-12-31 | STB    | 1             | Short                  |
| 2019-12-31 | SHB    | 1             | Short                  |
| 2020-01-31 | VCB    | 0             | Long                   |
| 2020-01-31 | BID    | 1             | Neutral                |
| 2020-01-31 | CTG    | 1             | Neutral                |
| 2020-01-31 | MBB    | 2             | Neutral                |
| 2020-01-31 | ACB    | 2             | Neutral                |
| 2020-01-31 | STB    | 3             | Short                  |
| 2020-01-31 | SHB    | 3             | Short                  |
| 2020-02-29 | VCB    | 1             | Neutral                |
| 2020-02-29 | BID    | 2             | Long                   |
| 2020-02-29 | CTG    | 0             | Neutral                |
| 2020-02-29 | MBB    | 1             | Neutral                |
| 2020-02-29 | ACB    | 1             | Neutral                |
| 2020-02-29 | STB    | 3             | Short                  |
| 2020-02-29 | SHB    | 0             | Neutral                |
| 2020-03-31 | VCB    | 0             | Neutral                |
| 2020-03-31 | BID    | 2             | Long                   |
| 2020-03-31 | CTG    | 3             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2020-03-31 | MBB    | 1             | Neutral                |
| 2020-03-31 | ACB    | 1             | Neutral                |
| 2020-03-31 | VIB    | 0             | Neutral                |
| 2020-03-31 | STB    | 1             | Neutral                |
| 2020-03-31 | SHB    | 3             | Short                  |
| 2020-04-30 | VCB    | 0             | Neutral                |
| 2020-04-30 | BID    | 2             | Long                   |
| 2020-04-30 | CTG    | 0             | Neutral                |
| 2020-04-30 | MBB    | 1             | Short                  |
| 2020-04-30 | ACB    | 1             | Short                  |
| 2020-04-30 | VIB    | 0             | Neutral                |
| 2020-04-30 | STB    | 1             | Short                  |
| 2020-04-30 | SHB    | 3             | Neutral                |
| 2020-05-31 | VCB    | 2             | Neutral                |
| 2020-05-31 | BID    | 0             | Neutral                |
| 2020-05-31 | CTG    | 0             | Neutral                |
| 2020-05-31 | MBB    | 1             | Short                  |
| 2020-05-31 | ACB    | 2             | Neutral                |
| 2020-05-31 | VIB    | 1             | Short                  |
| 2020-05-31 | STB    | 1             | Short                  |
| 2020-05-31 | SHB    | 3             | Long                   |
| 2020-06-30 | VCB    | 2             | Neutral                |
| 2020-06-30 | BID    | 0             | Neutral                |
| 2020-06-30 | CTG    | 0             | Neutral                |
| 2020-06-30 | MBB    | 3             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2020-06-30 | ACB    | 2             | Neutral                |
| 2020-06-30 | VIB    | 3             | Short                  |
| 2020-06-30 | STB    | 0             | Neutral                |
| 2020-06-30 | SHB    | 1             | Long                   |
| 2020-07-31 | VCB    | 2             | Neutral                |
| 2020-07-31 | BID    | 0             | Neutral                |
| 2020-07-31 | CTG    | 0             | Neutral                |
| 2020-07-31 | MBB    | 3             | Short                  |
| 2020-07-31 | ACB    | 2             | Neutral                |
| 2020-07-31 | VIB    | 3             | Short                  |
| 2020-07-31 | STB    | 0             | Neutral                |
| 2020-07-31 | SHB    | 1             | Long                   |
| 2020-08-31 | VCB    | 3             | Neutral                |
| 2020-08-31 | BID    | 2             | Neutral                |
| 2020-08-31 | CTG    | 2             | Neutral                |
| 2020-08-31 | MBB    | 0             | Short                  |
| 2020-08-31 | ACB    | 3             | Neutral                |
| 2020-08-31 | VIB    | 0             | Short                  |
| 2020-08-31 | STB    | 2             | Neutral                |
| 2020-08-31 | SHB    | 1             | Long                   |
| 2020-08-31 | KLB    | 3             | Neutral                |
| 2020-09-30 | VCB    | 0             | Neutral                |
| 2020-09-30 | BID    | 2             | Short                  |
| 2020-09-30 | CTG    | 3             | Neutral                |
| 2020-09-30 | MBB    | 0             | Neutral                |



| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2020-09-30 | ACB    | 0             | Neutral                |
| 2020-09-30 | VIB    | 3             | Neutral                |
| 2020-09-30 | STB    | 3             | Neutral                |
| 2020-09-30 | SHB    | 1             | Long                   |
| 2020-09-30 | KLB    | 0             | Neutral                |
| 2020-10-31 | VCB    | 0             | Neutral                |
| 2020-10-31 | BID    | 1             | Short                  |
| 2020-10-31 | CTG    | 1             | Short                  |
| 2020-10-31 | MBB    | 0             | Neutral                |
| 2020-10-31 | ACB    | 0             | Neutral                |
| 2020-10-31 | VPB    | 1             | Short                  |
| 2020-10-31 | VIB    | 3             | Neutral                |
| 2020-10-31 | STB    | 3             | Neutral                |
| 2020-10-31 | SHB    | 2             | Long                   |
| 2020-10-31 | KLB    | 0             | Neutral                |
| 2020-11-30 | VCB    | 0             | Neutral                |
| 2020-11-30 | BID    | 1             | Short                  |
| 2020-11-30 | CTG    | 1             | Short                  |
| 2020-11-30 | MBB    | 0             | Neutral                |
| 2020-11-30 | ACB    | 0             | Neutral                |
| 2020-11-30 | VPB    | 1             | Short                  |
| 2020-11-30 | VIB    | 2             | Neutral                |
| 2020-11-30 | STB    | 1             | Short                  |
| 2020-11-30 | SHB    | 3             | Long                   |
| 2020-11-30 | KLB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2020-12-31 | VCB    | 1             | Neutral                |
| 2020-12-31 | BID    | 2             | Neutral                |
| 2020-12-31 | CTG    | 2             | Neutral                |
| 2020-12-31 | MBB    | 1             | Neutral                |
| 2020-12-31 | ACB    | 1             | Neutral                |
| 2020-12-31 | VPB    | 2             | Neutral                |
| 2020-12-31 | VIB    | 0             | Long                   |
| 2020-12-31 | STB    | 0             | Long                   |
| 2020-12-31 | SHB    | 2             | Neutral                |
| 2020-12-31 | LPB    | 0             | Long                   |
| 2020-12-31 | KLB    | 3             | Short                  |
| 2021-01-31 | VCB    | 0             | Neutral                |
| 2021-01-31 | BID    | 1             | Short                  |
| 2021-01-31 | CTG    | 2             | Neutral                |
| 2021-01-31 | MBB    | 0             | Neutral                |
| 2021-01-31 | ACB    | 0             | Neutral                |
| 2021-01-31 | VPB    | 1             | Short                  |
| 2021-01-31 | VIB    | 3             | Long                   |
| 2021-01-31 | STB    | 2             | Neutral                |
| 2021-01-31 | SHB    | 1             | Short                  |
| 2021-01-31 | LPB    | 2             | Neutral                |
| 2021-01-31 | KLB    | 0             | Neutral                |
| 2021-02-28 | VCB    | 1             | Short                  |
| 2021-02-28 | BID    | 1             | Short                  |
| 2021-02-28 | CTG    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2021-02-28 | MBB    | 0             | Neutral                |
| 2021-02-28 | ACB    | 0             | Neutral                |
| 2021-02-28 | VPB    | 1             | Short                  |
| 2021-02-28 | VIB    | 3             | Long                   |
| 2021-02-28 | STB    | 0             | Neutral                |
| 2021-02-28 | SHB    | 1             | Short                  |
| 2021-02-28 | LPB    | 0             | Neutral                |
| 2021-02-28 | BAB    | 2             | Neutral                |
| 2021-02-28 | KLB    | 0             | Neutral                |
| 2021-03-31 | VCB    | 2             | Short                  |
| 2021-03-31 | BID    | 2             | Short                  |
| 2021-03-31 | CTG    | 3             | Neutral                |
| 2021-03-31 | MBB    | 3             | Neutral                |
| 2021-03-31 | ACB    | 3             | Neutral                |
| 2021-03-31 | VPB    | 3             | Neutral                |
| 2021-03-31 | HDB    | 3             | Neutral                |
| 2021-03-31 | VIB    | 0             | Long                   |
| 2021-03-31 | STB    | 3             | Neutral                |
| 2021-03-31 | SHB    | 2             | Short                  |
| 2021-03-31 | LPB    | 3             | Neutral                |
| 2021-03-31 | BAB    | 1             | Neutral                |
| 2021-03-31 | KLB    | 0             | Long                   |
| 2021-04-30 | VCB    | 1             | Short                  |
| 2021-04-30 | BID    | 1             | Short                  |
| 2021-04-30 | CTG    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2021-04-30 | MBB    | 2             | Neutral                |
| 2021-04-30 | ACB    | 2             | Neutral                |
| 2021-04-30 | VPB    | 2             | Neutral                |
| 2021-04-30 | HDB    | 2             | Neutral                |
| 2021-04-30 | VIB    | 0             | Long                   |
| 2021-04-30 | STB    | 2             | Neutral                |
| 2021-04-30 | SHB    | 1             | Short                  |
| 2021-04-30 | LPB    | 2             | Neutral                |
| 2021-04-30 | BAB    | 3             | Neutral                |
| 2021-04-30 | KLB    | 2             | Neutral                |
| 2021-05-31 | VCB    | 2             | Neutral                |
| 2021-05-31 | BID    | 1             | Short                  |
| 2021-05-31 | CTG    | 0             | Neutral                |
| 2021-05-31 | MBB    | 0             | Neutral                |
| 2021-05-31 | ACB    | 0             | Neutral                |
| 2021-05-31 | VPB    | 3             | Long                   |
| 2021-05-31 | HDB    | 0             | Neutral                |
| 2021-05-31 | VIB    | 3             | Long                   |
| 2021-05-31 | STB    | 0             | Neutral                |
| 2021-05-31 | SHB    | 1             | Short                  |
| 2021-05-31 | LPB    | 0             | Neutral                |
| 2021-05-31 | BAB    | 2             | Neutral                |
| 2021-05-31 | KLB    | 0             | Neutral                |
| 2021-06-30 | VCB    | 0             | Short                  |
| 2021-06-30 | BID    | 0             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2021-06-30 | CTG    | 1             | Neutral                |
| 2021-06-30 | MBB    | 1             | Neutral                |
| 2021-06-30 | ACB    | 3             | Neutral                |
| 2021-06-30 | VPB    | 2             | Long                   |
| 2021-06-30 | HDB    | 1             | Neutral                |
| 2021-06-30 | VIB    | 2             | Long                   |
| 2021-06-30 | TPB    | 3             | Neutral                |
| 2021-06-30 | STB    | 1             | Neutral                |
| 2021-06-30 | SHB    | 2             | Long                   |
| 2021-06-30 | LPB    | 1             | Neutral                |
| 2021-06-30 | BAB    | 3             | Neutral                |
| 2021-06-30 | KLB    | 1             | Neutral                |
| 2021-07-31 | VCB    | 2             | Neutral                |
| 2021-07-31 | BID    | 1             | Short                  |
| 2021-07-31 | CTG    | 1             | Short                  |
| 2021-07-31 | MBB    | 3             | Neutral                |
| 2021-07-31 | ACB    | 2             | Neutral                |
| 2021-07-31 | VPB    | 0             | Long                   |
| 2021-07-31 | HDB    | 1             | Short                  |
| 2021-07-31 | VIB    | 3             | Neutral                |
| 2021-07-31 | TPB    | 2             | Neutral                |
| 2021-07-31 | STB    | 3             | Neutral                |
| 2021-07-31 | SHB    | 0             | Long                   |
| 2021-07-31 | LPB    | 3             | Neutral                |
| 2021-07-31 | BAB    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2021-07-31 | KLB    | 3             | Neutral                |
| 2021-08-31 | VCB    | 0             | Neutral                |
| 2021-08-31 | BID    | 2             | Short                  |
| 2021-08-31 | CTG    | 2             | Short                  |
| 2021-08-31 | MBB    | 1             | Long                   |
| 2021-08-31 | TCB    | 1             | Long                   |
| 2021-08-31 | ACB    | 0             | Neutral                |
| 2021-08-31 | VPB    | 3             | Neutral                |
| 2021-08-31 | HDB    | 2             | Short                  |
| 2021-08-31 | VIB    | 1             | Long                   |
| 2021-08-31 | TPB    | 0             | Neutral                |
| 2021-08-31 | STB    | 1             | Long                   |
| 2021-08-31 | SHB    | 3             | Neutral                |
| 2021-08-31 | LPB    | 1             | Long                   |
| 2021-08-31 | BAB    | 0             | Neutral                |
| 2021-08-31 | KLB    | 1             | Long                   |
| 2021-09-30 | VCB    | 0             | Neutral                |
| 2021-09-30 | BID    | 1             | Short                  |
| 2021-09-30 | CTG    | 1             | Short                  |
| 2021-09-30 | MBB    | 2             | Long                   |
| 2021-09-30 | TCB    | 1             | Short                  |
| 2021-09-30 | ACB    | 0             | Neutral                |
| 2021-09-30 | VPB    | 3             | Neutral                |
| 2021-09-30 | HDB    | 1             | Short                  |
| 2021-09-30 | VIB    | 2             | Long                   |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2021-09-30 | TPB    | 0             | Neutral                |
| 2021-09-30 | STB    | 3             | Neutral                |
| 2021-09-30 | SHB    | 3             | Neutral                |
| 2021-09-30 | LPB    | 2             | Long                   |
| 2021-09-30 | BAB    | 0             | Neutral                |
| 2021-09-30 | KLB    | 1             | Short                  |
| 2021-10-31 | VCB    | 0             | Neutral                |
| 2021-10-31 | BID    | 2             | Short                  |
| 2021-10-31 | CTG    | 2             | Short                  |
| 2021-10-31 | MBB    | 1             | Neutral                |
| 2021-10-31 | TCB    | 1             | Neutral                |
| 2021-10-31 | ACB    | 1             | Neutral                |
| 2021-10-31 | VPB    | 3             | Long                   |
| 2021-10-31 | HDB    | 1             | Neutral                |
| 2021-10-31 | VIB    | 1             | Neutral                |
| 2021-10-31 | TPB    | 0             | Neutral                |
| 2021-10-31 | STB    | 3             | Long                   |
| 2021-10-31 | SHB    | 3             | Long                   |
| 2021-10-31 | LPB    | 1             | Neutral                |
| 2021-10-31 | BAB    | 0             | Neutral                |
| 2021-10-31 | KLB    | 2             | Short                  |
| 2021-11-30 | VCB    | 3             | Neutral                |
| 2021-11-30 | BID    | 0             | Short                  |
| 2021-11-30 | CTG    | 2             | Neutral                |
| 2021-11-30 | MBB    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2021-11-30 | TCB    | 2             | Neutral                |
| 2021-11-30 | ACB    | 3             | Neutral                |
| 2021-11-30 | VPB    | 1             | Long                   |
| 2021-11-30 | HDB    | 2             | Neutral                |
| 2021-11-30 | VIB    | 2             | Neutral                |
| 2021-11-30 | TPB    | 3             | Neutral                |
| 2021-11-30 | STB    | 1             | Long                   |
| 2021-11-30 | SHB    | 1             | Long                   |
| 2021-11-30 | LPB    | 1             | Long                   |
| 2021-11-30 | BAB    | 3             | Neutral                |
| 2021-11-30 | KLB    | 2             | Neutral                |
| 2021-12-31 | VCB    | 1             | Short                  |
| 2021-12-31 | BID    | 3             | Neutral                |
| 2021-12-31 | CTG    | 3             | Neutral                |
| 2021-12-31 | MBB    | 0             | Long                   |
| 2021-12-31 | TCB    | 0             | Long                   |
| 2021-12-31 | ACB    | 0             | Long                   |
| 2021-12-31 | VPB    | 2             | Neutral                |
| 2021-12-31 | HDB    | 0             | Long                   |
| 2021-12-31 | VIB    | 3             | Neutral                |
| 2021-12-31 | TPB    | 0             | Long                   |
| 2021-12-31 | STB    | 2             | Neutral                |
| 2021-12-31 | SHB    | 2             | Neutral                |
| 2021-12-31 | LPB    | 0             | Long                   |
| 2021-12-31 | BAB    | 1             | Short                  |



| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2021-12-31 | KLB    | 3             | Neutral                |
| 2022-01-31 | VCB    | 1             | Short                  |
| 2022-01-31 | BID    | 0             | Neutral                |
| 2022-01-31 | CTG    | 0             | Neutral                |
| 2022-01-31 | MBB    | 3             | Neutral                |
| 2022-01-31 | TCB    | 3             | Neutral                |
| 2022-01-31 | ACB    | 3             | Neutral                |
| 2022-01-31 | VPB    | 0             | Neutral                |
| 2022-01-31 | HDB    | 3             | Neutral                |
| 2022-01-31 | VIB    | 0             | Neutral                |
| 2022-01-31 | TPB    | 2             | Long                   |
| 2022-01-31 | STB    | 0             | Neutral                |
| 2022-01-31 | SHB    | 0             | Neutral                |
| 2022-01-31 | LPB    | 3             | Neutral                |
| 2022-01-31 | BAB    | 1             | Short                  |
| 2022-01-31 | KLB    | 0             | Neutral                |
| 2022-02-28 | VCB    | 0             | Long                   |
| 2022-02-28 | BID    | 1             | Neutral                |
| 2022-02-28 | CTG    | 1             | Neutral                |
| 2022-02-28 | MBB    | 0             | Long                   |
| 2022-02-28 | TCB    | 1             | Neutral                |
| 2022-02-28 | ACB    | 0             | Long                   |
| 2022-02-28 | VPB    | 2             | Neutral                |
| 2022-02-28 | HDB    | 0             | Long                   |
| 2022-02-28 | VIB    | 1             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2022-02-28 | TPB    | 0             | Long                   |
| 2022-02-28 | STB    | 1             | Neutral                |
| 2022-02-28 | SHB    | 2             | Neutral                |
| 2022-02-28 | LPB    | 1             | Neutral                |
| 2022-02-28 | BAB    | 3             | Short                  |
| 2022-02-28 | KLB    | 1             | Neutral                |
| 2022-03-31 | VCB    | 0             | Neutral                |
| 2022-03-31 | BID    | 1             | Neutral                |
| 2022-03-31 | CTG    | 2             | Short                  |
| 2022-03-31 | MBB    | 2             | Short                  |
| 2022-03-31 | TCB    | 1             | Neutral                |
| 2022-03-31 | ACB    | 0             | Neutral                |
| 2022-03-31 | VPB    | 1             | Neutral                |
| 2022-03-31 | HDB    | 1             | Neutral                |
| 2022-03-31 | VIB    | 1             | Neutral                |
| 2022-03-31 | TPB    | 3             | Long                   |
| 2022-03-31 | STB    | 1             | Neutral                |
| 2022-03-31 | SHB    | 1             | Neutral                |
| 2022-03-31 | LPB    | 2             | Short                  |
| 2022-03-31 | BAB    | 0             | Neutral                |
| 2022-03-31 | KLB    | 1             | Neutral                |
| 2022-04-30 | VCB    | 0             | Neutral                |
| 2022-04-30 | BID    | 1             | Long                   |
| 2022-04-30 | CTG    | 2             | Neutral                |
| 2022-04-30 | MBB    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2022-04-30 | TCB    | 2             | Neutral                |
| 2022-04-30 | ACB    | 0             | Neutral                |
| 2022-04-30 | VPB    | 3             | Short                  |
| 2022-04-30 | HDB    | 2             | Neutral                |
| 2022-04-30 | VIB    | 2             | Neutral                |
| 2022-04-30 | TPB    | 1             | Long                   |
| 2022-04-30 | STB    | 3             | Short                  |
| 2022-04-30 | SHB    | 3             | Short                  |
| 2022-04-30 | LPB    | 2             | Neutral                |
| 2022-04-30 | BAB    | 0             | Neutral                |
| 2022-04-30 | KLB    | 1             | Long                   |
| 2022-05-31 | VCB    | 0             | Neutral                |
| 2022-05-31 | BID    | 2             | Neutral                |
| 2022-05-31 | CTG    | 1             | Short                  |
| 2022-05-31 | MBB    | 2             | Neutral                |
| 2022-05-31 | TCB    | 1             | Short                  |
| 2022-05-31 | ACB    | 0             | Neutral                |
| 2022-05-31 | VPB    | 1             | Short                  |
| 2022-05-31 | HDB    | 1             | Short                  |
| 2022-05-31 | VIB    | 2             | Neutral                |
| 2022-05-31 | TPB    | 3             | Long                   |
| 2022-05-31 | STB    | 2             | Neutral                |
| 2022-05-31 | SHB    | 1             | Short                  |
| 2022-05-31 | LPB    | 1             | Short                  |
| 2022-05-31 | BAB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2022-05-31 | KLB    | 2             | Neutral                |
| 2022-06-30 | VCB    | 0             | Neutral                |
| 2022-06-30 | BID    | 1             | Neutral                |
| 2022-06-30 | CTG    | 2             | Short                  |
| 2022-06-30 | MBB    | 1             | Neutral                |
| 2022-06-30 | TCB    | 2             | Short                  |
| 2022-06-30 | ACB    | 0             | Neutral                |
| 2022-06-30 | VPB    | 2             | Short                  |
| 2022-06-30 | HDB    | 1             | Neutral                |
| 2022-06-30 | VIB    | 1             | Neutral                |
| 2022-06-30 | TPB    | 1             | Neutral                |
| 2022-06-30 | STB    | 1             | Neutral                |
| 2022-06-30 | SHB    | 2             | Short                  |
| 2022-06-30 | LPB    | 2             | Short                  |
| 2022-06-30 | BAB    | 0             | Neutral                |
| 2022-06-30 | KLB    | 3             | Long                   |
| 2022-07-31 | VCB    | 2             | Neutral                |
| 2022-07-31 | BID    | 0             | Long                   |
| 2022-07-31 | CTG    | 3             | Neutral                |
| 2022-07-31 | MBB    | 2             | Neutral                |
| 2022-07-31 | TCB    | 3             | Neutral                |
| 2022-07-31 | ACB    | 2             | Neutral                |
| 2022-07-31 | VPB    | 3             | Neutral                |
| 2022-07-31 | HDB    | 3             | Neutral                |
| 2022-07-31 | VIB    | 0             | Long                   |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2022-07-31 | TPB    | 2             | Neutral                |
| 2022-07-31 | STB    | 0             | Long                   |
| 2022-07-31 | SHB    | 1             | Short                  |
| 2022-07-31 | LPB    | 3             | Neutral                |
| 2022-07-31 | BAB    | 2             | Neutral                |
| 2022-07-31 | KLB    | 0             | Long                   |
| 2022-08-31 | VCB    | 0             | Neutral                |
| 2022-08-31 | BID    | 2             | Long                   |
| 2022-08-31 | CTG    | 3             | Neutral                |
| 2022-08-31 | MBB    | 2             | Long                   |
| 2022-08-31 | TCB    | 3             | Neutral                |
| 2022-08-31 | ACB    | 0             | Neutral                |
| 2022-08-31 | VPB    | 3             | Neutral                |
| 2022-08-31 | HDB    | 3             | Neutral                |
| 2022-08-31 | VIB    | 3             | Neutral                |
| 2022-08-31 | TPB    | 2             | Long                   |
| 2022-08-31 | STB    | 3             | Neutral                |
| 2022-08-31 | SHB    | 1             | Short                  |
| 2022-08-31 | LPB    | 3             | Neutral                |
| 2022-08-31 | BAB    | 0             | Neutral                |
| 2022-08-31 | KLB    | 2             | Long                   |
| 2022-09-30 | VCB    | 2             | Long                   |
| 2022-09-30 | BID    | 0             | Neutral                |
| 2022-09-30 | CTG    | 3             | Neutral                |
| 2022-09-30 | MBB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2022-09-30 | TCB    | 3             | Neutral                |
| 2022-09-30 | ACB    | 2             | Long                   |
| 2022-09-30 | VPB    | 0             | Neutral                |
| 2022-09-30 | HDB    | 3             | Neutral                |
| 2022-09-30 | VIB    | 1             | Short                  |
| 2022-09-30 | TPB    | 3             | Neutral                |
| 2022-09-30 | STB    | 1             | Short                  |
| 2022-09-30 | SHB    | 1             | Short                  |
| 2022-09-30 | LPB    | 3             | Neutral                |
| 2022-09-30 | BAB    | 2             | Long                   |
| 2022-09-30 | KLB    | 0             | Neutral                |
| 2022-09-30 | VBB    | 1             | Short                  |
| 2022-10-31 | VCB    | 0             | Long                   |
| 2022-10-31 | BID    | 2             | Neutral                |
| 2022-10-31 | CTG    | 2             | Neutral                |
| 2022-10-31 | MBB    | 2             | Neutral                |
| 2022-10-31 | TCB    | 2             | Neutral                |
| 2022-10-31 | ACB    | 0             | Long                   |
| 2022-10-31 | VPB    | 2             | Neutral                |
| 2022-10-31 | HDB    | 2             | Neutral                |
| 2022-10-31 | VIB    | 2             | Neutral                |
| 2022-10-31 | TPB    | 1             | Short                  |
| 2022-10-31 | STB    | 3             | Neutral                |
| 2022-10-31 | SHB    | 3             | Neutral                |
| 2022-10-31 | LPB    | 1             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2022-10-31 | BAB    | 0             | Long                   |
| 2022-10-31 | KLB    | 2             | Neutral                |
| 2022-10-31 | VBB    | 1             | Short                  |
| 2022-11-30 | VCB    | 0             | Long                   |
| 2022-11-30 | BID    | 3             | Neutral                |
| 2022-11-30 | CTG    | 3             | Neutral                |
| 2022-11-30 | MBB    | 3             | Neutral                |
| 2022-11-30 | TCB    | 1             | Neutral                |
| 2022-11-30 | ACB    | 0             | Long                   |
| 2022-11-30 | VPB    | 1             | Neutral                |
| 2022-11-30 | HDB    | 3             | Neutral                |
| 2022-11-30 | VIB    | 1             | Neutral                |
| 2022-11-30 | TPB    | 2             | Short                  |
| 2022-11-30 | STB    | 1             | Neutral                |
| 2022-11-30 | SHB    | 1             | Neutral                |
| 2022-11-30 | LPB    | 1             | Neutral                |
| 2022-11-30 | BAB    | 0             | Long                   |
| 2022-11-30 | KLB    | 3             | Neutral                |
| 2022-11-30 | VBB    | 2             | Short                  |
| 2022-12-31 | VCB    | 0             | Long                   |
| 2022-12-31 | BID    | 2             | Neutral                |
| 2022-12-31 | CTG    | 2             | Neutral                |
| 2022-12-31 | MBB    | 2             | Neutral                |
| 2022-12-31 | TCB    | 1             | Neutral                |
| 2022-12-31 | ACB    | 0             | Long                   |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2022-12-31 | VPB    | 1             | Neutral                |
| 2022-12-31 | HDB    | 2             | Neutral                |
| 2022-12-31 | VIB    | 1             | Neutral                |
| 2022-12-31 | TPB    | 3             | Short                  |
| 2022-12-31 | STB    | 1             | Neutral                |
| 2022-12-31 | SHB    | 1             | Neutral                |
| 2022-12-31 | LPB    | 1             | Neutral                |
| 2022-12-31 | BAB    | 0             | Long                   |
| 2022-12-31 | KLB    | 1             | Neutral                |
| 2022-12-31 | VBB    | 3             | Short                  |
| 2023-01-31 | VCB    | 0             | Long                   |
| 2023-01-31 | BID    | 2             | Neutral                |
| 2023-01-31 | CTG    | 2             | Neutral                |
| 2023-01-31 | MBB    | 3             | Neutral                |
| 2023-01-31 | TCB    | 1             | Short                  |
| 2023-01-31 | ACB    | 0             | Long                   |
| 2023-01-31 | VPB    | 1             | Short                  |
| 2023-01-31 | HDB    | 2             | Neutral                |
| 2023-01-31 | VIB    | 1             | Short                  |
| 2023-01-31 | TPB    | 3             | Neutral                |
| 2023-01-31 | STB    | 1             | Short                  |
| 2023-01-31 | SHB    | 1             | Short                  |
| 2023-01-31 | LPB    | 3             | Neutral                |
| 2023-01-31 | BAB    | 0             | Long                   |
| 2023-01-31 | KLB    | 1             | Short                  |



| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2023-01-31 | VBB    | 3             | Neutral                |
| 2023-02-28 | VCB    | 3             | Neutral                |
| 2023-02-28 | BID    | 0             | Long                   |
| 2023-02-28 | CTG    | 0             | Long                   |
| 2023-02-28 | MBB    | 1             | Short                  |
| 2023-02-28 | TCB    | 1             | Short                  |
| 2023-02-28 | ACB    | 3             | Neutral                |
| 2023-02-28 | VPB    | 2             | Neutral                |
| 2023-02-28 | HDB    | 1             | Short                  |
| 2023-02-28 | VIB    | 2             | Neutral                |
| 2023-02-28 | TPB    | 1             | Short                  |
| 2023-02-28 | STB    | 2             | Neutral                |
| 2023-02-28 | SHB    | 2             | Neutral                |
| 2023-02-28 | LPB    | 0             | Long                   |
| 2023-02-28 | BAB    | 3             | Neutral                |
| 2023-02-28 | KLB    | 2             | Neutral                |
| 2023-02-28 | VBB    | 1             | Short                  |
| 2023-03-31 | VCB    | 3             | Neutral                |
| 2023-03-31 | BID    | 1             | Long                   |
| 2023-03-31 | CTG    | 1             | Long                   |
| 2023-03-31 | MBB    | 0             | Neutral                |
| 2023-03-31 | TCB    | 0             | Neutral                |
| 2023-03-31 | ACB    | 3             | Neutral                |
| 2023-03-31 | VPB    | 1             | Long                   |
| 2023-03-31 | HDB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2023-03-31 | VIB    | 1             | Long                   |
| 2023-03-31 | TPB    | 0             | Neutral                |
| 2023-03-31 | STB    | 1             | Long                   |
| 2023-03-31 | SHB    | 1             | Long                   |
| 2023-03-31 | LPB    | 1             | Long                   |
| 2023-03-31 | BAB    | 3             | Neutral                |
| 2023-03-31 | KLB    | 2             | Short                  |
| 2023-03-31 | VBB    | 0             | Neutral                |
| 2023-04-30 | VCB    | 0             | Long                   |
| 2023-04-30 | BID    | 2             | Neutral                |
| 2023-04-30 | CTG    | 2             | Neutral                |
| 2023-04-30 | MBB    | 2             | Neutral                |
| 2023-04-30 | TCB    | 1             | Neutral                |
| 2023-04-30 | ACB    | 0             | Long                   |
| 2023-04-30 | VPB    | 1             | Neutral                |
| 2023-04-30 | HDB    | 2             | Neutral                |
| 2023-04-30 | VIB    | 1             | Neutral                |
| 2023-04-30 | TPB    | 2             | Neutral                |
| 2023-04-30 | STB    | 1             | Neutral                |
| 2023-04-30 | SHB    | 1             | Neutral                |
| 2023-04-30 | LPB    | 2             | Neutral                |
| 2023-04-30 | BAB    | 0             | Long                   |
| 2023-04-30 | KLB    | 3             | Short                  |
| 2023-04-30 | VBB    | 2             | Neutral                |
| 2023-05-31 | VCB    | 0             | Long                   |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2023-05-31 | BID    | 0             | Long                   |
| 2023-05-31 | CTG    | 0             | Long                   |
| 2023-05-31 | MBB    | 3             | Neutral                |
| 2023-05-31 | TCB    | 3             | Neutral                |
| 2023-05-31 | ACB    | 0             | Long                   |
| 2023-05-31 | VPB    | 3             | Neutral                |
| 2023-05-31 | HDB    | 0             | Long                   |
| 2023-05-31 | VIB    | 1             | Neutral                |
| 2023-05-31 | TPB    | 3             | Neutral                |
| 2023-05-31 | STB    | 1             | Neutral                |
| 2023-05-31 | SHB    | 3             | Neutral                |
| 2023-05-31 | LPB    | 1             | Neutral                |
| 2023-05-31 | BAB    | 0             | Long                   |
| 2023-05-31 | KLB    | 2             | Short                  |
| 2023-05-31 | VBB    | 1             | Neutral                |
| 2023-06-30 | VCB    | 3             | Neutral                |
| 2023-06-30 | BID    | 2             | Long                   |
| 2023-06-30 | CTG    | 2             | Long                   |
| 2023-06-30 | MBB    | 3             | Neutral                |
| 2023-06-30 | TCB    | 0             | Neutral                |
| 2023-06-30 | ACB    | 3             | Neutral                |
| 2023-06-30 | VPB    | 2             | Long                   |
| 2023-06-30 | HDB    | 3             | Neutral                |
| 2023-06-30 | VIB    | 0             | Neutral                |
| 2023-06-30 | TPB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2023-06-30 | STB    | 0             | Neutral                |
| 2023-06-30 | SHB    | 0             | Neutral                |
| 2023-06-30 | LPB    | 2             | Long                   |
| 2023-06-30 | BAB    | 3             | Neutral                |
| 2023-06-30 | KLB    | 1             | Short                  |
| 2023-06-30 | VBB    | 0             | Neutral                |
| 2023-07-31 | VCB    | 0             | Neutral                |
| 2023-07-31 | BID    | 2             | Neutral                |
| 2023-07-31 | CTG    | 2             | Neutral                |
| 2023-07-31 | MBB    | 0             | Neutral                |
| 2023-07-31 | TCB    | 2             | Neutral                |
| 2023-07-31 | ACB    | 0             | Neutral                |
| 2023-07-31 | VPB    | 2             | Neutral                |
| 2023-07-31 | HDB    | 0             | Neutral                |
| 2023-07-31 | VIB    | 1             | Long                   |
| 2023-07-31 | TPB    | 2             | Neutral                |
| 2023-07-31 | STB    | 1             | Long                   |
| 2023-07-31 | SHB    | 1             | Long                   |
| 2023-07-31 | LPB    | 1             | Long                   |
| 2023-07-31 | BAB    | 0             | Neutral                |
| 2023-07-31 | KLB    | 3             | Short                  |
| 2023-07-31 | VBB    | 1             | Long                   |
| 2023-08-31 | VCB    | 0             | Long                   |
| 2023-08-31 | BID    | 2             | Neutral                |
| 2023-08-31 | CTG    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2023-08-31 | MBB    | 2             | Neutral                |
| 2023-08-31 | TCB    | 1             | Neutral                |
| 2023-08-31 | ACB    | 0             | Long                   |
| 2023-08-31 | VPB    | 1             | Neutral                |
| 2023-08-31 | HDB    | 0             | Long                   |
| 2023-08-31 | VIB    | 1             | Neutral                |
| 2023-08-31 | TPB    | 1             | Neutral                |
| 2023-08-31 | STB    | 1             | Neutral                |
| 2023-08-31 | SHB    | 1             | Neutral                |
| 2023-08-31 | LPB    | 1             | Neutral                |
| 2023-08-31 | BAB    | 2             | Neutral                |
| 2023-08-31 | KLB    | 3             | Short                  |
| 2023-08-31 | VBB    | 1             | Neutral                |
| 2023-09-30 | VCB    | 2             | Long                   |
| 2023-09-30 | BID    | 0             | Neutral                |
| 2023-09-30 | CTG    | 0             | Neutral                |
| 2023-09-30 | MBB    | 2             | Long                   |
| 2023-09-30 | TCB    | 1             | Neutral                |
| 2023-09-30 | ACB    | 2             | Long                   |
| 2023-09-30 | VPB    | 0             | Neutral                |
| 2023-09-30 | HDB    | 2             | Long                   |
| 2023-09-30 | VIB    | 1             | Neutral                |
| 2023-09-30 | TPB    | 1             | Neutral                |
| 2023-09-30 | STB    | 1             | Neutral                |
| 2023-09-30 | SHB    | 1             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2023-09-30 | LPB    | 0             | Neutral                |
| 2023-09-30 | BAB    | 0             | Neutral                |
| 2023-09-30 | KLB    | 3             | Short                  |
| 2023-09-30 | VBB    | 1             | Neutral                |
| 2023-09-30 | BVB    | 1             | Neutral                |
| 2023-10-31 | VCB    | 1             | Neutral                |
| 2023-10-31 | BID    | 3             | Short                  |
| 2023-10-31 | CTG    | 3             | Short                  |
| 2023-10-31 | MBB    | 1             | Neutral                |
| 2023-10-31 | TCB    | 1             | Neutral                |
| 2023-10-31 | ACB    | 3             | Short                  |
| 2023-10-31 | VPB    | 1             | Neutral                |
| 2023-10-31 | HDB    | 1             | Neutral                |
| 2023-10-31 | VIB    | 0             | Neutral                |
| 2023-10-31 | TPB    | 1             | Neutral                |
| 2023-10-31 | STB    | 0             | Neutral                |
| 2023-10-31 | SHB    | 2             | Long                   |
| 2023-10-31 | LPB    | 0             | Neutral                |
| 2023-10-31 | BAB    | 3             | Short                  |
| 2023-10-31 | KLB    | 0             | Neutral                |
| 2023-10-31 | VBB    | 0             | Neutral                |
| 2023-10-31 | BVB    | 0             | Neutral                |
| 2023-11-30 | VCB    | 1             | Neutral                |
| 2023-11-30 | BID    | 1             | Neutral                |
| 2023-11-30 | CTG    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2023-11-30 | MBB    | 0             | Neutral                |
| 2023-11-30 | TCB    | 0             | Neutral                |
| 2023-11-30 | ACB    | 1             | Neutral                |
| 2023-11-30 | VPB    | 0             | Neutral                |
| 2023-11-30 | HDB    | 1             | Neutral                |
| 2023-11-30 | VIB    | 0             | Neutral                |
| 2023-11-30 | TPB    | 0             | Neutral                |
| 2023-11-30 | STB    | 2             | Long                   |
| 2023-11-30 | SHB    | 2             | Long                   |
| 2023-11-30 | LPB    | 2             | Long                   |
| 2023-11-30 | BAB    | 1             | Neutral                |
| 2023-11-30 | KLB    | 3             | Short                  |
| 2023-11-30 | VBB    | 3             | Short                  |
| 2023-11-30 | BVB    | 3             | Short                  |
| 2023-12-31 | VCB    | 0             | Neutral                |
| 2023-12-31 | BID    | 0             | Neutral                |
| 2023-12-31 | CTG    | 0             | Neutral                |
| 2023-12-31 | MBB    | 0             | Neutral                |
| 2023-12-31 | TCB    | 2             | Neutral                |
| 2023-12-31 | ACB    | 0             | Neutral                |
| 2023-12-31 | VPB    | 2             | Neutral                |
| 2023-12-31 | HDB    | 0             | Neutral                |
| 2023-12-31 | VIB    | 2             | Neutral                |
| 2023-12-31 | TPB    | 2             | Neutral                |
| 2023-12-31 | STB    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2023-12-31 | SHB    | 2             | Neutral                |
| 2023-12-31 | LPB    | 2             | Neutral                |
| 2023-12-31 | NAB    | 3             | Long                   |
| 2023-12-31 | BAB    | 0             | Neutral                |
| 2023-12-31 | KLB    | 1             | Short                  |
| 2023-12-31 | VBB    | 1             | Short                  |
| 2023-12-31 | SGB    | 2             | Neutral                |
| 2023-12-31 | BVB    | 1             | Short                  |
| 2024-01-31 | VCB    | 0             | Neutral                |
| 2024-01-31 | BID    | 3             | Neutral                |
| 2024-01-31 | CTG    | 3             | Neutral                |
| 2024-01-31 | MBB    | 0             | Neutral                |
| 2024-01-31 | TCB    | 3             | Neutral                |
| 2024-01-31 | ACB    | 0             | Neutral                |
| 2024-01-31 | VPB    | 3             | Neutral                |
| 2024-01-31 | HDB    | 0             | Neutral                |
| 2024-01-31 | VIB    | 1             | Short                  |
| 2024-01-31 | TPB    | 3             | Neutral                |
| 2024-01-31 | STB    | 1             | Short                  |
| 2024-01-31 | SHB    | 1             | Short                  |
| 2024-01-31 | LPB    | 2             | Long                   |
| 2024-01-31 | NAB    | 2             | Long                   |
| 2024-01-31 | BAB    | 3             | Neutral                |
| 2024-01-31 | KLB    | 1             | Short                  |
| 2024-01-31 | VBB    | 1             | Short                  |



| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-01-31 | SGB    | 3             | Neutral                |
| 2024-01-31 | BVB    | 1             | Short                  |
| 2024-02-29 | VCB    | 0             | Neutral                |
| 2024-02-29 | BID    | 3             | Neutral                |
| 2024-02-29 | CTG    | 0             | Neutral                |
| 2024-02-29 | MBB    | 0             | Neutral                |
| 2024-02-29 | TCB    | 3             | Neutral                |
| 2024-02-29 | ACB    | 0             | Neutral                |
| 2024-02-29 | VPB    | 3             | Neutral                |
| 2024-02-29 | HDB    | 0             | Neutral                |
| 2024-02-29 | VIB    | 3             | Neutral                |
| 2024-02-29 | TPB    | 3             | Neutral                |
| 2024-02-29 | STB    | 1             | Short                  |
| 2024-02-29 | SHB    | 1             | Short                  |
| 2024-02-29 | LPB    | 2             | Long                   |
| 2024-02-29 | MSB    | 0             | Neutral                |
| 2024-02-29 | NAB    | 2             | Long                   |
| 2024-02-29 | BAB    | 3             | Neutral                |
| 2024-02-29 | KLB    | 1             | Short                  |
| 2024-02-29 | PGB    | 1             | Short                  |
| 2024-02-29 | VBB    | 1             | Short                  |
| 2024-02-29 | SGB    | 3             | Neutral                |
| 2024-02-29 | ABB    | 3             | Neutral                |
| 2024-02-29 | BVB    | 1             | Short                  |
| 2024-03-31 | VCB    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-03-31 | BID    | 2             | Neutral                |
| 2024-03-31 | CTG    | 2             | Neutral                |
| 2024-03-31 | MBB    | 2             | Neutral                |
| 2024-03-31 | TCB    | 0             | Neutral                |
| 2024-03-31 | ACB    | 2             | Neutral                |
| 2024-03-31 | VPB    | 0             | Neutral                |
| 2024-03-31 | HDB    | 1             | Long                   |
| 2024-03-31 | VIB    | 0             | Neutral                |
| 2024-03-31 | TPB    | 0             | Neutral                |
| 2024-03-31 | STB    | 3             | Short                  |
| 2024-03-31 | SHB    | 0             | Neutral                |
| 2024-03-31 | LPB    | 1             | Long                   |
| 2024-03-31 | MSB    | 2             | Neutral                |
| 2024-03-31 | OCB    | 2             | Neutral                |
| 2024-03-31 | NAB    | 1             | Long                   |
| 2024-03-31 | BAB    | 2             | Neutral                |
| 2024-03-31 | KLB    | 3             | Short                  |
| 2024-03-31 | PGB    | 3             | Short                  |
| 2024-03-31 | VBB    | 3             | Short                  |
| 2024-03-31 | SGB    | 0             | Neutral                |
| 2024-03-31 | ABB    | 0             | Neutral                |
| 2024-03-31 | BVB    | 3             | Short                  |
| 2024-04-30 | VCB    | 0             | Neutral                |
| 2024-04-30 | BID    | 2             | Neutral                |
| 2024-04-30 | CTG    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-04-30 | MBB    | 2             | Neutral                |
| 2024-04-30 | TCB    | 0             | Neutral                |
| 2024-04-30 | ACB    | 2             | Neutral                |
| 2024-04-30 | VPB    | 0             | Neutral                |
| 2024-04-30 | HDB    | 2             | Neutral                |
| 2024-04-30 | VIB    | 0             | Neutral                |
| 2024-04-30 | TPB    | 0             | Neutral                |
| 2024-04-30 | STB    | 1             | Short                  |
| 2024-04-30 | SHB    | 1             | Short                  |
| 2024-04-30 | LPB    | 3             | Long                   |
| 2024-04-30 | MSB    | 0             | Neutral                |
| 2024-04-30 | OCB    | 2             | Neutral                |
| 2024-04-30 | NAB    | 3             | Long                   |
| 2024-04-30 | BAB    | 0             | Neutral                |
| 2024-04-30 | KLB    | 1             | Short                  |
| 2024-04-30 | PGB    | 1             | Short                  |
| 2024-04-30 | VBB    | 1             | Short                  |
| 2024-04-30 | SGB    | 0             | Neutral                |
| 2024-04-30 | ABB    | 0             | Neutral                |
| 2024-04-30 | BVB    | 1             | Short                  |
| 2024-05-31 | VCB    | 0             | Neutral                |
| 2024-05-31 | BID    | 2             | Long                   |
| 2024-05-31 | CTG    | 2             | Long                   |
| 2024-05-31 | MBB    | 2             | Long                   |
| 2024-05-31 | TCB    | 2             | Long                   |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-05-31 | ACB    | 0             | Neutral                |
| 2024-05-31 | VPB    | 3             | Neutral                |
| 2024-05-31 | HDB    | 2             | Long                   |
| 2024-05-31 | VIB    | 2             | Long                   |
| 2024-05-31 | TPB    | 3             | Neutral                |
| 2024-05-31 | STB    | 1             | Short                  |
| 2024-05-31 | SHB    | 3             | Neutral                |
| 2024-05-31 | LPB    | 1             | Short                  |
| 2024-05-31 | SSB    | 3             | Neutral                |
| 2024-05-31 | MSB    | 2             | Long                   |
| 2024-05-31 | OCB    | 2             | Long                   |
| 2024-05-31 | NAB    | 2             | Long                   |
| 2024-05-31 | BAB    | 0             | Neutral                |
| 2024-05-31 | KLB    | 1             | Short                  |
| 2024-05-31 | PGB    | 1             | Short                  |
| 2024-05-31 | VBB    | 1             | Short                  |
| 2024-05-31 | SGB    | 3             | Neutral                |
| 2024-05-31 | ABB    | 3             | Neutral                |
| 2024-05-31 | BVB    | 1             | Short                  |
| 2024-06-30 | VCB    | 0             | Neutral                |
| 2024-06-30 | BID    | 2             | Long                   |
| 2024-06-30 | CTG    | 2             | Long                   |
| 2024-06-30 | MBB    | 2             | Long                   |
| 2024-06-30 | TCB    | 2             | Long                   |
| 2024-06-30 | ACB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-06-30 | VPB    | 3             | Neutral                |
| 2024-06-30 | HDB    | 2             | Long                   |
| 2024-06-30 | VIB    | 2             | Long                   |
| 2024-06-30 | TPB    | 3             | Neutral                |
| 2024-06-30 | STB    | 1             | Short                  |
| 2024-06-30 | SHB    | 3             | Neutral                |
| 2024-06-30 | LPB    | 2             | Long                   |
| 2024-06-30 | SSB    | 3             | Neutral                |
| 2024-06-30 | MSB    | 3             | Neutral                |
| 2024-06-30 | OCB    | 3             | Neutral                |
| 2024-06-30 | NAB    | 2             | Long                   |
| 2024-06-30 | BAB    | 0             | Neutral                |
| 2024-06-30 | KLB    | 1             | Short                  |
| 2024-06-30 | PGB    | 1             | Short                  |
| 2024-06-30 | VBB    | 1             | Short                  |
| 2024-06-30 | SGB    | 3             | Neutral                |
| 2024-06-30 | ABB    | 3             | Neutral                |
| 2024-06-30 | BVB    | 1             | Short                  |
| 2024-07-31 | VCB    | 0             | Neutral                |
| 2024-07-31 | BID    | 1             | Long                   |
| 2024-07-31 | CTG    | 1             | Long                   |
| 2024-07-31 | MBB    | 1             | Long                   |
| 2024-07-31 | TCB    | 1             | Long                   |
| 2024-07-31 | ACB    | 3             | Neutral                |
| 2024-07-31 | VPB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-07-31 | HDB    | 1             | Long                   |
| 2024-07-31 | VIB    | 1             | Long                   |
| 2024-07-31 | TPB    | 0             | Neutral                |
| 2024-07-31 | STB    | 2             | Short                  |
| 2024-07-31 | SHB    | 0             | Neutral                |
| 2024-07-31 | LPB    | 1             | Long                   |
| 2024-07-31 | SSB    | 0             | Neutral                |
| 2024-07-31 | MSB    | 0             | Neutral                |
| 2024-07-31 | OCB    | 0             | Neutral                |
| 2024-07-31 | NAB    | 1             | Long                   |
| 2024-07-31 | BAB    | 3             | Neutral                |
| 2024-07-31 | KLB    | 2             | Short                  |
| 2024-07-31 | PGB    | 2             | Short                  |
| 2024-07-31 | VBB    | 1             | Long                   |
| 2024-07-31 | SGB    | 3             | Neutral                |
| 2024-07-31 | ABB    | 0             | Neutral                |
| 2024-07-31 | BVB    | 0             | Neutral                |
| 2024-08-31 | VCB    | 3             | Neutral                |
| 2024-08-31 | BID    | 0             | Neutral                |
| 2024-08-31 | CTG    | 0             | Neutral                |
| 2024-08-31 | MBB    | 0             | Neutral                |
| 2024-08-31 | TCB    | 2             | Long                   |
| 2024-08-31 | ACB    | 0             | Neutral                |
| 2024-08-31 | VPB    | 3             | Neutral                |
| 2024-08-31 | HDB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-08-31 | VIB    | 0             | Neutral                |
| 2024-08-31 | TPB    | 0             | Neutral                |
| 2024-08-31 | STB    | 1             | Short                  |
| 2024-08-31 | SHB    | 0             | Neutral                |
| 2024-08-31 | LPB    | 2             | Long                   |
| 2024-08-31 | SSB    | 3             | Neutral                |
| 2024-08-31 | MSB    | 0             | Neutral                |
| 2024-08-31 | OCB    | 0             | Neutral                |
| 2024-08-31 | NAB    | 0             | Neutral                |
| 2024-08-31 | BAB    | 3             | Neutral                |
| 2024-08-31 | KLB    | 1             | Short                  |
| 2024-08-31 | PGB    | 1             | Short                  |
| 2024-08-31 | VBB    | 0             | Neutral                |
| 2024-08-31 | SGB    | 3             | Neutral                |
| 2024-08-31 | ABB    | 0             | Neutral                |
| 2024-08-31 | BVB    | 0             | Neutral                |
| 2024-09-30 | VCB    | 3             | Neutral                |
| 2024-09-30 | BID    | 0             | Neutral                |
| 2024-09-30 | CTG    | 0             | Neutral                |
| 2024-09-30 | MBB    | 0             | Neutral                |
| 2024-09-30 | TCB    | 1             | Long                   |
| 2024-09-30 | ACB    | 3             | Neutral                |
| 2024-09-30 | VPB    | 0             | Neutral                |
| 2024-09-30 | HDB    | 0             | Neutral                |
| 2024-09-30 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-09-30 | TPB    | 0             | Neutral                |
| 2024-09-30 | STB    | 2             | Short                  |
| 2024-09-30 | SHB    | 0             | Neutral                |
| 2024-09-30 | LPB    | 1             | Long                   |
| 2024-09-30 | SSB    | 3             | Neutral                |
| 2024-09-30 | MSB    | 0             | Neutral                |
| 2024-09-30 | OCB    | 0             | Neutral                |
| 2024-09-30 | NAB    | 0             | Neutral                |
| 2024-09-30 | BAB    | 3             | Neutral                |
| 2024-09-30 | KLB    | 2             | Short                  |
| 2024-09-30 | PGB    | 2             | Short                  |
| 2024-09-30 | VAB    | 1             | Long                   |
| 2024-09-30 | VBB    | 0             | Neutral                |
| 2024-09-30 | SGB    | 3             | Neutral                |
| 2024-09-30 | ABB    | 0             | Neutral                |
| 2024-09-30 | BVB    | 0             | Neutral                |
| 2024-10-31 | VCB    | 1             | Neutral                |
| 2024-10-31 | BID    | 3             | Neutral                |
| 2024-10-31 | CTG    | 1             | Neutral                |
| 2024-10-31 | MBB    | 2             | Long                   |
| 2024-10-31 | TCB    | 2             | Long                   |
| 2024-10-31 | ACB    | 1             | Neutral                |
| 2024-10-31 | VPB    | 1             | Neutral                |
| 2024-10-31 | HDB    | 2             | Long                   |
| 2024-10-31 | VIB    | 3             | Neutral                |



| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-10-31 | TPB    | 3             | Neutral                |
| 2024-10-31 | STB    | 3             | Neutral                |
| 2024-10-31 | SHB    | 3             | Neutral                |
| 2024-10-31 | LPB    | 2             | Long                   |
| 2024-10-31 | SSB    | 1             | Neutral                |
| 2024-10-31 | MSB    | 1             | Neutral                |
| 2024-10-31 | OCB    | 3             | Neutral                |
| 2024-10-31 | NAB    | 2             | Long                   |
| 2024-10-31 | BAB    | 1             | Neutral                |
| 2024-10-31 | KLB    | 3             | Neutral                |
| 2024-10-31 | PGB    | 0             | Short                  |
| 2024-10-31 | VAB    | 2             | Long                   |
| 2024-10-31 | VBB    | 3             | Neutral                |
| 2024-10-31 | SGB    | 1             | Neutral                |
| 2024-10-31 | ABB    | 1             | Neutral                |
| 2024-10-31 | BVB    | 2             | Long                   |
| 2024-11-30 | VCB    | 3             | Neutral                |
| 2024-11-30 | BID    | 0             | Neutral                |
| 2024-11-30 | CTG    | 0             | Neutral                |
| 2024-11-30 | MBB    | 0             | Neutral                |
| 2024-11-30 | TCB    | 0             | Neutral                |
| 2024-11-30 | ACB    | 3             | Neutral                |
| 2024-11-30 | VPB    | 0             | Neutral                |
| 2024-11-30 | HDB    | 0             | Neutral                |
| 2024-11-30 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-11-30 | TPB    | 0             | Neutral                |
| 2024-11-30 | STB    | 0             | Neutral                |
| 2024-11-30 | SHB    | 0             | Neutral                |
| 2024-11-30 | LPB    | 2             | Long                   |
| 2024-11-30 | SSB    | 3             | Neutral                |
| 2024-11-30 | MSB    | 0             | Neutral                |
| 2024-11-30 | OCB    | 0             | Neutral                |
| 2024-11-30 | NAB    | 0             | Neutral                |
| 2024-11-30 | BAB    | 3             | Neutral                |
| 2024-11-30 | KLB    | 0             | Neutral                |
| 2024-11-30 | PGB    | 1             | Short                  |
| 2024-11-30 | VAB    | 0             | Neutral                |
| 2024-11-30 | VBB    | 0             | Neutral                |
| 2024-11-30 | SGB    | 3             | Neutral                |
| 2024-11-30 | ABB    | 0             | Neutral                |
| 2024-11-30 | BVB    | 0             | Neutral                |
| 2024-12-31 | VCB    | 2             | Neutral                |
| 2024-12-31 | BID    | 0             | Neutral                |
| 2024-12-31 | CTG    | 0             | Neutral                |
| 2024-12-31 | MBB    | 0             | Neutral                |
| 2024-12-31 | TCB    | 0             | Neutral                |
| 2024-12-31 | ACB    | 2             | Neutral                |
| 2024-12-31 | VPB    | 0             | Neutral                |
| 2024-12-31 | HDB    | 0             | Neutral                |
| 2024-12-31 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2024-12-31 | TPB    | 0             | Neutral                |
| 2024-12-31 | STB    | 0             | Neutral                |
| 2024-12-31 | SHB    | 0             | Neutral                |
| 2024-12-31 | LPB    | 1             | Long                   |
| 2024-12-31 | SSB    | 2             | Neutral                |
| 2024-12-31 | MSB    | 0             | Neutral                |
| 2024-12-31 | OCB    | 0             | Neutral                |
| 2024-12-31 | NAB    | 0             | Neutral                |
| 2024-12-31 | BAB    | 2             | Neutral                |
| 2024-12-31 | KLB    | 0             | Neutral                |
| 2024-12-31 | PGB    | 3             | Short                  |
| 2024-12-31 | VAB    | 0             | Neutral                |
| 2024-12-31 | VBB    | 0             | Neutral                |
| 2024-12-31 | SGB    | 2             | Neutral                |
| 2024-12-31 | ABB    | 0             | Neutral                |
| 2024-12-31 | BVB    | 0             | Neutral                |
| 2025-01-31 | VCB    | 3             | Neutral                |
| 2025-01-31 | BID    | 0             | Short                  |
| 2025-01-31 | CTG    | 2             | Neutral                |
| 2025-01-31 | MBB    | 2             | Neutral                |
| 2025-01-31 | TCB    | 0             | Short                  |
| 2025-01-31 | ACB    | 3             | Neutral                |
| 2025-01-31 | VPB    | 2             | Neutral                |
| 2025-01-31 | HDB    | 2             | Neutral                |
| 2025-01-31 | VIB    | 0             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-01-31 | TPB    | 2             | Neutral                |
| 2025-01-31 | STB    | 1             | Long                   |
| 2025-01-31 | SHB    | 0             | Short                  |
| 2025-01-31 | LPB    | 1             | Long                   |
| 2025-01-31 | SSB    | 3             | Neutral                |
| 2025-01-31 | MSB    | 2             | Neutral                |
| 2025-01-31 | OCB    | 0             | Short                  |
| 2025-01-31 | NAB    | 2             | Neutral                |
| 2025-01-31 | BAB    | 3             | Neutral                |
| 2025-01-31 | KLB    | 0             | Short                  |
| 2025-01-31 | PGB    | 0             | Short                  |
| 2025-01-31 | VAB    | 0             | Short                  |
| 2025-01-31 | VBB    | 0             | Short                  |
| 2025-01-31 | SGB    | 3             | Neutral                |
| 2025-01-31 | ABB    | 3             | Neutral                |
| 2025-01-31 | BVB    | 2             | Neutral                |
| 2025-02-28 | VCB    | 1             | Neutral                |
| 2025-02-28 | BID    | 0             | Neutral                |
| 2025-02-28 | CTG    | 3             | Long                   |
| 2025-02-28 | MBB    | 3             | Long                   |
| 2025-02-28 | TCB    | 0             | Neutral                |
| 2025-02-28 | ACB    | 1             | Neutral                |
| 2025-02-28 | VPB    | 1             | Neutral                |
| 2025-02-28 | HDB    | 3             | Long                   |
| 2025-02-28 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-02-28 | TPB    | 3             | Long                   |
| 2025-02-28 | STB    | 3             | Long                   |
| 2025-02-28 | SHB    | 2             | Short                  |
| 2025-02-28 | LPB    | 3             | Long                   |
| 2025-02-28 | SSB    | 2             | Short                  |
| 2025-02-28 | MSB    | 1             | Neutral                |
| 2025-02-28 | OCB    | 2             | Short                  |
| 2025-02-28 | NAB    | 3             | Long                   |
| 2025-02-28 | BAB    | 1             | Neutral                |
| 2025-02-28 | KLB    | 0             | Neutral                |
| 2025-02-28 | PGB    | 0             | Neutral                |
| 2025-02-28 | VAB    | 2             | Short                  |
| 2025-02-28 | VBB    | 0             | Neutral                |
| 2025-02-28 | SGB    | 2             | Short                  |
| 2025-02-28 | ABB    | 2             | Short                  |
| 2025-02-28 | BVB    | 2             | Short                  |
| 2025-03-31 | VCB    | 2             | Neutral                |
| 2025-03-31 | BID    | 1             | Neutral                |
| 2025-03-31 | CTG    | 3             | Long                   |
| 2025-03-31 | MBB    | 1             | Neutral                |
| 2025-03-31 | TCB    | 1             | Neutral                |
| 2025-03-31 | ACB    | 2             | Neutral                |
| 2025-03-31 | VPB    | 1             | Neutral                |
| 2025-03-31 | HDB    | 3             | Long                   |
| 2025-03-31 | VIB    | 1             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-03-31 | TPB    | 3             | Long                   |
| 2025-03-31 | STB    | 3             | Long                   |
| 2025-03-31 | SHB    | 1             | Neutral                |
| 2025-03-31 | LPB    | 3             | Long                   |
| 2025-03-31 | SSB    | 2             | Neutral                |
| 2025-03-31 | MSB    | 1             | Neutral                |
| 2025-03-31 | OCB    | 1             | Neutral                |
| 2025-03-31 | NAB    | 3             | Long                   |
| 2025-03-31 | BAB    | 2             | Neutral                |
| 2025-03-31 | KLB    | 0             | Short                  |
| 2025-03-31 | PGB    | 0             | Short                  |
| 2025-03-31 | VAB    | 1             | Neutral                |
| 2025-03-31 | VBB    | 3             | Long                   |
| 2025-03-31 | SGB    | 2             | Neutral                |
| 2025-03-31 | ABB    | 1             | Neutral                |
| 2025-03-31 | BVB    | 1             | Neutral                |
| 2025-04-30 | VCB    | 2             | Neutral                |
| 2025-04-30 | BID    | 0             | Neutral                |
| 2025-04-30 | CTG    | 0             | Neutral                |
| 2025-04-30 | MBB    | 0             | Neutral                |
| 2025-04-30 | TCB    | 0             | Neutral                |
| 2025-04-30 | ACB    | 2             | Neutral                |
| 2025-04-30 | VPB    | 0             | Neutral                |
| 2025-04-30 | HDB    | 0             | Neutral                |
| 2025-04-30 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-04-30 | TPB    | 0             | Neutral                |
| 2025-04-30 | STB    | 3             | Long                   |
| 2025-04-30 | SHB    | 1             | Short                  |
| 2025-04-30 | LPB    | 3             | Long                   |
| 2025-04-30 | SSB    | 2             | Neutral                |
| 2025-04-30 | MSB    | 0             | Neutral                |
| 2025-04-30 | OCB    | 1             | Short                  |
| 2025-04-30 | NAB    | 0             | Neutral                |
| 2025-04-30 | BAB    | 2             | Neutral                |
| 2025-04-30 | KLB    | 1             | Short                  |
| 2025-04-30 | PGB    | 1             | Short                  |
| 2025-04-30 | VAB    | 1             | Short                  |
| 2025-04-30 | VBB    | 0             | Neutral                |
| 2025-04-30 | SGB    | 2             | Neutral                |
| 2025-04-30 | ABB    | 1             | Short                  |
| 2025-04-30 | BVB    | 0             | Neutral                |
| 2025-05-31 | VCB    | 2             | Neutral                |
| 2025-05-31 | BID    | 0             | Neutral                |
| 2025-05-31 | CTG    | 1             | Long                   |
| 2025-05-31 | MBB    | 0             | Neutral                |
| 2025-05-31 | TCB    | 1             | Long                   |
| 2025-05-31 | ACB    | 2             | Neutral                |
| 2025-05-31 | VPB    | 0             | Neutral                |
| 2025-05-31 | HDB    | 0             | Neutral                |
| 2025-05-31 | VIB    | 1             | Long                   |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-05-31 | TPB    | 1             | Long                   |
| 2025-05-31 | STB    | 1             | Long                   |
| 2025-05-31 | SHB    | 0             | Neutral                |
| 2025-05-31 | LPB    | 1             | Long                   |
| 2025-05-31 | SSB    | 2             | Neutral                |
| 2025-05-31 | MSB    | 0             | Neutral                |
| 2025-05-31 | OCB    | 0             | Neutral                |
| 2025-05-31 | NAB    | 0             | Neutral                |
| 2025-05-31 | BAB    | 2             | Neutral                |
| 2025-05-31 | KLB    | 3             | Short                  |
| 2025-05-31 | PGB    | 3             | Short                  |
| 2025-05-31 | VAB    | 1             | Long                   |
| 2025-05-31 | VBB    | 1             | Long                   |
| 2025-05-31 | SGB    | 2             | Neutral                |
| 2025-05-31 | ABB    | 0             | Neutral                |
| 2025-05-31 | BVB    | 1             | Long                   |
| 2025-06-30 | VCB    | 2             | Neutral                |
| 2025-06-30 | BID    | 0             | Neutral                |
| 2025-06-30 | CTG    | 1             | Long                   |
| 2025-06-30 | MBB    | 1             | Long                   |
| 2025-06-30 | TCB    | 1             | Long                   |
| 2025-06-30 | ACB    | 2             | Neutral                |
| 2025-06-30 | VPB    | 0             | Neutral                |
| 2025-06-30 | HDB    | 0             | Neutral                |
| 2025-06-30 | VIB    | 1             | Long                   |



| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-06-30 | TPB    | 0             | Neutral                |
| 2025-06-30 | STB    | 1             | Long                   |
| 2025-06-30 | SHB    | 1             | Long                   |
| 2025-06-30 | LPB    | 1             | Long                   |
| 2025-06-30 | SSB    | 2             | Neutral                |
| 2025-06-30 | MSB    | 0             | Neutral                |
| 2025-06-30 | OCB    | 0             | Neutral                |
| 2025-06-30 | NAB    | 0             | Neutral                |
| 2025-06-30 | BAB    | 2             | Neutral                |
| 2025-06-30 | KLB    | 3             | Short                  |
| 2025-06-30 | PGB    | 3             | Short                  |
| 2025-06-30 | VAB    | 1             | Long                   |
| 2025-06-30 | VBB    | 1             | Long                   |
| 2025-06-30 | SGB    | 2             | Neutral                |
| 2025-06-30 | ABB    | 0             | Neutral                |
| 2025-06-30 | BVB    | 1             | Long                   |
| 2025-07-31 | VCB    | 1             | Neutral                |
| 2025-07-31 | BID    | 0             | Short                  |
| 2025-07-31 | CTG    | 1             | Neutral                |
| 2025-07-31 | MBB    | 2             | Long                   |
| 2025-07-31 | TCB    | 2             | Long                   |
| 2025-07-31 | ACB    | 1             | Neutral                |
| 2025-07-31 | VPB    | 0             | Short                  |
| 2025-07-31 | HDB    | 0             | Short                  |
| 2025-07-31 | VIB    | 0             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-07-31 | TPB    | 0             | Short                  |
| 2025-07-31 | STB    | 2             | Long                   |
| 2025-07-31 | SHB    | 2             | Long                   |
| 2025-07-31 | LPB    | 2             | Long                   |
| 2025-07-31 | SSB    | 1             | Neutral                |
| 2025-07-31 | MSB    | 0             | Short                  |
| 2025-07-31 | OCB    | 0             | Short                  |
| 2025-07-31 | NAB    | 2             | Long                   |
| 2025-07-31 | BAB    | 1             | Neutral                |
| 2025-07-31 | KLB    | 3             | Neutral                |
| 2025-07-31 | PGB    | 3             | Neutral                |
| 2025-07-31 | VAB    | 2             | Long                   |
| 2025-07-31 | VBB    | 2             | Long                   |
| 2025-07-31 | SGB    | 1             | Neutral                |
| 2025-07-31 | ABB    | 1             | Neutral                |
| 2025-07-31 | BVB    | 2             | Long                   |
| 2025-08-31 | VCB    | 2             | Neutral                |
| 2025-08-31 | BID    | 1             | Short                  |
| 2025-08-31 | CTG    | 1             | Short                  |
| 2025-08-31 | MBB    | 1             | Short                  |
| 2025-08-31 | TCB    | 0             | Neutral                |
| 2025-08-31 | ACB    | 2             | Neutral                |
| 2025-08-31 | VPB    | 1             | Short                  |
| 2025-08-31 | HDB    | 1             | Short                  |
| 2025-08-31 | VIB    | 1             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-08-31 | TPB    | 1             | Short                  |
| 2025-08-31 | STB    | 0             | Neutral                |
| 2025-08-31 | SHB    | 3             | Long                   |
| 2025-08-31 | LPB    | 0             | Neutral                |
| 2025-08-31 | SSB    | 2             | Neutral                |
| 2025-08-31 | MSB    | 1             | Short                  |
| 2025-08-31 | OCB    | 1             | Short                  |
| 2025-08-31 | NAB    | 1             | Short                  |
| 2025-08-31 | BAB    | 2             | Neutral                |
| 2025-08-31 | KLB    | 3             | Long                   |
| 2025-08-31 | PGB    | 0             | Neutral                |
| 2025-08-31 | VAB    | 3             | Long                   |
| 2025-08-31 | VBB    | 0             | Neutral                |
| 2025-08-31 | SGB    | 2             | Neutral                |
| 2025-08-31 | ABB    | 1             | Short                  |
| 2025-08-31 | BVB    | 1             | Short                  |
| 2025-09-30 | VCB    | 3             | Neutral                |
| 2025-09-30 | BID    | 2             | Short                  |
| 2025-09-30 | CTG    | 0             | Neutral                |
| 2025-09-30 | MBB    | 0             | Neutral                |
| 2025-09-30 | TCB    | 1             | Long                   |
| 2025-09-30 | ACB    | 3             | Neutral                |
| 2025-09-30 | VPB    | 2             | Short                  |
| 2025-09-30 | HDB    | 2             | Short                  |
| 2025-09-30 | VIB    | 2             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-09-30 | TPB    | 2             | Short                  |
| 2025-09-30 | STB    | 1             | Long                   |
| 2025-09-30 | SHB    | 1             | Long                   |
| 2025-09-30 | LPB    | 0             | Neutral                |
| 2025-09-30 | SSB    | 3             | Neutral                |
| 2025-09-30 | MSB    | 0             | Neutral                |
| 2025-09-30 | OCB    | 0             | Neutral                |
| 2025-09-30 | NAB    | 0             | Neutral                |
| 2025-09-30 | BAB    | 3             | Neutral                |
| 2025-09-30 | KLB    | 1             | Long                   |
| 2025-09-30 | PGB    | 2             | Short                  |
| 2025-09-30 | VAB    | 1             | Long                   |
| 2025-09-30 | VBB    | 1             | Long                   |
| 2025-09-30 | SGB    | 3             | Neutral                |
| 2025-09-30 | ABB    | 0             | Neutral                |
| 2025-09-30 | BVB    | 0             | Neutral                |
| 2025-10-31 | VCB    | 1             | Short                  |
| 2025-10-31 | BID    | 0             | Neutral                |
| 2025-10-31 | CTG    | 0             | Neutral                |
| 2025-10-31 | MBB    | 0             | Neutral                |
| 2025-10-31 | TCB    | 3             | Neutral                |
| 2025-10-31 | ACB    | 1             | Short                  |
| 2025-10-31 | VPB    | 3             | Neutral                |
| 2025-10-31 | HDB    | 0             | Neutral                |
| 2025-10-31 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-10-31 | TPB    | 0             | Neutral                |
| 2025-10-31 | STB    | 3             | Neutral                |
| 2025-10-31 | SHB    | 2             | Long                   |
| 2025-10-31 | LPB    | 0             | Neutral                |
| 2025-10-31 | SSB    | 1             | Short                  |
| 2025-10-31 | MSB    | 0             | Neutral                |
| 2025-10-31 | OCB    | 0             | Neutral                |
| 2025-10-31 | NAB    | 0             | Neutral                |
| 2025-10-31 | BAB    | 1             | Short                  |
| 2025-10-31 | KLB    | 2             | Long                   |
| 2025-10-31 | PGB    | 3             | Neutral                |
| 2025-10-31 | VAB    | 2             | Long                   |
| 2025-10-31 | VBB    | 3             | Neutral                |
| 2025-10-31 | SGB    | 1             | Short                  |
| 2025-10-31 | ABB    | 2             | Long                   |
| 2025-10-31 | BVB    | 0             | Neutral                |
| 2025-11-30 | VCB    | 3             | Neutral                |
| 2025-11-30 | BID    | 2             | Short                  |
| 2025-11-30 | CTG    | 3             | Neutral                |
| 2025-11-30 | MBB    | 0             | Neutral                |
| 2025-11-30 | TCB    | 0             | Neutral                |
| 2025-11-30 | ACB    | 3             | Neutral                |
| 2025-11-30 | VPB    | 1             | Long                   |
| 2025-11-30 | HDB    | 0             | Neutral                |
| 2025-11-30 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-11-30 | TPB    | 0             | Neutral                |
| 2025-11-30 | STB    | 0             | Neutral                |
| 2025-11-30 | SHB    | 1             | Long                   |
| 2025-11-30 | LPB    | 0             | Neutral                |
| 2025-11-30 | SSB    | 3             | Neutral                |
| 2025-11-30 | MSB    | 0             | Neutral                |
| 2025-11-30 | OCB    | 0             | Neutral                |
| 2025-11-30 | NAB    | 2             | Short                  |
| 2025-11-30 | BAB    | 3             | Neutral                |
| 2025-11-30 | KLB    | 1             | Long                   |
| 2025-11-30 | PGB    | 2             | Short                  |
| 2025-11-30 | VAB    | 1             | Long                   |
| 2025-11-30 | VBB    | 0             | Neutral                |
| 2025-11-30 | SGB    | 3             | Neutral                |
| 2025-11-30 | ABB    | 1             | Long                   |
| 2025-11-30 | BVB    | 2             | Short                  |
| 2025-12-31 | VCB    | 1             | Short                  |
| 2025-12-31 | BID    | 1             | Short                  |
| 2025-12-31 | CTG    | 1             | Short                  |
| 2025-12-31 | MBB    | 0             | Neutral                |
| 2025-12-31 | TCB    | 0             | Neutral                |
| 2025-12-31 | ACB    | 1             | Short                  |
| 2025-12-31 | VPB    | 2             | Neutral                |
| 2025-12-31 | HDB    | 0             | Neutral                |
| 2025-12-31 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2025-12-31 | TPB    | 0             | Neutral                |
| 2025-12-31 | STB    | 0             | Neutral                |
| 2025-12-31 | SHB    | 2             | Neutral                |
| 2025-12-31 | LPB    | 2             | Neutral                |
| 2025-12-31 | SSB    | 1             | Short                  |
| 2025-12-31 | MSB    | 0             | Neutral                |
| 2025-12-31 | OCB    | 0             | Neutral                |
| 2025-12-31 | NAB    | 1             | Short                  |
| 2025-12-31 | BAB    | 1             | Short                  |
| 2025-12-31 | KLB    | 3             | Long                   |
| 2025-12-31 | PGB    | 0             | Neutral                |
| 2025-12-31 | VAB    | 2             | Neutral                |
| 2025-12-31 | VBB    | 0             | Neutral                |
| 2025-12-31 | SGB    | 1             | Short                  |
| 2025-12-31 | ABB    | 2             | Neutral                |
| 2025-12-31 | BVB    | 1             | Short                  |
| 2026-01-31 | VCB    | 0             | Short                  |
| 2026-01-31 | BID    | 0             | Short                  |
| 2026-01-31 | CTG    | 2             | Neutral                |
| 2026-01-31 | MBB    | 2             | Neutral                |
| 2026-01-31 | TCB    | 2             | Neutral                |
| 2026-01-31 | ACB    | 0             | Short                  |
| 2026-01-31 | VPB    | 1             | Neutral                |
| 2026-01-31 | HDB    | 1             | Neutral                |
| 2026-01-31 | VIB    | 2             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2026-01-31 | TPB    | 1             | Neutral                |
| 2026-01-31 | STB    | 2             | Neutral                |
| 2026-01-31 | SHB    | 1             | Neutral                |
| 2026-01-31 | LPB    | 1             | Neutral                |
| 2026-01-31 | SSB    | 0             | Short                  |
| 2026-01-31 | MSB    | 2             | Neutral                |
| 2026-01-31 | OCB    | 2             | Neutral                |
| 2026-01-31 | NAB    | 2             | Neutral                |
| 2026-01-31 | BAB    | 0             | Short                  |
| 2026-01-31 | KLB    | 3             | Long                   |
| 2026-01-31 | PGB    | 1             | Neutral                |
| 2026-01-31 | VAB    | 1             | Neutral                |
| 2026-01-31 | VBB    | 1             | Neutral                |
| 2026-01-31 | SGB    | 0             | Short                  |
| 2026-01-31 | ABB    | 1             | Neutral                |
| 2026-01-31 | BVB    | 0             | Short                  |
| 2026-02-28 | VCB    | 0             | Short                  |
| 2026-02-28 | BID    | 0             | Short                  |
| 2026-02-28 | CTG    | 2             | Neutral                |
| 2026-02-28 | MBB    | 2             | Neutral                |
| 2026-02-28 | TCB    | 2             | Neutral                |
| 2026-02-28 | ACB    | 0             | Short                  |
| 2026-02-28 | VPB    | 1             | Long                   |
| 2026-02-28 | HDB    | 1             | Long                   |
| 2026-02-28 | VIB    | 2             | Neutral                |



| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2026-02-28 | TPB    | 1             | Long                   |
| 2026-02-28 | STB    | 2             | Neutral                |
| 2026-02-28 | SHB    | 1             | Long                   |
| 2026-02-28 | LPB    | 1             | Long                   |
| 2026-02-28 | SSB    | 0             | Short                  |
| 2026-02-28 | MSB    | 1             | Long                   |
| 2026-02-28 | OCB    | 2             | Neutral                |
| 2026-02-28 | NAB    | 2             | Neutral                |
| 2026-02-28 | BAB    | 0             | Short                  |
| 2026-02-28 | KLB    | 3             | Neutral                |
| 2026-02-28 | PGB    | 3             | Neutral                |
| 2026-02-28 | VAB    | 2             | Neutral                |
| 2026-02-28 | VBB    | 3             | Neutral                |
| 2026-02-28 | SGB    | 0             | Short                  |
| 2026-02-28 | ABB    | 1             | Long                   |
| 2026-02-28 | BVB    | 0             | Short                  |
| 2026-03-31 | VCB    | 2             | Short                  |
| 2026-03-31 | BID    | 0             | Neutral                |
| 2026-03-31 | CTG    | 0             | Neutral                |
| 2026-03-31 | MBB    | 3             | Neutral                |
| 2026-03-31 | TCB    | 0             | Neutral                |
| 2026-03-31 | ACB    | 2             | Short                  |
| 2026-03-31 | VPB    | 1             | Long                   |
| 2026-03-31 | HDB    | 1             | Long                   |
| 2026-03-31 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2026-03-31 | TPB    | 3             | Neutral                |
| 2026-03-31 | STB    | 3             | Neutral                |
| 2026-03-31 | SHB    | 3             | Neutral                |
| 2026-03-31 | LPB    | 3             | Neutral                |
| 2026-03-31 | SSB    | 2             | Short                  |
| 2026-03-31 | MSB    | 3             | Neutral                |
| 2026-03-31 | OCB    | 0             | Neutral                |
| 2026-03-31 | NAB    | 0             | Neutral                |
| 2026-03-31 | BAB    | 2             | Short                  |
| 2026-03-31 | KLB    | 1             | Long                   |
| 2026-03-31 | PGB    | 3             | Neutral                |
| 2026-03-31 | VAB    | 0             | Neutral                |
| 2026-03-31 | VBB    | 0             | Neutral                |
| 2026-03-31 | SGB    | 2             | Short                  |
| 2026-03-31 | ABB    | 1             | Long                   |
| 2026-03-31 | BVB    | 0             | Neutral                |
| 2026-04-30 | VCB    | 1             | Long                   |
| 2026-04-30 | BID    | 1             | Long                   |
| 2026-04-30 | CTG    | 1             | Long                   |
| 2026-04-30 | MBB    | 1             | Long                   |
| 2026-04-30 | TCB    | 0             | Neutral                |
| 2026-04-30 | ACB    | 3             | Short                  |
| 2026-04-30 | VPB    | 2             | Neutral                |
| 2026-04-30 | HDB    | 1             | Long                   |
| 2026-04-30 | VIB    | 3             | Short                  |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2026-04-30 | TPB    | 0             | Neutral                |
| 2026-04-30 | STB    | 1             | Long                   |
| 2026-04-30 | SHB    | 0             | Neutral                |
| 2026-04-30 | LPB    | 2             | Neutral                |
| 2026-04-30 | SSB    | 3             | Short                  |
| 2026-04-30 | MSB    | 0             | Neutral                |
| 2026-04-30 | OCB    | 3             | Short                  |
| 2026-04-30 | NAB    | 3             | Short                  |
| 2026-04-30 | BAB    | 3             | Short                  |
| 2026-04-30 | KLB    | 2             | Neutral                |
| 2026-04-30 | PGB    | 2             | Neutral                |
| 2026-04-30 | VAB    | 0             | Neutral                |
| 2026-04-30 | VBB    | 0             | Neutral                |
| 2026-04-30 | SGB    | 3             | Short                  |
| 2026-04-30 | ABB    | 2             | Neutral                |
| 2026-04-30 | BVB    | 3             | Short                  |
| 2026-05-31 | VCB    | 0             | Neutral                |
| 2026-05-31 | BID    | 3             | Neutral                |
| 2026-05-31 | CTG    | 2             | Long                   |
| 2026-05-31 | MBB    | 2             | Long                   |
| 2026-05-31 | TCB    | 1             | Short                  |
| 2026-05-31 | ACB    | 0             | Neutral                |
| 2026-05-31 | VPB    | 1             | Short                  |
| 2026-05-31 | HDB    | 3             | Neutral                |
| 2026-05-31 | VIB    | 0             | Neutral                |

| Date       | Ticker | Cluster_Label | Assigned_View_Position |
|------------|--------|---------------|------------------------|
| 2026-05-31 | TPB    | 1             | Short                  |
| 2026-05-31 | STB    | 2             | Long                   |
| 2026-05-31 | SHB    | 1             | Short                  |
| 2026-05-31 | LPB    | 1             | Short                  |
| 2026-05-31 | SSB    | 0             | Neutral                |
| 2026-05-31 | MSB    | 1             | Short                  |
| 2026-05-31 | OCB    | 0             | Neutral                |
| 2026-05-31 | NAB    | 0             | Neutral                |
| 2026-05-31 | BAB    | 0             | Neutral                |
| 2026-05-31 | KLB    | 3             | Neutral                |
| 2026-05-31 | PGB    | 1             | Short                  |
| 2026-05-31 | VAB    | 1             | Short                  |
| 2026-05-31 | VBB    | 0             | Neutral                |
| 2026-05-31 | SGB    | 0             | Neutral                |
| 2026-05-31 | ABB    | 3             | Neutral                |
| 2026-05-31 | BVB    | 0             | Neutral                |

9. Lịch sử Trọng Danh mục

Trọng của 6 danh mục cho từng cổ phiếu tại mỗi tái cân bằng. w\_mkt = vốn hóa thị trường; w\_TAN = Tangency; w\_MV = Min-Variance; w\_BL = Black-Litterman; w\_EW = Equal-Weight; w\_RP = Risk-Parity.

Tổng số quan sát: 1,561 dòng

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2017-08-31 | VCB    | 0.373349 | 0.300000 | 0.228374 | 0.000000 | 0.142857 | 0.143445 |
| 2017-08-31 | BID    | 0.235950 | 0.078146 | 0.000000 | 0.100000 | 0.142857 | 0.116484 |
| 2017-08-31 | CTG    | 0.138339 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.126106 |
| 2017-08-31 | MBB    | 0.085414 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.200617 |
| 2017-08-31 | ACB    | 0.067046 | 0.300000 | 0.180592 | 0.000000 | 0.142857 | 0.148225 |
| 2017-08-31 | STB    | 0.070691 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.126525 |
| 2017-08-31 | SHB    | 0.029211 | 0.021854 | 0.291034 | 0.300000 | 0.142857 | 0.138600 |
| 2017-09-30 | VCB    | 0.376914 | 0.300000 | 0.237058 | 0.000000 | 0.142857 | 0.144085 |
| 2017-09-30 | BID    | 0.233311 | 0.035950 | 0.000000 | 0.100000 | 0.142857 | 0.116089 |
| 2017-09-30 | CTG    | 0.132644 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.126566 |
| 2017-09-30 | MBB    | 0.087568 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.200615 |
| 2017-09-30 | ACB    | 0.076128 | 0.300000 | 0.176242 | 0.000000 | 0.142857 | 0.148797 |
| 2017-09-30 | STB    | 0.064861 | 0.000000 | 0.005906 | 0.300000 | 0.142857 | 0.126844 |
| 2017-09-30 | SHB    | 0.028574 | 0.064050 | 0.280794 | 0.300000 | 0.142857 | 0.137005 |
| 2017-10-31 | VCB    | 0.387099 | 0.300000 | 0.246125 | 0.000000 | 0.142857 | 0.146093 |
| 2017-10-31 | BID    | 0.219039 | 0.079333 | 0.000000 | 0.100000 | 0.142857 | 0.116211 |
| 2017-10-31 | CTG    | 0.132609 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.125730 |
| 2017-10-31 | MBB    | 0.083441 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.201082 |
| 2017-10-31 | ACB    | 0.080421 | 0.300000 | 0.168708 | 0.000000 | 0.142857 | 0.147171 |
| 2017-10-31 | STB    | 0.068705 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.124942 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2017-10-31 | SHB    | 0.028686 | 0.020667 | 0.285167 | 0.300000 | 0.142857 | 0.138770 |
| 2017-11-30 | VCB    | 0.401989 | 0.300000 | 0.258089 | 0.000000 | 0.142857 | 0.148904 |
| 2017-11-30 | BID    | 0.230218 | 0.034857 | 0.000000 | 0.100000 | 0.142857 | 0.116873 |
| 2017-11-30 | CTG    | 0.126737 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.126059 |
| 2017-11-30 | MBB    | 0.081745 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.198075 |
| 2017-11-30 | ACB    | 0.075875 | 0.300000 | 0.160293 | 0.000000 | 0.142857 | 0.147259 |
| 2017-11-30 | STB    | 0.057042 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.123783 |
| 2017-11-30 | SHB    | 0.026393 | 0.065143 | 0.281618 | 0.300000 | 0.142857 | 0.139047 |
| 2017-12-31 | VCB    | 0.397617 | 0.300000 | 0.248045 | 0.000000 | 0.142857 | 0.148158 |
| 2017-12-31 | BID    | 0.228292 | 0.070882 | 0.000000 | 0.120614 | 0.142857 | 0.116260 |
| 2017-12-31 | CTG    | 0.138682 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.126748 |
| 2017-12-31 | MBB    | 0.078230 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.198885 |
| 2017-12-31 | ACB    | 0.073008 | 0.300000 | 0.167856 | 0.000000 | 0.142857 | 0.147921 |
| 2017-12-31 | STB    | 0.057355 | 0.000000 | 0.000000 | 0.279386 | 0.142857 | 0.121493 |
| 2017-12-31 | SHB    | 0.026816 | 0.029118 | 0.284100 | 0.300000 | 0.142857 | 0.140535 |
| 2018-01-31 | VCB    | 0.424734 | 0.300000 | 0.234821 | 0.300000 | 0.142857 | 0.148700 |
| 2018-01-31 | BID    | 0.216572 | 0.070629 | 0.000000 | 0.300000 | 0.142857 | 0.119427 |
| 2018-01-31 | CTG    | 0.131843 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.120997 |
| 2018-01-31 | MBB    | 0.074099 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.201702 |
| 2018-01-31 | ACB    | 0.073756 | 0.300000 | 0.208134 | 0.100000 | 0.142857 | 0.151086 |
| 2018-01-31 | STB    | 0.053679 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.120734 |
| 2018-01-31 | SHB    | 0.025316 | 0.029371 | 0.257046 | 0.000000 | 0.142857 | 0.137354 |
| 2018-02-28 | VCB    | 0.418904 | 0.300000 | 0.232274 | 0.198273 | 0.142857 | 0.150310 |
| 2018-02-28 | BID    | 0.233422 | 0.033103 | 0.000000 | 0.300000 | 0.142857 | 0.119047 |
| 2018-02-28 | CTG    | 0.119111 | 0.000000 | 0.000000 | 0.201727 | 0.142857 | 0.120776 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2018-02-28 | MBB    | 0.076429 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.200325 |
| 2018-02-28 | ACB    | 0.066989 | 0.300000 | 0.209142 | 0.000000 | 0.142857 | 0.150444 |
| 2018-02-28 | STB    | 0.054579 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.120510 |
| 2018-02-28 | SHB    | 0.030568 | 0.066897 | 0.258584 | 0.000000 | 0.142857 | 0.138587 |
| 2018-03-31 | VCB    | 0.414303 | 0.300000 | 0.266719 | 0.300000 | 0.142857 | 0.151868 |
| 2018-03-31 | BID    | 0.237604 | 0.022044 | 0.000000 | 0.187710 | 0.142857 | 0.129991 |
| 2018-03-31 | CTG    | 0.131338 | 0.000000 | 0.000000 | 0.212290 | 0.142857 | 0.140028 |
| 2018-03-31 | MBB    | 0.073556 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.181390 |
| 2018-03-31 | ACB    | 0.067390 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.159768 |
| 2018-03-31 | STB    | 0.048267 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.119400 |
| 2018-03-31 | SHB    | 0.027543 | 0.077956 | 0.133281 | 0.000000 | 0.142857 | 0.117555 |
| 2018-04-30 | VCB    | 0.389328 | 0.300000 | 0.254140 | 0.300000 | 0.142857 | 0.152819 |
| 2018-04-30 | BID    | 0.260056 | 0.019358 | 0.000000 | 0.300000 | 0.142857 | 0.129780 |
| 2018-04-30 | CTG    | 0.132959 | 0.000000 | 0.000000 | 0.100000 | 0.142857 | 0.133971 |
| 2018-04-30 | MBB    | 0.072920 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.183512 |
| 2018-04-30 | ACB    | 0.072388 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.159655 |
| 2018-04-30 | STB    | 0.045704 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.121077 |
| 2018-04-30 | SHB    | 0.026645 | 0.080642 | 0.145860 | 0.000000 | 0.142857 | 0.119185 |
| 2018-05-31 | VCB    | 0.388273 | 0.300000 | 0.261662 | 0.300000 | 0.142857 | 0.154197 |
| 2018-05-31 | BID    | 0.253551 | 0.056793 | 0.000000 | 0.300000 | 0.142857 | 0.128754 |
| 2018-05-31 | CTG    | 0.132002 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.136729 |
| 2018-05-31 | MBB    | 0.073715 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.183561 |
| 2018-05-31 | ACB    | 0.077021 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.158183 |
| 2018-05-31 | STB    | 0.047862 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.119996 |
| 2018-05-31 | SHB    | 0.027575 | 0.043207 | 0.138338 | 0.000000 | 0.142857 | 0.118579 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2018-06-30 | VCB    | 0.395582 | 0.300000 | 0.242855 | 0.247144 | 0.142857 | 0.151310 |
| 2018-06-30 | BID    | 0.232078 | 0.021900 | 0.000000 | 0.300000 | 0.142857 | 0.127641 |
| 2018-06-30 | CTG    | 0.137903 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.136343 |
| 2018-06-30 | MBB    | 0.081008 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.177291 |
| 2018-06-30 | ACB    | 0.082125 | 0.300000 | 0.300000 | 0.152856 | 0.142857 | 0.160742 |
| 2018-06-30 | STB    | 0.045875 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.124615 |
| 2018-06-30 | SHB    | 0.025429 | 0.078100 | 0.157145 | 0.000000 | 0.142857 | 0.122059 |
| 2018-07-31 | VCB    | 0.438421 | 0.300000 | 0.300000 | 0.181967 | 0.142857 | 0.155666 |
| 2018-07-31 | BID    | 0.213290 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.122501 |
| 2018-07-31 | CTG    | 0.127604 | 0.020642 | 0.000000 | 0.300000 | 0.142857 | 0.136709 |
| 2018-07-31 | MBB    | 0.075376 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.177940 |
| 2018-07-31 | ACB    | 0.075545 | 0.300000 | 0.300000 | 0.009195 | 0.142857 | 0.167837 |
| 2018-07-31 | STB    | 0.046613 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.121973 |
| 2018-07-31 | SHB    | 0.023152 | 0.079358 | 0.100000 | 0.208838 | 0.142857 | 0.117373 |
| 2018-08-31 | VCB    | 0.438906 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.161942 |
| 2018-08-31 | BID    | 0.214099 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.122746 |
| 2018-08-31 | CTG    | 0.123236 | 0.047303 | 0.000000 | 0.300000 | 0.142857 | 0.135822 |
| 2018-08-31 | MBB    | 0.079746 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.175114 |
| 2018-08-31 | ACB    | 0.075422 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.163707 |
| 2018-08-31 | STB    | 0.045868 | 0.000000 | 0.012860 | 0.000000 | 0.142857 | 0.123873 |
| 2018-08-31 | SHB    | 0.022722 | 0.052697 | 0.087140 | 0.000000 | 0.142857 | 0.116795 |
| 2018-09-30 | VCB    | 0.417540 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.160807 |
| 2018-09-30 | BID    | 0.249364 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.124611 |
| 2018-09-30 | CTG    | 0.126297 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.139271 |
| 2018-09-30 | MBB    | 0.071702 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.173356 |



| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2018-09-30 | ACB    | 0.073380 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.162861 |
| 2018-09-30 | STB    | 0.040486 | 0.000000 | 0.013247 | 0.000000 | 0.142857 | 0.123124 |
| 2018-09-30 | SHB    | 0.021232 | 0.100000 | 0.086753 | 0.000000 | 0.142857 | 0.115971 |
| 2018-10-31 | VCB    | 0.413195 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.167176 |
| 2018-10-31 | BID    | 0.250562 | 0.000000 | 0.000000 | 0.100000 | 0.142857 | 0.113130 |
| 2018-10-31 | CTG    | 0.125792 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.137628 |
| 2018-10-31 | MBB    | 0.068582 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.173066 |
| 2018-10-31 | ACB    | 0.072234 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.170543 |
| 2018-10-31 | STB    | 0.047278 | 0.000000 | 0.007476 | 0.000000 | 0.142857 | 0.122569 |
| 2018-10-31 | SHB    | 0.022356 | 0.100000 | 0.092524 | 0.000000 | 0.142857 | 0.115889 |
| 2018-11-30 | VCB    | 0.420658 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.167815 |
| 2018-11-30 | BID    | 0.239121 | 0.000000 | 0.000000 | 0.100000 | 0.142857 | 0.113567 |
| 2018-11-30 | CTG    | 0.121656 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.138167 |
| 2018-11-30 | MBB    | 0.074092 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.174392 |
| 2018-11-30 | ACB    | 0.073125 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.171472 |
| 2018-11-30 | STB    | 0.049847 | 0.000000 | 0.013261 | 0.000000 | 0.142857 | 0.118590 |
| 2018-11-30 | SHB    | 0.021501 | 0.100000 | 0.086739 | 0.000000 | 0.142857 | 0.115998 |
| 2018-12-31 | VCB    | 0.415930 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.168719 |
| 2018-12-31 | BID    | 0.252662 | 0.000000 | 0.000000 | 0.100000 | 0.142857 | 0.112393 |
| 2018-12-31 | CTG    | 0.118716 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.136075 |
| 2018-12-31 | MBB    | 0.072385 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.178076 |
| 2018-12-31 | ACB    | 0.071391 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.168932 |
| 2018-12-31 | STB    | 0.048089 | 0.000000 | 0.039476 | 0.000000 | 0.142857 | 0.120820 |
| 2018-12-31 | SHB    | 0.020828 | 0.100000 | 0.060524 | 0.000000 | 0.142857 | 0.114985 |
| 2019-01-31 | VCB    | 0.409325 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.170686 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2019-01-31 | BID    | 0.281927 | 0.073694 | 0.000000 | 0.300000 | 0.142857 | 0.113303 |
| 2019-01-31 | CTG    | 0.101434 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.136392 |
| 2019-01-31 | MBB    | 0.066575 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.174570 |
| 2019-01-31 | ACB    | 0.072193 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.167835 |
| 2019-01-31 | STB    | 0.048193 | 0.000000 | 0.077855 | 0.000000 | 0.142857 | 0.124695 |
| 2019-01-31 | SHB    | 0.020352 | 0.026306 | 0.022145 | 0.000000 | 0.142857 | 0.112519 |
| 2019-02-28 | VCB    | 0.428960 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.171268 |
| 2019-02-28 | BID    | 0.254887 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.113703 |
| 2019-02-28 | CTG    | 0.102289 | 0.000000 | 0.000000 | 0.100000 | 0.142857 | 0.131303 |
| 2019-02-28 | MBB    | 0.073970 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.170248 |
| 2019-02-28 | ACB    | 0.070834 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.169107 |
| 2019-02-28 | STB    | 0.048968 | 0.000000 | 0.093891 | 0.300000 | 0.142857 | 0.130938 |
| 2019-02-28 | SHB    | 0.020091 | 0.000000 | 0.006109 | 0.000000 | 0.142857 | 0.113433 |
| 2019-03-31 | VCB    | 0.438656 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.171248 |
| 2019-03-31 | BID    | 0.252572 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.117281 |
| 2019-03-31 | CTG    | 0.103662 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.131273 |
| 2019-03-31 | MBB    | 0.069664 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.165414 |
| 2019-03-31 | ACB    | 0.068640 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.168791 |
| 2019-03-31 | STB    | 0.047019 | 0.000000 | 0.100000 | 0.000000 | 0.142857 | 0.133603 |
| 2019-03-31 | SHB    | 0.019786 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.112391 |
| 2019-04-30 | VCB    | 0.448710 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.170996 |
| 2019-04-30 | BID    | 0.255822 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.117509 |
| 2019-04-30 | CTG    | 0.102834 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.131118 |
| 2019-04-30 | MBB    | 0.066792 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.165002 |
| 2019-04-30 | ACB    | 0.064460 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.168978 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2019-04-30 | STB    | 0.042886 | 0.000000 | 0.100000 | 0.000000 | 0.142857 | 0.133895 |
| 2019-04-30 | SHB    | 0.018498 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.112502 |
| 2019-05-31 | VCB    | 0.457295 | 0.300000 | 0.300000 | 0.283801 | 0.142857 | 0.167831 |
| 2019-05-31 | BID    | 0.251111 | 0.100000 | 0.000000 | 0.000000 | 0.142857 | 0.117203 |
| 2019-05-31 | CTG    | 0.098402 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.131355 |
| 2019-05-31 | MBB    | 0.067460 | 0.300000 | 0.300000 | 0.121459 | 0.142857 | 0.164976 |
| 2019-05-31 | ACB    | 0.064621 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.172002 |
| 2019-05-31 | STB    | 0.042428 | 0.000000 | 0.100000 | 0.300000 | 0.142857 | 0.134375 |
| 2019-05-31 | SHB    | 0.018683 | 0.000000 | 0.000000 | 0.294740 | 0.142857 | 0.112260 |
| 2019-06-30 | VCB    | 0.471576 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.170680 |
| 2019-06-30 | BID    | 0.236853 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.116605 |
| 2019-06-30 | CTG    | 0.097897 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.130236 |
| 2019-06-30 | MBB    | 0.067179 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.164749 |
| 2019-06-30 | ACB    | 0.064915 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.171922 |
| 2019-06-30 | STB    | 0.043469 | 0.000000 | 0.100000 | 0.000000 | 0.142857 | 0.133870 |
| 2019-06-30 | SHB    | 0.018111 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.111938 |
| 2019-07-31 | VCB    | 0.485559 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.171102 |
| 2019-07-31 | BID    | 0.233966 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.115394 |
| 2019-07-31 | CTG    | 0.092262 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.131000 |
| 2019-07-31 | MBB    | 0.066174 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.164099 |
| 2019-07-31 | ACB    | 0.063519 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.171285 |
| 2019-07-31 | STB    | 0.041214 | 0.000000 | 0.100000 | 0.000000 | 0.142857 | 0.135196 |
| 2019-07-31 | SHB    | 0.017307 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.111924 |
| 2019-08-31 | VCB    | 0.500510 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.170530 |
| 2019-08-31 | BID    | 0.237375 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.115587 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2019-08-31 | CTG    | 0.089039 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.131112 |
| 2019-08-31 | MBB    | 0.063691 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.165396 |
| 2019-08-31 | ACB    | 0.058992 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.170782 |
| 2019-08-31 | STB    | 0.035207 | 0.000000 | 0.100000 | 0.000000 | 0.142857 | 0.134846 |
| 2019-08-31 | SHB    | 0.015186 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.111747 |
| 2019-09-30 | VCB    | 0.483369 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.170916 |
| 2019-09-30 | BID    | 0.258367 | 0.100000 | 0.000000 | 0.000000 | 0.142857 | 0.114564 |
| 2019-09-30 | CTG    | 0.086328 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.131017 |
| 2019-09-30 | MBB    | 0.066037 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.164890 |
| 2019-09-30 | ACB    | 0.057285 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.172767 |
| 2019-09-30 | STB    | 0.034111 | 0.000000 | 0.100000 | 0.000000 | 0.142857 | 0.134191 |
| 2019-09-30 | SHB    | 0.014501 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.111655 |
| 2019-10-31 | VCB    | 0.489136 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.169621 |
| 2019-10-31 | BID    | 0.251482 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.114901 |
| 2019-10-31 | CTG    | 0.086596 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.130658 |
| 2019-10-31 | MBB    | 0.067399 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.164797 |
| 2019-10-31 | ACB    | 0.057760 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.171762 |
| 2019-10-31 | STB    | 0.033133 | 0.000000 | 0.100000 | 0.000000 | 0.142857 | 0.135085 |
| 2019-10-31 | SHB    | 0.014494 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.113176 |
| 2019-11-30 | VCB    | 0.498635 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.169133 |
| 2019-11-30 | BID    | 0.247408 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.114349 |
| 2019-11-30 | CTG    | 0.085986 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.129952 |
| 2019-11-30 | MBB    | 0.064914 | 0.300000 | 0.284686 | 0.000000 | 0.142857 | 0.163545 |
| 2019-11-30 | ACB    | 0.056848 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.170605 |
| 2019-11-30 | STB    | 0.032375 | 0.000000 | 0.115314 | 0.000000 | 0.142857 | 0.136952 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2019-11-30 | SHB    | 0.013835 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.115465 |
| 2019-12-31 | VCB    | 0.494942 | 0.300000 | 0.300000 | 0.300000 | 0.142857 | 0.170360 |
| 2019-12-31 | BID    | 0.260411 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.113886 |
| 2019-12-31 | CTG    | 0.081548 | 0.000000 | 0.000000 | 0.100000 | 0.142857 | 0.129036 |
| 2019-12-31 | MBB    | 0.063759 | 0.300000 | 0.280297 | 0.300000 | 0.142857 | 0.162529 |
| 2019-12-31 | ACB    | 0.055586 | 0.300000 | 0.300000 | 0.000000 | 0.142857 | 0.169617 |
| 2019-12-31 | STB    | 0.030892 | 0.000000 | 0.119703 | 0.000000 | 0.142857 | 0.136794 |
| 2019-12-31 | SHB    | 0.012863 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.117778 |
| 2020-01-31 | VCB    | 0.494655 | 0.300000 | 0.290018 | 0.300000 | 0.142857 | 0.169474 |
| 2020-01-31 | BID    | 0.277999 | 0.100000 | 0.000000 | 0.300000 | 0.142857 | 0.113946 |
| 2020-01-31 | CTG    | 0.078027 | 0.000000 | 0.000000 | 0.300000 | 0.142857 | 0.127837 |
| 2020-01-31 | MBB    | 0.056113 | 0.300000 | 0.269295 | 0.000000 | 0.142857 | 0.163685 |
| 2020-01-31 | ACB    | 0.051371 | 0.300000 | 0.300000 | 0.100000 | 0.142857 | 0.167361 |
| 2020-01-31 | STB    | 0.028797 | 0.000000 | 0.140687 | 0.000000 | 0.142857 | 0.141447 |
| 2020-01-31 | SHB    | 0.013039 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.116250 |
| 2020-02-29 | VCB    | 0.467669 | 0.300000 | 0.276845 | 0.300000 | 0.142857 | 0.167074 |
| 2020-02-29 | BID    | 0.298281 | 0.255886 | 0.000000 | 0.300000 | 0.142857 | 0.112808 |
| 2020-02-29 | CTG    | 0.087657 | 0.000000 | 0.000000 | 0.150484 | 0.142857 | 0.127413 |
| 2020-02-29 | MBB    | 0.053952 | 0.280408 | 0.237207 | 0.015377 | 0.142857 | 0.159390 |
| 2020-02-29 | ACB    | 0.049843 | 0.163706 | 0.300000 | 0.234139 | 0.142857 | 0.164978 |
| 2020-02-29 | STB    | 0.028101 | 0.000000 | 0.185949 | 0.000000 | 0.142857 | 0.153602 |
| 2020-02-29 | SHB    | 0.014497 | 0.000000 | 0.000000 | 0.000000 | 0.142857 | 0.114736 |
| 2020-03-31 | VCB    | 0.448047 | 0.300000 | 0.293476 | 0.300000 | 0.125000 | 0.141392 |
| 2020-03-31 | BID    | 0.271740 | 0.300000 | 0.000000 | 0.300000 | 0.125000 | 0.095978 |
| 2020-03-31 | CTG    | 0.095336 | 0.000000 | 0.000000 | 0.000000 | 0.125000 | 0.112612 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2020-03-31 | MBB    | 0.054536 | 0.300000 | 0.139178 | 0.000000 | 0.125000 | 0.133415 |
| 2020-03-31 | ACB    | 0.056280 | 0.100000 | 0.300000 | 0.297898 | 0.125000 | 0.172055 |
| 2020-03-31 | VIB    | 0.018004 | 0.000000 | 0.061800 | 0.102102 | 0.125000 | 0.120064 |
| 2020-03-31 | STB    | 0.032958 | 0.000000 | 0.205545 | 0.000000 | 0.125000 | 0.130132 |
| 2020-03-31 | SHB    | 0.023100 | 0.000000 | 0.000000 | 0.000000 | 0.125000 | 0.094351 |
| 2020-04-30 | VCB    | 0.465684 | 0.300000 | 0.300000 | 0.300000 | 0.125000 | 0.143503 |
| 2020-04-30 | BID    | 0.254497 | 0.191172 | 0.000000 | 0.300000 | 0.125000 | 0.096175 |
| 2020-04-30 | CTG    | 0.087985 | 0.000000 | 0.000000 | 0.180847 | 0.125000 | 0.115229 |
| 2020-04-30 | MBB    | 0.050055 | 0.192232 | 0.147313 | 0.000000 | 0.125000 | 0.136300 |
| 2020-04-30 | ACB    | 0.054967 | 0.251651 | 0.300000 | 0.031903 | 0.125000 | 0.174011 |
| 2020-04-30 | VIB    | 0.017614 | 0.000000 | 0.062582 | 0.187249 | 0.125000 | 0.122770 |
| 2020-04-30 | STB    | 0.028651 | 0.000000 | 0.190105 | 0.000000 | 0.125000 | 0.129179 |
| 2020-04-30 | SHB    | 0.040545 | 0.064945 | 0.000000 | 0.000000 | 0.125000 | 0.082833 |
| 2020-05-31 | VCB    | 0.448980 | 0.300000 | 0.300000 | 0.143694 | 0.125000 | 0.148615 |
| 2020-05-31 | BID    | 0.258161 | 0.300000 | 0.000000 | 0.000000 | 0.125000 | 0.102299 |
| 2020-05-31 | CTG    | 0.089926 | 0.000000 | 0.000000 | 0.233272 | 0.125000 | 0.116232 |
| 2020-05-31 | MBB    | 0.051820 | 0.000000 | 0.129241 | 0.000000 | 0.125000 | 0.128187 |
| 2020-05-31 | ACB    | 0.055653 | 0.100000 | 0.300000 | 0.137009 | 0.125000 | 0.157648 |
| 2020-05-31 | VIB    | 0.017556 | 0.000000 | 0.003073 | 0.186025 | 0.125000 | 0.126971 |
| 2020-05-31 | STB    | 0.031397 | 0.000000 | 0.000000 | 0.000000 | 0.125000 | 0.120569 |
| 2020-05-31 | SHB    | 0.046506 | 0.300000 | 0.267685 | 0.300000 | 0.125000 | 0.099479 |
| 2020-06-30 | VCB    | 0.481368 | 0.300000 | 0.300000 | 0.094670 | 0.125000 | 0.150942 |
| 2020-06-30 | BID    | 0.248812 | 0.300000 | 0.000000 | 0.000000 | 0.125000 | 0.103320 |
| 2020-06-30 | CTG    | 0.086612 | 0.000000 | 0.000000 | 0.005330 | 0.125000 | 0.115924 |
| 2020-06-30 | MBB    | 0.047778 | 0.000000 | 0.092300 | 0.000000 | 0.125000 | 0.126739 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2020-06-30 | ACB    | 0.053168 | 0.100000 | 0.300000 | 0.300000 | 0.125000 | 0.156363 |
| 2020-06-30 | VIB    | 0.016841 | 0.000000 | 0.056661 | 0.000000 | 0.125000 | 0.128923 |
| 2020-06-30 | STB    | 0.030406 | 0.000000 | 0.000000 | 0.300000 | 0.125000 | 0.116151 |
| 2020-06-30 | SHB    | 0.035016 | 0.300000 | 0.251040 | 0.300000 | 0.125000 | 0.101638 |
| 2020-07-31 | VCB    | 0.475964 | 0.300000 | 0.300000 | 0.000000 | 0.125000 | 0.143114 |
| 2020-07-31 | BID    | 0.246758 | 0.300000 | 0.000000 | 0.000000 | 0.125000 | 0.104278 |
| 2020-07-31 | CTG    | 0.086994 | 0.000000 | 0.000000 | 0.100000 | 0.125000 | 0.116403 |
| 2020-07-31 | MBB    | 0.048426 | 0.000000 | 0.110347 | 0.000000 | 0.125000 | 0.131954 |
| 2020-07-31 | ACB    | 0.055532 | 0.100000 | 0.300000 | 0.300000 | 0.125000 | 0.156527 |
| 2020-07-31 | VIB    | 0.018894 | 0.000000 | 0.023637 | 0.000000 | 0.125000 | 0.130081 |
| 2020-07-31 | STB    | 0.033297 | 0.000000 | 0.000000 | 0.300000 | 0.125000 | 0.116346 |
| 2020-07-31 | SHB    | 0.034135 | 0.300000 | 0.266016 | 0.300000 | 0.125000 | 0.101298 |
| 2020-08-31 | VCB    | 0.470276 | 0.300000 | 0.239297 | 0.000000 | 0.111111 | 0.123321 |
| 2020-08-31 | BID    | 0.247840 | 0.255999 | 0.000000 | 0.000000 | 0.111111 | 0.090577 |
| 2020-08-31 | CTG    | 0.088866 | 0.000000 | 0.000000 | 0.062077 | 0.111111 | 0.101343 |
| 2020-08-31 | MBB    | 0.047681 | 0.000000 | 0.000000 | 0.000000 | 0.111111 | 0.118399 |
| 2020-08-31 | ACB    | 0.056874 | 0.004559 | 0.300000 | 0.296788 | 0.111111 | 0.136027 |
| 2020-08-31 | VIB    | 0.020279 | 0.000000 | 0.000000 | 0.000000 | 0.111111 | 0.112517 |
| 2020-08-31 | STB    | 0.032484 | 0.000000 | 0.000000 | 0.041135 | 0.111111 | 0.101895 |
| 2020-08-31 | SHB    | 0.032185 | 0.300000 | 0.160703 | 0.300000 | 0.111111 | 0.087421 |
| 2020-08-31 | KLB    | 0.003515 | 0.139442 | 0.300000 | 0.300000 | 0.111111 | 0.128499 |
| 2020-09-30 | VCB    | 0.455432 | 0.300000 | 0.246477 | 0.000000 | 0.111111 | 0.122445 |
| 2020-09-30 | BID    | 0.246771 | 0.144144 | 0.000000 | 0.000000 | 0.111111 | 0.090502 |
| 2020-09-30 | CTG    | 0.096041 | 0.000000 | 0.000000 | 0.000000 | 0.111111 | 0.101076 |
| 2020-09-30 | MBB    | 0.048444 | 0.000000 | 0.000000 | 0.000000 | 0.111111 | 0.118361 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2020-09-30 | ACB    | 0.062104 | 0.060901 | 0.300000 | 0.100000 | 0.111111 | 0.135591 |
| 2020-09-30 | VIB    | 0.022110 | 0.000000 | 0.000000 | 0.000000 | 0.111111 | 0.112176 |
| 2020-09-30 | STB    | 0.032111 | 0.000000 | 0.000000 | 0.300000 | 0.111111 | 0.101555 |
| 2020-09-30 | SHB    | 0.033760 | 0.300000 | 0.153523 | 0.300000 | 0.111111 | 0.086662 |
| 2020-09-30 | KLB    | 0.003228 | 0.194955 | 0.300000 | 0.300000 | 0.111111 | 0.131632 |
| 2020-10-31 | VCB    | 0.409789 | 0.300000 | 0.300000 | 0.000000 | 0.100000 | 0.113448 |
| 2020-10-31 | BID    | 0.218408 | 0.121667 | 0.000000 | 0.000000 | 0.100000 | 0.083748 |
| 2020-10-31 | CTG    | 0.088126 | 0.000000 | 0.000000 | 0.000000 | 0.100000 | 0.091362 |
| 2020-10-31 | MBB    | 0.047310 | 0.000000 | 0.000000 | 0.000000 | 0.100000 | 0.107570 |
| 2020-10-31 | ACB    | 0.058557 | 0.000000 | 0.240280 | 0.100000 | 0.100000 | 0.120093 |
| 2020-10-31 | VPB    | 0.081657 | 0.000000 | 0.000000 | 0.000000 | 0.100000 | 0.086028 |
| 2020-10-31 | VIB    | 0.024706 | 0.043184 | 0.000000 | 0.000000 | 0.100000 | 0.100957 |
| 2020-10-31 | STB    | 0.035139 | 0.000000 | 0.000000 | 0.300000 | 0.100000 | 0.094401 |
| 2020-10-31 | SHB    | 0.033224 | 0.300000 | 0.159720 | 0.300000 | 0.100000 | 0.079652 |
| 2020-10-31 | KLB    | 0.003084 | 0.235149 | 0.300000 | 0.300000 | 0.100000 | 0.122740 |
| 2020-11-30 | VCB    | 0.404752 | 0.300000 | 0.300000 | 0.105812 | 0.100000 | 0.114370 |
| 2020-11-30 | BID    | 0.206215 | 0.013720 | 0.000000 | 0.000000 | 0.100000 | 0.084757 |
| 2020-11-30 | CTG    | 0.096313 | 0.000000 | 0.000000 | 0.000000 | 0.100000 | 0.092055 |
| 2020-11-30 | MBB    | 0.049074 | 0.000000 | 0.000000 | 0.000000 | 0.100000 | 0.107561 |
| 2020-11-30 | ACB    | 0.062761 | 0.000000 | 0.240837 | 0.264574 | 0.100000 | 0.121131 |
| 2020-11-30 | VPB    | 0.081175 | 0.000000 | 0.000000 | 0.029614 | 0.100000 | 0.086776 |
| 2020-11-30 | VIB    | 0.029273 | 0.146524 | 0.000000 | 0.000000 | 0.100000 | 0.097078 |
| 2020-11-30 | STB    | 0.033889 | 0.000000 | 0.000000 | 0.000000 | 0.100000 | 0.091718 |
| 2020-11-30 | SHB    | 0.033246 | 0.269958 | 0.159163 | 0.300000 | 0.100000 | 0.079960 |
| 2020-11-30 | KLB    | 0.003303 | 0.269798 | 0.300000 | 0.300000 | 0.100000 | 0.124595 |



| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2020-12-31 | VCB    | 0.400213 | 0.300000 | 0.300000 | 0.038394 | 0.090909 | 0.104254 |
| 2020-12-31 | BID    | 0.197126 | 0.000000 | 0.000000 | 0.300000 | 0.090909 | 0.076899 |
| 2020-12-31 | CTG    | 0.098182 | 0.000000 | 0.000000 | 0.156124 | 0.090909 | 0.082941 |
| 2020-12-31 | MBB    | 0.049490 | 0.000000 | 0.000000 | 0.101321 | 0.090909 | 0.097452 |
| 2020-12-31 | ACB    | 0.062549 | 0.000000 | 0.240434 | 0.000000 | 0.090909 | 0.109236 |
| 2020-12-31 | VPB    | 0.081282 | 0.000000 | 0.000000 | 0.140821 | 0.090909 | 0.079042 |
| 2020-12-31 | VIB    | 0.026080 | 0.236992 | 0.000000 | 0.001454 | 0.090909 | 0.086133 |
| 2020-12-31 | STB    | 0.032939 | 0.000000 | 0.000000 | 0.002530 | 0.090909 | 0.084303 |
| 2020-12-31 | SHB    | 0.032392 | 0.252185 | 0.159566 | 0.000000 | 0.090909 | 0.072532 |
| 2020-12-31 | LPB    | 0.016567 | 0.000000 | 0.000000 | 0.259355 | 0.090909 | 0.094387 |
| 2020-12-31 | KLB    | 0.003180 | 0.210822 | 0.300000 | 0.000000 | 0.090909 | 0.112820 |
| 2021-01-31 | VCB    | 0.385892 | 0.269339 | 0.300000 | 0.000000 | 0.090909 | 0.104911 |
| 2021-01-31 | BID    | 0.209155 | 0.000000 | 0.000000 | 0.000000 | 0.090909 | 0.077250 |
| 2021-01-31 | CTG    | 0.093336 | 0.000000 | 0.000000 | 0.000000 | 0.090909 | 0.085707 |
| 2021-01-31 | MBB    | 0.050850 | 0.000000 | 0.000000 | 0.236123 | 0.090909 | 0.096346 |
| 2021-01-31 | ACB    | 0.058667 | 0.000000 | 0.221427 | 0.000000 | 0.090909 | 0.108619 |
| 2021-01-31 | VPB    | 0.089583 | 0.000000 | 0.000000 | 0.000000 | 0.090909 | 0.077337 |
| 2021-01-31 | VIB    | 0.027827 | 0.224999 | 0.000000 | 0.300000 | 0.090909 | 0.085355 |
| 2021-01-31 | STB    | 0.034524 | 0.000000 | 0.000000 | 0.000000 | 0.090909 | 0.085217 |
| 2021-01-31 | SHB    | 0.029430 | 0.205662 | 0.164210 | 0.000000 | 0.090909 | 0.072922 |
| 2021-01-31 | LPB    | 0.016509 | 0.000000 | 0.014363 | 0.163877 | 0.090909 | 0.094501 |
| 2021-01-31 | KLB    | 0.004227 | 0.300000 | 0.300000 | 0.300000 | 0.090909 | 0.111836 |
| 2021-02-28 | VCB    | 0.391361 | 0.189547 | 0.255318 | 0.000000 | 0.083333 | 0.084670 |
| 2021-02-28 | BID    | 0.186043 | 0.000000 | 0.000000 | 0.000000 | 0.083333 | 0.060766 |
| 2021-02-28 | CTG    | 0.087982 | 0.000000 | 0.000000 | 0.000000 | 0.083333 | 0.068346 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2021-02-28 | MBB    | 0.055356 | 0.000000 | 0.000000 | 0.003297 | 0.083333 | 0.075686 |
| 2021-02-28 | ACB    | 0.059791 | 0.000000 | 0.092282 | 0.086887 | 0.083333 | 0.086925 |
| 2021-02-28 | VPB    | 0.088312 | 0.000000 | 0.000000 | 0.000000 | 0.083333 | 0.060042 |
| 2021-02-28 | VIB    | 0.028990 | 0.300000 | 0.000000 | 0.300000 | 0.083333 | 0.066998 |
| 2021-02-28 | STB    | 0.036864 | 0.000000 | 0.000000 | 0.000000 | 0.083333 | 0.066845 |
| 2021-02-28 | SHB    | 0.027003 | 0.210453 | 0.121874 | 0.000000 | 0.083333 | 0.058239 |
| 2021-02-28 | LPB    | 0.017974 | 0.000000 | 0.000000 | 0.152701 | 0.083333 | 0.074926 |
| 2021-02-28 | BAB    | 0.016306 | 0.000000 | 0.300000 | 0.300000 | 0.083333 | 0.220327 |
| 2021-02-28 | KLB    | 0.004019 | 0.300000 | 0.230526 | 0.157115 | 0.083333 | 0.076230 |
| 2021-03-31 | VCB    | 0.355770 | 0.261641 | 0.300000 | 0.000000 | 0.076923 | 0.079908 |
| 2021-03-31 | BID    | 0.175039 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.056312 |
| 2021-03-31 | CTG    | 0.092474 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.060973 |
| 2021-03-31 | MBB    | 0.056158 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.073280 |
| 2021-03-31 | ACB    | 0.063439 | 0.000000 | 0.048220 | 0.000000 | 0.076923 | 0.079084 |
| 2021-03-31 | VPB    | 0.101383 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.056538 |
| 2021-03-31 | HDB    | 0.028668 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.068571 |
| 2021-03-31 | VIB    | 0.031333 | 0.300000 | 0.000000 | 0.300000 | 0.076923 | 0.063679 |
| 2021-03-31 | STB    | 0.034455 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.063188 |
| 2021-03-31 | SHB    | 0.025309 | 0.135702 | 0.161904 | 0.000000 | 0.076923 | 0.057594 |
| 2021-03-31 | LPB    | 0.017971 | 0.000000 | 0.015017 | 0.100000 | 0.076923 | 0.076171 |
| 2021-03-31 | BAB    | 0.014561 | 0.002657 | 0.300000 | 0.300000 | 0.076923 | 0.197437 |
| 2021-03-31 | KLB    | 0.003442 | 0.300000 | 0.174860 | 0.300000 | 0.076923 | 0.067263 |
| 2021-04-30 | VCB    | 0.329000 | 0.035307 | 0.300000 | 0.000000 | 0.076923 | 0.081472 |
| 2021-04-30 | BID    | 0.165127 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.057198 |
| 2021-04-30 | CTG    | 0.095499 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.061755 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2021-04-30 | MBB    | 0.054785 | 0.000000 | 0.000000 | 0.256608 | 0.076923 | 0.071828 |
| 2021-04-30 | ACB    | 0.061210 | 0.000000 | 0.040075 | 0.000000 | 0.076923 | 0.076345 |
| 2021-04-30 | VPB    | 0.108396 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.053904 |
| 2021-04-30 | HDB    | 0.028801 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.068030 |
| 2021-04-30 | VIB    | 0.036609 | 0.300000 | 0.000000 | 0.300000 | 0.076923 | 0.061689 |
| 2021-04-30 | STB    | 0.038581 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.063291 |
| 2021-04-30 | SHB    | 0.039178 | 0.098117 | 0.165276 | 0.000000 | 0.076923 | 0.057778 |
| 2021-04-30 | LPB    | 0.020321 | 0.000000 | 0.006097 | 0.300000 | 0.076923 | 0.074917 |
| 2021-04-30 | BAB    | 0.018600 | 0.266577 | 0.300000 | 0.035611 | 0.076923 | 0.203911 |
| 2021-04-30 | KLB    | 0.003893 | 0.300000 | 0.188552 | 0.107781 | 0.076923 | 0.067882 |
| 2021-05-31 | VCB    | 0.324410 | 0.004001 | 0.300000 | 0.095120 | 0.076923 | 0.089779 |
| 2021-05-31 | BID    | 0.147356 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.063484 |
| 2021-05-31 | CTG    | 0.090715 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.067636 |
| 2021-05-31 | MBB    | 0.055244 | 0.000000 | 0.000000 | 0.300000 | 0.076923 | 0.078996 |
| 2021-05-31 | ACB    | 0.059559 | 0.000000 | 0.132389 | 0.000000 | 0.076923 | 0.084757 |
| 2021-05-31 | VPB    | 0.132753 | 0.000000 | 0.000000 | 0.271848 | 0.076923 | 0.059045 |
| 2021-05-31 | HDB    | 0.027801 | 0.000000 | 0.000000 | 0.033032 | 0.076923 | 0.074725 |
| 2021-05-31 | VIB    | 0.039326 | 0.300000 | 0.000000 | 0.300000 | 0.076923 | 0.068381 |
| 2021-05-31 | STB    | 0.040187 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.068193 |
| 2021-05-31 | SHB    | 0.039055 | 0.131285 | 0.074042 | 0.000000 | 0.076923 | 0.054371 |
| 2021-05-31 | LPB    | 0.023553 | 0.000000 | 0.000000 | 0.000000 | 0.076923 | 0.079760 |
| 2021-05-31 | BAB    | 0.015663 | 0.300000 | 0.300000 | 0.000000 | 0.076923 | 0.138032 |
| 2021-05-31 | KLB    | 0.004379 | 0.264714 | 0.193569 | 0.000000 | 0.076923 | 0.072840 |
| 2021-06-30 | VCB    | 0.269885 | 0.249672 | 0.292735 | 0.000000 | 0.071429 | 0.085073 |
| 2021-06-30 | BID    | 0.147687 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.058816 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2021-06-30 | CTG    | 0.099616 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.062739 |
| 2021-06-30 | MBB    | 0.058322 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.073093 |
| 2021-06-30 | ACB    | 0.060033 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.077584 |
| 2021-06-30 | VPB    | 0.132490 | 0.000000 | 0.000000 | 0.170277 | 0.071429 | 0.052031 |
| 2021-06-30 | HDB    | 0.029084 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.067594 |
| 2021-06-30 | VIB    | 0.040871 | 0.300000 | 0.000000 | 0.000000 | 0.071429 | 0.062825 |
| 2021-06-30 | TPB    | 0.025708 | 0.000000 | 0.230208 | 0.000000 | 0.071429 | 0.096516 |
| 2021-06-30 | STB    | 0.047953 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.061774 |
| 2021-06-30 | SHB    | 0.040204 | 0.236911 | 0.096537 | 0.300000 | 0.071429 | 0.049623 |
| 2021-06-30 | LPB    | 0.027674 | 0.000000 | 0.000000 | 0.217232 | 0.071429 | 0.068165 |
| 2021-06-30 | BAB    | 0.015926 | 0.071808 | 0.300000 | 0.185210 | 0.071429 | 0.118059 |
| 2021-06-30 | KLB    | 0.004547 | 0.141609 | 0.080520 | 0.127281 | 0.071429 | 0.066107 |
| 2021-07-31 | VCB    | 0.304913 | 0.288372 | 0.300000 | 0.000000 | 0.071429 | 0.090883 |
| 2021-07-31 | BID    | 0.137137 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.062504 |
| 2021-07-31 | CTG    | 0.094596 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.061947 |
| 2021-07-31 | MBB    | 0.063680 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.071966 |
| 2021-07-31 | ACB    | 0.061927 | 0.000000 | 0.011659 | 0.000000 | 0.071429 | 0.078219 |
| 2021-07-31 | VPB    | 0.124049 | 0.000000 | 0.000000 | 0.300000 | 0.071429 | 0.054760 |
| 2021-07-31 | HDB    | 0.029248 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.067663 |
| 2021-07-31 | VIB    | 0.039521 | 0.300000 | 0.000000 | 0.000000 | 0.071429 | 0.068067 |
| 2021-07-31 | TPB    | 0.024145 | 0.004652 | 0.167387 | 0.000000 | 0.071429 | 0.085073 |
| 2021-07-31 | STB    | 0.041546 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.060176 |
| 2021-07-31 | SHB    | 0.036308 | 0.280543 | 0.122372 | 0.300000 | 0.071429 | 0.052677 |
| 2021-07-31 | LPB    | 0.026484 | 0.000000 | 0.000000 | 0.000000 | 0.071429 | 0.067580 |
| 2021-07-31 | BAB    | 0.012647 | 0.000000 | 0.300000 | 0.234981 | 0.071429 | 0.111472 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2021-07-31 | KLB    | 0.003799 | 0.126433 | 0.098583 | 0.165019 | 0.071429 | 0.067014 |
| 2021-08-31 | VCB    | 0.254886 | 0.300000 | 0.300000 | 0.000000 | 0.066667 | 0.083184 |
| 2021-08-31 | BID    | 0.118439 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.060272 |
| 2021-08-31 | CTG    | 0.079140 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.059753 |
| 2021-08-31 | MBB    | 0.056944 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.068713 |
| 2021-08-31 | TCB    | 0.119425 | 0.000000 | 0.000000 | 0.034881 | 0.066667 | 0.060118 |
| 2021-08-31 | ACB    | 0.062258 | 0.000000 | 0.054315 | 0.000000 | 0.066667 | 0.076169 |
| 2021-08-31 | VPB    | 0.110969 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.051924 |
| 2021-08-31 | HDB    | 0.027559 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.064352 |
| 2021-08-31 | VIB    | 0.033019 | 0.300000 | 0.000000 | 0.078364 | 0.066667 | 0.065097 |
| 2021-08-31 | TPB    | 0.022821 | 0.022450 | 0.137321 | 0.000000 | 0.066667 | 0.080784 |
| 2021-08-31 | STB    | 0.040307 | 0.000000 | 0.000000 | 0.032210 | 0.066667 | 0.056445 |
| 2021-08-31 | SHB    | 0.034945 | 0.300000 | 0.120161 | 0.000000 | 0.066667 | 0.050663 |
| 2021-08-31 | LPB    | 0.025078 | 0.000000 | 0.000000 | 0.300000 | 0.066667 | 0.065151 |
| 2021-08-31 | BAB    | 0.011077 | 0.000000 | 0.300000 | 0.254546 | 0.066667 | 0.095168 |
| 2021-08-31 | KLB    | 0.003134 | 0.077550 | 0.088203 | 0.300000 | 0.066667 | 0.062208 |
| 2021-09-30 | VCB    | 0.267803 | 0.089171 | 0.300000 | 0.000000 | 0.066667 | 0.079395 |
| 2021-09-30 | BID    | 0.116290 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.059862 |
| 2021-09-30 | CTG    | 0.075773 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.058820 |
| 2021-09-30 | MBB    | 0.056907 | 0.000000 | 0.000000 | 0.300000 | 0.066667 | 0.068566 |
| 2021-09-30 | TCB    | 0.117428 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.061713 |
| 2021-09-30 | ACB    | 0.057110 | 0.000000 | 0.065548 | 0.033406 | 0.066667 | 0.077441 |
| 2021-09-30 | VPB    | 0.116065 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.052402 |
| 2021-09-30 | HDB    | 0.027482 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065155 |
| 2021-09-30 | VIB    | 0.029790 | 0.300000 | 0.000000 | 0.300000 | 0.066667 | 0.062940 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2021-09-30 | TPB    | 0.023033 | 0.175842 | 0.167326 | 0.000000 | 0.066667 | 0.082789 |
| 2021-09-30 | STB    | 0.038191 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.057308 |
| 2021-09-30 | SHB    | 0.035807 | 0.300000 | 0.132744 | 0.066594 | 0.066667 | 0.051402 |
| 2021-09-30 | LPB    | 0.023477 | 0.134987 | 0.000000 | 0.300000 | 0.066667 | 0.067647 |
| 2021-09-30 | BAB    | 0.011567 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.094015 |
| 2021-09-30 | KLB    | 0.003277 | 0.000000 | 0.034382 | 0.000000 | 0.066667 | 0.060544 |
| 2021-10-31 | VCB    | 0.261601 | 0.107695 | 0.300000 | 0.000000 | 0.066667 | 0.079705 |
| 2021-10-31 | BID    | 0.117629 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.063715 |
| 2021-10-31 | CTG    | 0.072360 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.059165 |
| 2021-10-31 | MBB    | 0.056760 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.068276 |
| 2021-10-31 | TCB    | 0.119978 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.061707 |
| 2021-10-31 | ACB    | 0.056255 | 0.000000 | 0.053615 | 0.029498 | 0.066667 | 0.075514 |
| 2021-10-31 | VPB    | 0.124052 | 0.000000 | 0.000000 | 0.300000 | 0.066667 | 0.052542 |
| 2021-10-31 | HDB    | 0.026883 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065189 |
| 2021-10-31 | VIB    | 0.029033 | 0.220406 | 0.000000 | 0.000000 | 0.066667 | 0.060967 |
| 2021-10-31 | TPB    | 0.028073 | 0.118432 | 0.183153 | 0.000000 | 0.066667 | 0.082597 |
| 2021-10-31 | STB    | 0.035852 | 0.000000 | 0.000000 | 0.251191 | 0.066667 | 0.056699 |
| 2021-10-31 | SHB    | 0.034598 | 0.300000 | 0.136118 | 0.300000 | 0.066667 | 0.051281 |
| 2021-10-31 | LPB    | 0.022039 | 0.226779 | 0.000000 | 0.119312 | 0.066667 | 0.066871 |
| 2021-10-31 | BAB    | 0.011555 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.093989 |
| 2021-10-31 | KLB    | 0.003333 | 0.026688 | 0.027115 | 0.000000 | 0.066667 | 0.061782 |
| 2021-11-30 | VCB    | 0.255940 | 0.089712 | 0.300000 | 0.000000 | 0.066667 | 0.079781 |
| 2021-11-30 | BID    | 0.120088 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.063926 |
| 2021-11-30 | CTG    | 0.072758 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.059175 |
| 2021-11-30 | MBB    | 0.056347 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.068621 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2021-11-30 | TCB    | 0.121085 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.062363 |
| 2021-11-30 | ACB    | 0.057094 | 0.000000 | 0.058296 | 0.000000 | 0.066667 | 0.075649 |
| 2021-11-30 | VPB    | 0.125100 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.052633 |
| 2021-11-30 | HDB    | 0.025934 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065328 |
| 2021-11-30 | VIB    | 0.029476 | 0.300000 | 0.000000 | 0.000000 | 0.066667 | 0.060954 |
| 2021-11-30 | TPB    | 0.029365 | 0.300000 | 0.178534 | 0.000000 | 0.066667 | 0.078878 |
| 2021-11-30 | STB    | 0.036006 | 0.000000 | 0.000000 | 0.286983 | 0.066667 | 0.057662 |
| 2021-11-30 | SHB    | 0.035265 | 0.300000 | 0.136405 | 0.300000 | 0.066667 | 0.051284 |
| 2021-11-30 | LPB    | 0.020548 | 0.010288 | 0.000000 | 0.300000 | 0.066667 | 0.066839 |
| 2021-11-30 | BAB    | 0.011109 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.094272 |
| 2021-11-30 | KLB    | 0.003884 | 0.000000 | 0.026766 | 0.113017 | 0.066667 | 0.062636 |
| 2021-12-31 | VCB    | 0.250050 | 0.198537 | 0.300000 | 0.000000 | 0.066667 | 0.080619 |
| 2021-12-31 | BID    | 0.124413 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065129 |
| 2021-12-31 | CTG    | 0.075031 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.059885 |
| 2021-12-31 | MBB    | 0.055527 | 0.000000 | 0.000000 | 0.089245 | 0.066667 | 0.067943 |
| 2021-12-31 | TCB    | 0.117718 | 0.000000 | 0.000000 | 0.113233 | 0.066667 | 0.061886 |
| 2021-12-31 | ACB    | 0.055774 | 0.000000 | 0.086323 | 0.300000 | 0.066667 | 0.076545 |
| 2021-12-31 | VPB    | 0.120011 | 0.000000 | 0.000000 | 0.078775 | 0.066667 | 0.054333 |
| 2021-12-31 | HDB    | 0.030305 | 0.000000 | 0.000000 | 0.110549 | 0.066667 | 0.066643 |
| 2021-12-31 | VIB    | 0.032555 | 0.198610 | 0.000000 | 0.000000 | 0.066667 | 0.060892 |
| 2021-12-31 | TPB    | 0.028595 | 0.300000 | 0.151726 | 0.096226 | 0.066667 | 0.077485 |
| 2021-12-31 | STB    | 0.037873 | 0.000000 | 0.000000 | 0.069509 | 0.066667 | 0.057159 |
| 2021-12-31 | SHB    | 0.035498 | 0.300000 | 0.152327 | 0.000000 | 0.066667 | 0.052020 |
| 2021-12-31 | LPB    | 0.021410 | 0.000000 | 0.000000 | 0.142464 | 0.066667 | 0.066682 |
| 2021-12-31 | BAB    | 0.011494 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.092681 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2021-12-31 | KLB    | 0.003745 | 0.002853 | 0.009623 | 0.000000 | 0.066667 | 0.060097 |
| 2022-01-31 | VCB    | 0.252133 | 0.162588 | 0.300000 | 0.000000 | 0.066667 | 0.080777 |
| 2022-01-31 | BID    | 0.128671 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065175 |
| 2022-01-31 | CTG    | 0.076101 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.059908 |
| 2022-01-31 | MBB    | 0.054231 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.068159 |
| 2022-01-31 | TCB    | 0.111282 | 0.000000 | 0.000000 | 0.300000 | 0.066667 | 0.062124 |
| 2022-01-31 | ACB    | 0.056599 | 0.000000 | 0.083347 | 0.300000 | 0.066667 | 0.076816 |
| 2022-01-31 | VPB    | 0.111626 | 0.000000 | 0.000000 | 0.004191 | 0.066667 | 0.054490 |
| 2022-01-31 | HDB    | 0.030047 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065351 |
| 2022-01-31 | VIB    | 0.035747 | 0.237412 | 0.000000 | 0.000000 | 0.066667 | 0.060590 |
| 2022-01-31 | TPB    | 0.034418 | 0.300000 | 0.153425 | 0.300000 | 0.066667 | 0.077905 |
| 2022-01-31 | STB    | 0.040441 | 0.000000 | 0.000000 | 0.046411 | 0.066667 | 0.057183 |
| 2022-01-31 | SHB    | 0.033004 | 0.300000 | 0.152971 | 0.000000 | 0.066667 | 0.052269 |
| 2022-01-31 | LPB    | 0.020893 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.066682 |
| 2022-01-31 | BAB    | 0.010921 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.092193 |
| 2022-01-31 | KLB    | 0.003886 | 0.000000 | 0.010257 | 0.049398 | 0.066667 | 0.060376 |
| 2022-02-28 | VCB    | 0.257855 | 0.131970 | 0.300000 | 0.300000 | 0.066667 | 0.081088 |
| 2022-02-28 | BID    | 0.150601 | 0.000000 | 0.000000 | 0.011320 | 0.066667 | 0.065467 |
| 2022-02-28 | CTG    | 0.075010 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.062216 |
| 2022-02-28 | MBB    | 0.057240 | 0.000000 | 0.000000 | 0.242635 | 0.066667 | 0.069452 |
| 2022-02-28 | TCB    | 0.106439 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.061934 |
| 2022-02-28 | ACB    | 0.051318 | 0.000000 | 0.085852 | 0.300000 | 0.066667 | 0.076871 |
| 2022-02-28 | VPB    | 0.103484 | 0.000000 | 0.000000 | 0.060375 | 0.066667 | 0.054378 |
| 2022-02-28 | HDB    | 0.027254 | 0.000000 | 0.000000 | 0.085670 | 0.066667 | 0.065307 |
| 2022-02-28 | VIB    | 0.032537 | 0.268030 | 0.000000 | 0.000000 | 0.066667 | 0.060929 |



| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2022-02-28 | TPB    | 0.031853 | 0.300000 | 0.145731 | 0.000000 | 0.066667 | 0.074283 |
| 2022-02-28 | STB    | 0.041331 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.057047 |
| 2022-02-28 | SHB    | 0.030138 | 0.300000 | 0.154746 | 0.000000 | 0.066667 | 0.052158 |
| 2022-02-28 | LPB    | 0.021204 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.066540 |
| 2022-02-28 | BAB    | 0.009849 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.091761 |
| 2022-02-28 | KLB    | 0.003886 | 0.000000 | 0.013671 | 0.000000 | 0.066667 | 0.060567 |
| 2022-03-31 | VCB    | 0.255927 | 0.157162 | 0.300000 | 0.037671 | 0.066667 | 0.080258 |
| 2022-03-31 | BID    | 0.143447 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.062453 |
| 2022-03-31 | CTG    | 0.070451 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.062300 |
| 2022-03-31 | MBB    | 0.061096 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.069126 |
| 2022-03-31 | TCB    | 0.106214 | 0.000000 | 0.000000 | 0.173545 | 0.066667 | 0.062168 |
| 2022-03-31 | ACB    | 0.052653 | 0.000000 | 0.085562 | 0.000000 | 0.066667 | 0.077311 |
| 2022-03-31 | VPB    | 0.112170 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.054668 |
| 2022-03-31 | HDB    | 0.026544 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065626 |
| 2022-03-31 | VIB    | 0.033553 | 0.242838 | 0.000000 | 0.000000 | 0.066667 | 0.061373 |
| 2022-03-31 | TPB    | 0.033537 | 0.300000 | 0.147568 | 0.300000 | 0.066667 | 0.074652 |
| 2022-03-31 | STB    | 0.039925 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.056911 |
| 2022-03-31 | SHB    | 0.030327 | 0.300000 | 0.155947 | 0.000000 | 0.066667 | 0.052449 |
| 2022-03-31 | LPB    | 0.020087 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.067268 |
| 2022-03-31 | BAB    | 0.010111 | 0.000000 | 0.300000 | 0.271999 | 0.066667 | 0.092178 |
| 2022-03-31 | KLB    | 0.003959 | 0.000000 | 0.010922 | 0.216784 | 0.066667 | 0.061259 |
| 2022-04-30 | VCB    | 0.253889 | 0.102956 | 0.300000 | 0.300000 | 0.066667 | 0.080272 |
| 2022-04-30 | BID    | 0.145820 | 0.000000 | 0.000000 | 0.300000 | 0.066667 | 0.062008 |
| 2022-04-30 | CTG    | 0.070414 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.061614 |
| 2022-04-30 | MBB    | 0.059731 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.069679 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2022-04-30 | TCB    | 0.106589 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.062246 |
| 2022-04-30 | ACB    | 0.052409 | 0.000000 | 0.084006 | 0.000000 | 0.066667 | 0.077513 |
| 2022-04-30 | VPB    | 0.112128 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.054957 |
| 2022-04-30 | HDB    | 0.026687 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065459 |
| 2022-04-30 | VIB    | 0.034377 | 0.297044 | 0.000000 | 0.000000 | 0.066667 | 0.061496 |
| 2022-04-30 | TPB    | 0.032538 | 0.300000 | 0.152237 | 0.291952 | 0.066667 | 0.075116 |
| 2022-04-30 | STB    | 0.039338 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.056689 |
| 2022-04-30 | SHB    | 0.030590 | 0.300000 | 0.153891 | 0.000000 | 0.066667 | 0.052476 |
| 2022-04-30 | LPB    | 0.019605 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.066204 |
| 2022-04-30 | BAB    | 0.010279 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.092534 |
| 2022-04-30 | KLB    | 0.005605 | 0.000000 | 0.009866 | 0.108048 | 0.066667 | 0.061736 |
| 2022-05-31 | VCB    | 0.272787 | 0.000000 | 0.300000 | 0.300000 | 0.066667 | 0.081617 |
| 2022-05-31 | BID    | 0.136155 | 0.000000 | 0.000000 | 0.258623 | 0.066667 | 0.062757 |
| 2022-05-31 | CTG    | 0.065632 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.062048 |
| 2022-05-31 | MBB    | 0.058903 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.069648 |
| 2022-05-31 | TCB    | 0.103230 | 0.000000 | 0.000000 | 0.041063 | 0.066667 | 0.062693 |
| 2022-05-31 | ACB    | 0.056550 | 0.000000 | 0.091413 | 0.000000 | 0.066667 | 0.077649 |
| 2022-05-31 | VPB    | 0.120598 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.055279 |
| 2022-05-31 | HDB    | 0.025972 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065763 |
| 2022-05-31 | VIB    | 0.034251 | 0.284594 | 0.000000 | 0.000000 | 0.066667 | 0.061753 |
| 2022-05-31 | TPB    | 0.029692 | 0.300000 | 0.126780 | 0.300000 | 0.066667 | 0.074773 |
| 2022-05-31 | STB    | 0.037477 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.056877 |
| 2022-05-31 | SHB    | 0.025415 | 0.297768 | 0.142362 | 0.000000 | 0.066667 | 0.052679 |
| 2022-05-31 | LPB    | 0.017901 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.066118 |
| 2022-05-31 | BAB    | 0.010031 | 0.000000 | 0.300000 | 0.100314 | 0.066667 | 0.093008 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2022-05-31 | KLB    | 0.005406 | 0.117638 | 0.039445 | 0.000000 | 0.066667 | 0.057338 |
| 2022-06-30 | VCB    | 0.286788 | 0.012871 | 0.300000 | 0.297747 | 0.066667 | 0.083319 |
| 2022-06-30 | BID    | 0.139075 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.062499 |
| 2022-06-30 | CTG    | 0.071132 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.061886 |
| 2022-06-30 | MBB    | 0.059160 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.069885 |
| 2022-06-30 | TCB    | 0.093965 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.063264 |
| 2022-06-30 | ACB    | 0.058821 | 0.000000 | 0.187579 | 0.000000 | 0.066667 | 0.079346 |
| 2022-06-30 | VPB    | 0.109821 | 0.039369 | 0.000000 | 0.000000 | 0.066667 | 0.056790 |
| 2022-06-30 | HDB    | 0.028870 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.066442 |
| 2022-06-30 | VIB    | 0.031072 | 0.300000 | 0.000000 | 0.000000 | 0.066667 | 0.062144 |
| 2022-06-30 | TPB    | 0.030826 | 0.213955 | 0.047980 | 0.102253 | 0.066667 | 0.072131 |
| 2022-06-30 | STB    | 0.032601 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.057024 |
| 2022-06-30 | SHB    | 0.024883 | 0.235397 | 0.105450 | 0.000000 | 0.066667 | 0.050496 |
| 2022-06-30 | LPB    | 0.017659 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.064888 |
| 2022-06-30 | BAB    | 0.010489 | 0.000000 | 0.300000 | 0.300000 | 0.066667 | 0.092359 |
| 2022-06-30 | KLB    | 0.004839 | 0.198409 | 0.058991 | 0.300000 | 0.066667 | 0.057527 |
| 2022-07-31 | VCB    | 0.291893 | 0.014193 | 0.300000 | 0.300000 | 0.066667 | 0.083964 |
| 2022-07-31 | BID    | 0.141710 | 0.000000 | 0.000000 | 0.300000 | 0.066667 | 0.063572 |
| 2022-07-31 | CTG    | 0.071585 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.062688 |
| 2022-07-31 | MBB    | 0.055346 | 0.000000 | 0.000000 | 0.172353 | 0.066667 | 0.070183 |
| 2022-07-31 | TCB    | 0.096526 | 0.000000 | 0.000000 | 0.187960 | 0.066667 | 0.062174 |
| 2022-07-31 | ACB    | 0.060027 | 0.181105 | 0.205502 | 0.000000 | 0.066667 | 0.080094 |
| 2022-07-31 | VPB    | 0.110311 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.055793 |
| 2022-07-31 | HDB    | 0.028513 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.067213 |
| 2022-07-31 | VIB    | 0.026748 | 0.300000 | 0.000000 | 0.039687 | 0.066667 | 0.060653 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2022-07-31 | TPB    | 0.027758 | 0.083043 | 0.039997 | 0.000000 | 0.066667 | 0.073175 |
| 2022-07-31 | STB    | 0.033665 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.055198 |
| 2022-07-31 | SHB    | 0.024347 | 0.275985 | 0.107880 | 0.000000 | 0.066667 | 0.050780 |
| 2022-07-31 | LPB    | 0.016475 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.065587 |
| 2022-07-31 | BAB    | 0.010376 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.092936 |
| 2022-07-31 | KLB    | 0.004721 | 0.145674 | 0.046620 | 0.000000 | 0.066667 | 0.055989 |
| 2022-08-31 | VCB    | 0.279114 | 0.000000 | 0.300000 | 0.300000 | 0.066667 | 0.084761 |
| 2022-08-31 | BID    | 0.151057 | 0.000000 | 0.000000 | 0.300000 | 0.066667 | 0.064227 |
| 2022-08-31 | CTG    | 0.071282 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.063374 |
| 2022-08-31 | MBB    | 0.056202 | 0.000000 | 0.000000 | 0.191965 | 0.066667 | 0.069158 |
| 2022-08-31 | TCB    | 0.098500 | 0.000000 | 0.000000 | 0.132088 | 0.066667 | 0.063707 |
| 2022-08-31 | ACB    | 0.058911 | 0.256132 | 0.242274 | 0.000000 | 0.066667 | 0.080639 |
| 2022-08-31 | VPB    | 0.104496 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.056192 |
| 2022-08-31 | HDB    | 0.027471 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.067419 |
| 2022-08-31 | VIB    | 0.031703 | 0.300000 | 0.000000 | 0.000000 | 0.066667 | 0.058583 |
| 2022-08-31 | TPB    | 0.026055 | 0.000000 | 0.000000 | 0.075947 | 0.066667 | 0.071089 |
| 2022-08-31 | STB    | 0.037035 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.055961 |
| 2022-08-31 | SHB    | 0.025319 | 0.292083 | 0.104553 | 0.000000 | 0.066667 | 0.051104 |
| 2022-08-31 | LPB    | 0.018389 | 0.000000 | 0.000000 | 0.000000 | 0.066667 | 0.064823 |
| 2022-08-31 | BAB    | 0.010180 | 0.000000 | 0.300000 | 0.000000 | 0.066667 | 0.092779 |
| 2022-08-31 | KLB    | 0.004285 | 0.151785 | 0.053173 | 0.000000 | 0.066667 | 0.056183 |
| 2022-09-30 | VCB    | 0.291132 | 0.000000 | 0.300000 | 0.300000 | 0.062500 | 0.081789 |
| 2022-09-30 | BID    | 0.150266 | 0.000000 | 0.000000 | 0.300000 | 0.062500 | 0.060475 |
| 2022-09-30 | CTG    | 0.068693 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.059699 |
| 2022-09-30 | MBB    | 0.057938 | 0.000000 | 0.000000 | 0.247901 | 0.062500 | 0.065053 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2022-09-30 | TCB    | 0.093918 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.059790 |
| 2022-09-30 | ACB    | 0.054757 | 0.251442 | 0.201363 | 0.152099 | 0.062500 | 0.075841 |
| 2022-09-30 | VPB    | 0.107403 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.052874 |
| 2022-09-30 | HDB    | 0.028174 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.063798 |
| 2022-09-30 | VIB    | 0.027626 | 0.300000 | 0.000000 | 0.000000 | 0.062500 | 0.054117 |
| 2022-09-30 | TPB    | 0.025435 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.066779 |
| 2022-09-30 | STB    | 0.034491 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.052334 |
| 2022-09-30 | SHB    | 0.025015 | 0.300000 | 0.087826 | 0.000000 | 0.062500 | 0.048127 |
| 2022-09-30 | LPB    | 0.017718 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.060054 |
| 2022-09-30 | BAB    | 0.009386 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.087195 |
| 2022-09-30 | KLB    | 0.004279 | 0.148558 | 0.026055 | 0.000000 | 0.062500 | 0.052814 |
| 2022-09-30 | VBB    | 0.003771 | 0.000000 | 0.084756 | 0.000000 | 0.062500 | 0.059261 |
| 2022-10-31 | VCB    | 0.296212 | 0.000000 | 0.300000 | 0.300000 | 0.062500 | 0.079928 |
| 2022-10-31 | BID    | 0.148705 | 0.000000 | 0.000000 | 0.228056 | 0.062500 | 0.060349 |
| 2022-10-31 | CTG    | 0.065861 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.059513 |
| 2022-10-31 | MBB    | 0.056920 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.064245 |
| 2022-10-31 | TCB    | 0.091516 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.059558 |
| 2022-10-31 | ACB    | 0.057558 | 0.300000 | 0.185046 | 0.300000 | 0.062500 | 0.075723 |
| 2022-10-31 | VPB    | 0.106464 | 0.000000 | 0.000000 | 0.171944 | 0.062500 | 0.052566 |
| 2022-10-31 | HDB    | 0.029476 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.063079 |
| 2022-10-31 | VIB    | 0.028379 | 0.004408 | 0.000000 | 0.000000 | 0.062500 | 0.053652 |
| 2022-10-31 | TPB    | 0.026509 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.066392 |
| 2022-10-31 | STB    | 0.033448 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.052185 |
| 2022-10-31 | SHB    | 0.023287 | 0.300000 | 0.084268 | 0.000000 | 0.062500 | 0.048082 |
| 2022-10-31 | LPB    | 0.016904 | 0.186458 | 0.000000 | 0.000000 | 0.062500 | 0.060279 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2022-10-31 | BAB    | 0.010298 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.086782 |
| 2022-10-31 | KLB    | 0.004798 | 0.209134 | 0.007534 | 0.000000 | 0.062500 | 0.052504 |
| 2022-10-31 | VBB    | 0.003665 | 0.000000 | 0.123152 | 0.000000 | 0.062500 | 0.065164 |
| 2022-11-30 | VCB    | 0.314370 | 0.000000 | 0.300000 | 0.300000 | 0.062500 | 0.079523 |
| 2022-11-30 | BID    | 0.160176 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.060227 |
| 2022-11-30 | CTG    | 0.073815 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.058687 |
| 2022-11-30 | MBB    | 0.053273 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.063591 |
| 2022-11-30 | TCB    | 0.073188 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.059298 |
| 2022-11-30 | ACB    | 0.061396 | 0.300000 | 0.201554 | 0.300000 | 0.062500 | 0.076004 |
| 2022-11-30 | VPB    | 0.102197 | 0.000000 | 0.000000 | 0.300000 | 0.062500 | 0.052691 |
| 2022-11-30 | HDB    | 0.026681 | 0.000000 | 0.000000 | 0.100000 | 0.062500 | 0.063769 |
| 2022-11-30 | VIB    | 0.026740 | 0.091443 | 0.000000 | 0.000000 | 0.062500 | 0.054023 |
| 2022-11-30 | TPB    | 0.023209 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.067017 |
| 2022-11-30 | STB    | 0.027716 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.051952 |
| 2022-11-30 | SHB    | 0.022651 | 0.286587 | 0.060825 | 0.000000 | 0.062500 | 0.047367 |
| 2022-11-30 | LPB    | 0.016262 | 0.021969 | 0.000000 | 0.000000 | 0.062500 | 0.059033 |
| 2022-11-30 | BAB    | 0.009837 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.088404 |
| 2022-11-30 | KLB    | 0.004615 | 0.300000 | 0.033258 | 0.000000 | 0.062500 | 0.053734 |
| 2022-11-30 | VBB    | 0.003873 | 0.000000 | 0.104362 | 0.000000 | 0.062500 | 0.064680 |
| 2022-12-31 | VCB    | 0.319787 | 0.000000 | 0.300000 | 0.300000 | 0.062500 | 0.081312 |
| 2022-12-31 | BID    | 0.175901 | 0.000000 | 0.000000 | 0.068573 | 0.062500 | 0.061219 |
| 2022-12-31 | CTG    | 0.075501 | 0.000000 | 0.000000 | 0.031427 | 0.062500 | 0.059551 |
| 2022-12-31 | MBB    | 0.048360 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.063449 |
| 2022-12-31 | TCB    | 0.071129 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.056292 |
| 2022-12-31 | ACB    | 0.055946 | 0.300000 | 0.203962 | 0.300000 | 0.062500 | 0.077232 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2022-12-31 | VPB    | 0.096948 | 0.000000 | 0.000000 | 0.300000 | 0.062500 | 0.053100 |
| 2022-12-31 | HDB    | 0.023363 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.063110 |
| 2022-12-31 | VIB    | 0.025487 | 0.100000 | 0.000000 | 0.000000 | 0.062500 | 0.054134 |
| 2022-12-31 | TPB    | 0.022758 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.065088 |
| 2022-12-31 | STB    | 0.031687 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.050503 |
| 2022-12-31 | SHB    | 0.021219 | 0.300000 | 0.055329 | 0.000000 | 0.062500 | 0.047798 |
| 2022-12-31 | LPB    | 0.016820 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.059243 |
| 2022-12-31 | BAB    | 0.008824 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.088009 |
| 2022-12-31 | KLB    | 0.002975 | 0.300000 | 0.023589 | 0.000000 | 0.062500 | 0.054068 |
| 2022-12-31 | VBB    | 0.003294 | 0.000000 | 0.117120 | 0.000000 | 0.062500 | 0.065891 |
| 2023-01-31 | VCB    | 0.317999 | 0.000000 | 0.300000 | 0.300000 | 0.062500 | 0.080921 |
| 2023-01-31 | BID    | 0.166343 | 0.000000 | 0.000000 | 0.300000 | 0.062500 | 0.060148 |
| 2023-01-31 | CTG    | 0.076014 | 0.000000 | 0.000000 | 0.300000 | 0.062500 | 0.059978 |
| 2023-01-31 | MBB    | 0.047811 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.064027 |
| 2023-01-31 | TCB    | 0.071492 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.056665 |
| 2023-01-31 | ACB    | 0.055805 | 0.300000 | 0.207299 | 0.100000 | 0.062500 | 0.078125 |
| 2023-01-31 | VPB    | 0.104050 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.053774 |
| 2023-01-31 | HDB    | 0.024116 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.064209 |
| 2023-01-31 | VIB    | 0.024097 | 0.202438 | 0.000000 | 0.000000 | 0.062500 | 0.054722 |
| 2023-01-31 | TPB    | 0.021929 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.065514 |
| 2023-01-31 | STB    | 0.035890 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.049855 |
| 2023-01-31 | SHB    | 0.021019 | 0.300000 | 0.059810 | 0.000000 | 0.062500 | 0.048573 |
| 2023-01-31 | LPB    | 0.019386 | 0.157028 | 0.000000 | 0.000000 | 0.062500 | 0.059282 |
| 2023-01-31 | BAB    | 0.008560 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.088540 |
| 2023-01-31 | KLB    | 0.002337 | 0.040534 | 0.022268 | 0.000000 | 0.062500 | 0.049622 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2023-01-31 | VBB    | 0.003153 | 0.000000 | 0.110623 | 0.000000 | 0.062500 | 0.066046 |
| 2023-02-28 | VCB    | 0.318169 | 0.000000 | 0.300000 | 0.173711 | 0.062500 | 0.081291 |
| 2023-02-28 | BID    | 0.169254 | 0.000000 | 0.000000 | 0.300000 | 0.062500 | 0.060719 |
| 2023-02-28 | CTG    | 0.074083 | 0.000000 | 0.000000 | 0.300000 | 0.062500 | 0.059925 |
| 2023-02-28 | MBB    | 0.047624 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.064278 |
| 2023-02-28 | TCB    | 0.070808 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.056655 |
| 2023-02-28 | ACB    | 0.057804 | 0.300000 | 0.198868 | 0.000000 | 0.062500 | 0.078101 |
| 2023-02-28 | VPB    | 0.098462 | 0.072847 | 0.000000 | 0.000000 | 0.062500 | 0.053712 |
| 2023-02-28 | HDB    | 0.024575 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.064204 |
| 2023-02-28 | VIB    | 0.025994 | 0.027153 | 0.000000 | 0.000000 | 0.062500 | 0.054375 |
| 2023-02-28 | TPB    | 0.022682 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.065435 |
| 2023-02-28 | STB    | 0.037650 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.049557 |
| 2023-02-28 | SHB    | 0.020255 | 0.300000 | 0.059475 | 0.107355 | 0.062500 | 0.048542 |
| 2023-02-28 | LPB    | 0.019065 | 0.300000 | 0.000000 | 0.118934 | 0.062500 | 0.058702 |
| 2023-02-28 | BAB    | 0.007794 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.088277 |
| 2023-02-28 | KLB    | 0.002051 | 0.000000 | 0.021094 | 0.000000 | 0.062500 | 0.049526 |
| 2023-02-28 | VBB    | 0.003730 | 0.000000 | 0.120563 | 0.000000 | 0.062500 | 0.066700 |
| 2023-03-31 | VCB    | 0.339910 | 0.000000 | 0.300000 | 0.300000 | 0.062500 | 0.080403 |
| 2023-03-31 | BID    | 0.173758 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.061039 |
| 2023-03-31 | CTG    | 0.069895 | 0.000000 | 0.000000 | 0.190223 | 0.062500 | 0.061447 |
| 2023-03-31 | MBB    | 0.044104 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.064379 |
| 2023-03-31 | TCB    | 0.067025 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.057270 |
| 2023-03-31 | ACB    | 0.056841 | 0.300000 | 0.216046 | 0.098525 | 0.062500 | 0.076742 |
| 2023-03-31 | VPB    | 0.090619 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.054768 |
| 2023-03-31 | HDB    | 0.023577 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.063989 |



| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2023-03-31 | VIB    | 0.024898 | 0.138354 | 0.000000 | 0.000000 | 0.062500 | 0.053982 |
| 2023-03-31 | TPB    | 0.022202 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.064890 |
| 2023-03-31 | STB    | 0.034647 | 0.000000 | 0.000000 | 0.176741 | 0.062500 | 0.049583 |
| 2023-03-31 | SHB    | 0.018940 | 0.261646 | 0.072202 | 0.000000 | 0.062500 | 0.049560 |
| 2023-03-31 | LPB    | 0.018886 | 0.300000 | 0.000000 | 0.234512 | 0.062500 | 0.059525 |
| 2023-03-31 | BAB    | 0.008244 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.089336 |
| 2023-03-31 | KLB    | 0.001908 | 0.000000 | 0.058678 | 0.000000 | 0.062500 | 0.051691 |
| 2023-03-31 | VBB    | 0.004547 | 0.000000 | 0.053074 | 0.000000 | 0.062500 | 0.061395 |
| 2023-04-30 | VCB    | 0.317308 | 0.006709 | 0.300000 | 0.300000 | 0.062500 | 0.081219 |
| 2023-04-30 | BID    | 0.173867 | 0.000000 | 0.000000 | 0.100000 | 0.062500 | 0.061877 |
| 2023-04-30 | CTG    | 0.071110 | 0.000000 | 0.000000 | 0.300000 | 0.062500 | 0.060733 |
| 2023-04-30 | MBB    | 0.044597 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.063209 |
| 2023-04-30 | TCB    | 0.068495 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.056723 |
| 2023-04-30 | ACB    | 0.055646 | 0.300000 | 0.208118 | 0.300000 | 0.062500 | 0.076416 |
| 2023-04-30 | VPB    | 0.106821 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.055010 |
| 2023-04-30 | HDB    | 0.025420 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.063295 |
| 2023-04-30 | VIB    | 0.024766 | 0.300000 | 0.000000 | 0.000000 | 0.062500 | 0.053433 |
| 2023-04-30 | TPB    | 0.022412 | 0.047374 | 0.000000 | 0.000000 | 0.062500 | 0.064361 |
| 2023-04-30 | STB    | 0.036499 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.049179 |
| 2023-04-30 | SHB    | 0.019859 | 0.045917 | 0.080770 | 0.000000 | 0.062500 | 0.054659 |
| 2023-04-30 | LPB    | 0.019911 | 0.300000 | 0.000000 | 0.000000 | 0.062500 | 0.059102 |
| 2023-04-30 | BAB    | 0.007759 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.089289 |
| 2023-04-30 | KLB    | 0.001787 | 0.000000 | 0.052841 | 0.000000 | 0.062500 | 0.051399 |
| 2023-04-30 | VBB    | 0.003741 | 0.000000 | 0.058272 | 0.000000 | 0.062500 | 0.060097 |
| 2023-05-31 | VCB    | 0.320670 | 0.149606 | 0.300000 | 0.300000 | 0.062500 | 0.084141 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2023-05-31 | BID    | 0.167057 | 0.000000 | 0.000000 | 0.156299 | 0.062500 | 0.064744 |
| 2023-05-31 | CTG    | 0.071719 | 0.000000 | 0.000000 | 0.254238 | 0.062500 | 0.064222 |
| 2023-05-31 | MBB    | 0.045973 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.068858 |
| 2023-05-31 | TCB    | 0.072724 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.057588 |
| 2023-05-31 | ACB    | 0.054968 | 0.300000 | 0.300000 | 0.241506 | 0.062500 | 0.085659 |
| 2023-05-31 | VPB    | 0.103090 | 0.130830 | 0.000000 | 0.000000 | 0.062500 | 0.058118 |
| 2023-05-31 | HDB    | 0.025333 | 0.068560 | 0.030592 | 0.000000 | 0.062500 | 0.071602 |
| 2023-05-31 | VIB    | 0.024780 | 0.051004 | 0.000000 | 0.000000 | 0.062500 | 0.052449 |
| 2023-05-31 | TPB    | 0.024526 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.060195 |
| 2023-05-31 | STB    | 0.035969 | 0.000000 | 0.000000 | 0.000576 | 0.062500 | 0.050783 |
| 2023-05-31 | SHB    | 0.021095 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.049802 |
| 2023-05-31 | LPB    | 0.018473 | 0.300000 | 0.000000 | 0.047380 | 0.062500 | 0.055446 |
| 2023-05-31 | BAB    | 0.007803 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.078827 |
| 2023-05-31 | KLB    | 0.001928 | 0.000000 | 0.064249 | 0.000000 | 0.062500 | 0.045498 |
| 2023-05-31 | VBB    | 0.003893 | 0.000000 | 0.005159 | 0.000000 | 0.062500 | 0.052067 |
| 2023-06-30 | VCB    | 0.325965 | 0.139853 | 0.300000 | 0.300000 | 0.062500 | 0.083745 |
| 2023-06-30 | BID    | 0.165383 | 0.000000 | 0.000000 | 0.106952 | 0.062500 | 0.064621 |
| 2023-06-30 | CTG    | 0.068139 | 0.000000 | 0.000000 | 0.199169 | 0.062500 | 0.064528 |
| 2023-06-30 | MBB    | 0.045404 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.069785 |
| 2023-06-30 | TCB    | 0.072279 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.057516 |
| 2023-06-30 | ACB    | 0.056457 | 0.300000 | 0.300000 | 0.120888 | 0.062500 | 0.086207 |
| 2023-06-30 | VPB    | 0.097861 | 0.071718 | 0.000000 | 0.000000 | 0.062500 | 0.058291 |
| 2023-06-30 | HDB    | 0.025599 | 0.011254 | 0.039531 | 0.000000 | 0.062500 | 0.071888 |
| 2023-06-30 | VIB    | 0.025311 | 0.156098 | 0.000000 | 0.000000 | 0.062500 | 0.051843 |
| 2023-06-30 | TPB    | 0.025216 | 0.104135 | 0.000000 | 0.000000 | 0.062500 | 0.059129 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2023-06-30 | STB    | 0.038546 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.051263 |
| 2023-06-30 | SHB    | 0.021668 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.051712 |
| 2023-06-30 | LPB    | 0.018721 | 0.216942 | 0.000000 | 0.272992 | 0.062500 | 0.055572 |
| 2023-06-30 | BAB    | 0.007864 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.077756 |
| 2023-06-30 | KLB    | 0.001920 | 0.000000 | 0.060469 | 0.000000 | 0.062500 | 0.044745 |
| 2023-06-30 | VBB    | 0.003668 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.051401 |
| 2023-07-31 | VCB    | 0.331378 | 0.011376 | 0.300000 | 0.256949 | 0.062500 | 0.091596 |
| 2023-07-31 | BID    | 0.155696 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.064280 |
| 2023-07-31 | CTG    | 0.068591 | 0.000000 | 0.000000 | 0.048401 | 0.062500 | 0.063893 |
| 2023-07-31 | MBB    | 0.048290 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.068637 |
| 2023-07-31 | TCB    | 0.074593 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.058167 |
| 2023-07-31 | ACB    | 0.056082 | 0.300000 | 0.300000 | 0.146211 | 0.062500 | 0.085829 |
| 2023-07-31 | VPB    | 0.096136 | 0.018919 | 0.000000 | 0.000000 | 0.062500 | 0.057599 |
| 2023-07-31 | HDB    | 0.024708 | 0.000000 | 0.071517 | 0.000000 | 0.062500 | 0.072630 |
| 2023-07-31 | VIB    | 0.026643 | 0.243322 | 0.000000 | 0.000000 | 0.062500 | 0.051051 |
| 2023-07-31 | TPB    | 0.024149 | 0.139942 | 0.000000 | 0.000000 | 0.062500 | 0.059273 |
| 2023-07-31 | STB    | 0.039623 | 0.000000 | 0.000000 | 0.248438 | 0.062500 | 0.050422 |
| 2023-07-31 | SHB    | 0.022023 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.051331 |
| 2023-07-31 | LPB    | 0.018688 | 0.286441 | 0.000000 | 0.300000 | 0.062500 | 0.054836 |
| 2023-07-31 | BAB    | 0.007568 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.076053 |
| 2023-07-31 | KLB    | 0.002190 | 0.000000 | 0.028483 | 0.000000 | 0.062500 | 0.044116 |
| 2023-07-31 | VBB    | 0.003641 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.050287 |
| 2023-08-31 | VCB    | 0.334119 | 0.179641 | 0.300000 | 0.300000 | 0.062500 | 0.092067 |
| 2023-08-31 | BID    | 0.157990 | 0.000000 | 0.000000 | 0.126176 | 0.062500 | 0.064491 |
| 2023-08-31 | CTG    | 0.064987 | 0.000000 | 0.000000 | 0.133906 | 0.062500 | 0.063957 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2023-08-31 | MBB    | 0.048297 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.068062 |
| 2023-08-31 | TCB    | 0.073702 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.058101 |
| 2023-08-31 | ACB    | 0.054397 | 0.300000 | 0.300000 | 0.300000 | 0.062500 | 0.085773 |
| 2023-08-31 | VPB    | 0.099980 | 0.184233 | 0.000000 | 0.000000 | 0.062500 | 0.059106 |
| 2023-08-31 | HDB    | 0.024712 | 0.000000 | 0.079869 | 0.015573 | 0.062500 | 0.072481 |
| 2023-08-31 | VIB    | 0.026301 | 0.180189 | 0.000000 | 0.000000 | 0.062500 | 0.050824 |
| 2023-08-31 | TPB    | 0.023446 | 0.027478 | 0.000000 | 0.000000 | 0.062500 | 0.059171 |
| 2023-08-31 | STB    | 0.035877 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.050248 |
| 2023-08-31 | SHB    | 0.024428 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.051418 |
| 2023-08-31 | LPB    | 0.018793 | 0.128459 | 0.000000 | 0.124346 | 0.062500 | 0.054722 |
| 2023-08-31 | BAB    | 0.007053 | 0.000000 | 0.300000 | 0.000000 | 0.062500 | 0.075896 |
| 2023-08-31 | KLB    | 0.002055 | 0.000000 | 0.020131 | 0.000000 | 0.062500 | 0.043349 |
| 2023-08-31 | VBB    | 0.003862 | 0.000000 | 0.000000 | 0.000000 | 0.062500 | 0.050334 |
| 2023-09-30 | VCB    | 0.324600 | 0.300000 | 0.300000 | 0.246128 | 0.058824 | 0.088216 |
| 2023-09-30 | BID    | 0.156965 | 0.000000 | 0.000000 | 0.180050 | 0.058824 | 0.061490 |
| 2023-09-30 | CTG    | 0.070231 | 0.000000 | 0.000000 | 0.183906 | 0.058824 | 0.061343 |
| 2023-09-30 | MBB    | 0.047316 | 0.000000 | 0.000000 | 0.146603 | 0.058824 | 0.065590 |
| 2023-09-30 | TCB    | 0.074029 | 0.000000 | 0.000000 | 0.050742 | 0.058824 | 0.056028 |
| 2023-09-30 | ACB    | 0.053487 | 0.300000 | 0.300000 | 0.000000 | 0.058824 | 0.082388 |
| 2023-09-30 | VPB    | 0.094491 | 0.197311 | 0.000000 | 0.063640 | 0.058824 | 0.056284 |
| 2023-09-30 | HDB    | 0.023968 | 0.000000 | 0.079231 | 0.000000 | 0.058824 | 0.069504 |
| 2023-09-30 | VIB    | 0.025701 | 0.060356 | 0.000000 | 0.000000 | 0.058824 | 0.048648 |
| 2023-09-30 | TPB    | 0.024475 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.056678 |
| 2023-09-30 | STB    | 0.040412 | 0.000000 | 0.000000 | 0.074431 | 0.058824 | 0.048323 |
| 2023-09-30 | SHB    | 0.024104 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.048697 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2023-09-30 | LPB    | 0.023732 | 0.142333 | 0.000000 | 0.054499 | 0.058824 | 0.052347 |
| 2023-09-30 | BAB    | 0.006994 | 0.000000 | 0.300000 | 0.000000 | 0.058824 | 0.072697 |
| 2023-09-30 | KLB    | 0.002039 | 0.000000 | 0.020769 | 0.000000 | 0.058824 | 0.041538 |
| 2023-09-30 | VBB    | 0.003609 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.047591 |
| 2023-09-30 | BVB    | 0.003848 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.042637 |
| 2023-10-31 | VCB    | 0.329213 | 0.265677 | 0.300000 | 0.176175 | 0.058824 | 0.087415 |
| 2023-10-31 | BID    | 0.153150 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.061257 |
| 2023-10-31 | CTG    | 0.067140 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.062686 |
| 2023-10-31 | MBB    | 0.048864 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.065456 |
| 2023-10-31 | TCB    | 0.074582 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.056430 |
| 2023-10-31 | ACB    | 0.053391 | 0.300000 | 0.300000 | 0.000000 | 0.058824 | 0.088038 |
| 2023-10-31 | VPB    | 0.101025 | 0.078970 | 0.000000 | 0.189753 | 0.058824 | 0.055798 |
| 2023-10-31 | HDB    | 0.025489 | 0.000000 | 0.095045 | 0.057828 | 0.058824 | 0.070834 |
| 2023-10-31 | VIB    | 0.025229 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.048959 |
| 2023-10-31 | TPB    | 0.023018 | 0.055353 | 0.000000 | 0.000000 | 0.058824 | 0.055992 |
| 2023-10-31 | STB    | 0.039178 | 0.000000 | 0.000000 | 0.036175 | 0.058824 | 0.047410 |
| 2023-10-31 | SHB    | 0.022191 | 0.000000 | 0.000000 | 0.300000 | 0.058824 | 0.048758 |
| 2023-10-31 | LPB    | 0.021146 | 0.300000 | 0.000000 | 0.098145 | 0.058824 | 0.049854 |
| 2023-10-31 | BAB    | 0.007119 | 0.000000 | 0.300000 | 0.133611 | 0.058824 | 0.071600 |
| 2023-10-31 | KLB    | 0.001934 | 0.000000 | 0.004955 | 0.008312 | 0.058824 | 0.040849 |
| 2023-10-31 | VBB    | 0.003432 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.046709 |
| 2023-10-31 | BVB    | 0.003898 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.041953 |
| 2023-11-30 | VCB    | 0.347333 | 0.235729 | 0.300000 | 0.300000 | 0.058824 | 0.087045 |
| 2023-11-30 | BID    | 0.147656 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.060807 |
| 2023-11-30 | CTG    | 0.065836 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.061896 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2023-11-30 | MBB    | 0.048057 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.065889 |
| 2023-11-30 | TCB    | 0.065318 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.056412 |
| 2023-11-30 | ACB    | 0.055641 | 0.300000 | 0.300000 | 0.000000 | 0.058824 | 0.087678 |
| 2023-11-30 | VPB    | 0.099044 | 0.300000 | 0.000000 | 0.000000 | 0.058824 | 0.055717 |
| 2023-11-30 | HDB    | 0.026884 | 0.000000 | 0.099727 | 0.000000 | 0.058824 | 0.071626 |
| 2023-11-30 | VIB    | 0.024815 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.050591 |
| 2023-11-30 | TPB    | 0.021663 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.055113 |
| 2023-11-30 | STB    | 0.036708 | 0.000000 | 0.000000 | 0.294513 | 0.058824 | 0.048389 |
| 2023-11-30 | SHB    | 0.021463 | 0.000000 | 0.000000 | 0.105487 | 0.058824 | 0.048371 |
| 2023-11-30 | LPB    | 0.023784 | 0.164271 | 0.000000 | 0.300000 | 0.058824 | 0.049891 |
| 2023-11-30 | BAB    | 0.006964 | 0.000000 | 0.300000 | 0.000000 | 0.058824 | 0.071547 |
| 2023-11-30 | KLB    | 0.001812 | 0.000000 | 0.000273 | 0.000000 | 0.058824 | 0.040678 |
| 2023-11-30 | VBB    | 0.002997 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.046471 |
| 2023-11-30 | BVB    | 0.004025 | 0.000000 | 0.000000 | 0.000000 | 0.058824 | 0.041879 |
| 2023-12-31 | VCB    | 0.327663 | 0.300000 | 0.300000 | 0.227479 | 0.052632 | 0.079399 |
| 2023-12-31 | BID    | 0.155413 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.054902 |
| 2023-12-31 | CTG    | 0.067536 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.056263 |
| 2023-12-31 | MBB    | 0.048370 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.059196 |
| 2023-12-31 | TCB    | 0.067229 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.049295 |
| 2023-12-31 | ACB    | 0.055147 | 0.170580 | 0.300000 | 0.000000 | 0.052632 | 0.080287 |
| 2023-12-31 | VPB    | 0.095964 | 0.288639 | 0.000000 | 0.000000 | 0.052632 | 0.050198 |
| 2023-12-31 | HDB    | 0.027187 | 0.000000 | 0.053398 | 0.000000 | 0.052632 | 0.065236 |
| 2023-12-31 | VIB    | 0.025068 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.046765 |
| 2023-12-31 | TPB    | 0.022544 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.049278 |
| 2023-12-31 | STB    | 0.036142 | 0.000000 | 0.000000 | 0.208437 | 0.052632 | 0.043413 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2023-12-31 | SHB    | 0.022193 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.043701 |
| 2023-12-31 | LPB    | 0.024012 | 0.240781 | 0.000000 | 0.264084 | 0.052632 | 0.045630 |
| 2023-12-31 | NAB    | 0.007090 | 0.000000 | 0.000000 | 0.300000 | 0.052632 | 0.046208 |
| 2023-12-31 | BAB    | 0.006786 | 0.000000 | 0.300000 | 0.000000 | 0.052632 | 0.064472 |
| 2023-12-31 | KLB    | 0.001831 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.036790 |
| 2023-12-31 | VBB    | 0.003195 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.041427 |
| 2023-12-31 | SGB    | 0.002582 | 0.000000 | 0.046602 | 0.000000 | 0.052632 | 0.048951 |
| 2023-12-31 | BVB    | 0.004047 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.038588 |
| 2024-01-31 | VCB    | 0.303513 | 0.226881 | 0.300000 | 0.300000 | 0.052632 | 0.080546 |
| 2024-01-31 | BID    | 0.169631 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.054158 |
| 2024-01-31 | CTG    | 0.067995 | 0.000000 | 0.000000 | 0.037261 | 0.052632 | 0.056899 |
| 2024-01-31 | MBB    | 0.049503 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.059721 |
| 2024-01-31 | TCB    | 0.070785 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.049215 |
| 2024-01-31 | ACB    | 0.058673 | 0.173119 | 0.300000 | 0.212688 | 0.052632 | 0.082126 |
| 2024-01-31 | VPB    | 0.094245 | 0.300000 | 0.000000 | 0.000000 | 0.052632 | 0.050182 |
| 2024-01-31 | HDB    | 0.029957 | 0.000000 | 0.087687 | 0.000000 | 0.052632 | 0.066485 |
| 2024-01-31 | VIB    | 0.025669 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.046137 |
| 2024-01-31 | TPB    | 0.022548 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.049961 |
| 2024-01-31 | STB    | 0.035889 | 0.000000 | 0.000000 | 0.150052 | 0.052632 | 0.043092 |
| 2024-01-31 | SHB    | 0.021683 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.043339 |
| 2024-01-31 | LPB    | 0.024152 | 0.300000 | 0.000000 | 0.300000 | 0.052632 | 0.045079 |
| 2024-01-31 | NAB    | 0.007629 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.045871 |
| 2024-01-31 | BAB    | 0.006630 | 0.000000 | 0.300000 | 0.000000 | 0.052632 | 0.063698 |
| 2024-01-31 | KLB    | 0.001771 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.036455 |
| 2024-01-31 | VBB    | 0.003253 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.040634 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-01-31 | SGB    | 0.002483 | 0.000000 | 0.012313 | 0.000000 | 0.052632 | 0.048329 |
| 2024-01-31 | BVB    | 0.003992 | 0.000000 | 0.000000 | 0.000000 | 0.052632 | 0.038071 |
| 2024-02-29 | VCB    | 0.299357 | 0.049873 | 0.300000 | 0.300000 | 0.045455 | 0.069171 |
| 2024-02-29 | BID    | 0.166889 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.047376 |
| 2024-02-29 | CTG    | 0.070939 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.049094 |
| 2024-02-29 | MBB    | 0.051671 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.052405 |
| 2024-02-29 | TCB    | 0.068823 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.045858 |
| 2024-02-29 | ACB    | 0.056584 | 0.300000 | 0.300000 | 0.273400 | 0.045455 | 0.069684 |
| 2024-02-29 | VPB    | 0.085221 | 0.058465 | 0.000000 | 0.000000 | 0.045455 | 0.044877 |
| 2024-02-29 | HDB    | 0.028781 | 0.259594 | 0.065650 | 0.000000 | 0.045455 | 0.056419 |
| 2024-02-29 | VIB    | 0.025026 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.040588 |
| 2024-02-29 | TPB    | 0.020696 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.044596 |
| 2024-02-29 | STB    | 0.034362 | 0.000000 | 0.000000 | 0.126600 | 0.045455 | 0.037650 |
| 2024-02-29 | SHB    | 0.020844 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.037337 |
| 2024-02-29 | LPB    | 0.024220 | 0.300000 | 0.000000 | 0.300000 | 0.045455 | 0.039004 |
| 2024-02-29 | MSB    | 0.013821 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.052534 |
| 2024-02-29 | NAB    | 0.007003 | 0.032068 | 0.000000 | 0.000000 | 0.045455 | 0.039381 |
| 2024-02-29 | BAB    | 0.006278 | 0.000000 | 0.300000 | 0.000000 | 0.045455 | 0.059088 |
| 2024-02-29 | KLB    | 0.001657 | 0.000000 | 0.034350 | 0.000000 | 0.045455 | 0.034753 |
| 2024-02-29 | PGB    | 0.004209 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.026372 |
| 2024-02-29 | VBB    | 0.003016 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.035009 |
| 2024-02-29 | SGB    | 0.002328 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.041795 |
| 2024-02-29 | ABB    | 0.004669 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.043322 |
| 2024-02-29 | BVB    | 0.003607 | 0.000000 | 0.000000 | 0.000000 | 0.045455 | 0.033689 |
| 2024-03-31 | VCB    | 0.295866 | 0.229590 | 0.300000 | 0.300000 | 0.043478 | 0.064554 |



| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-03-31 | BID    | 0.166670 | 0.000000 | 0.000000 | 0.045852 | 0.043478 | 0.046818 |
| 2024-03-31 | CTG    | 0.071739 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.046158 |
| 2024-03-31 | MBB    | 0.051561 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.047958 |
| 2024-03-31 | TCB    | 0.075567 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.042984 |
| 2024-03-31 | ACB    | 0.055308 | 0.300000 | 0.300000 | 0.300000 | 0.043478 | 0.065892 |
| 2024-03-31 | VPB    | 0.078559 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.042909 |
| 2024-03-31 | HDB    | 0.027659 | 0.081629 | 0.039021 | 0.054148 | 0.043478 | 0.053644 |
| 2024-03-31 | VIB    | 0.023736 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.038206 |
| 2024-03-31 | TPB    | 0.020687 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.042489 |
| 2024-03-31 | STB    | 0.032592 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.035507 |
| 2024-03-31 | SHB    | 0.019070 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.035960 |
| 2024-03-31 | LPB    | 0.021100 | 0.255029 | 0.000000 | 0.300000 | 0.043478 | 0.036594 |
| 2024-03-31 | MSB    | 0.014518 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.050638 |
| 2024-03-31 | OCB    | 0.014984 | 0.000000 | 0.022387 | 0.000000 | 0.043478 | 0.047706 |
| 2024-03-31 | NAB    | 0.006618 | 0.042630 | 0.000000 | 0.000000 | 0.043478 | 0.038319 |
| 2024-03-31 | BAB    | 0.005691 | 0.000000 | 0.300000 | 0.000000 | 0.043478 | 0.055333 |
| 2024-03-31 | KLB    | 0.001501 | 0.000000 | 0.033466 | 0.000000 | 0.043478 | 0.033064 |
| 2024-03-31 | PGB    | 0.004091 | 0.091121 | 0.000000 | 0.000000 | 0.043478 | 0.026980 |
| 2024-03-31 | VBB    | 0.002738 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.033102 |
| 2024-03-31 | SGB    | 0.002107 | 0.000000 | 0.005125 | 0.000000 | 0.043478 | 0.040660 |
| 2024-03-31 | ABB    | 0.004305 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.041621 |
| 2024-03-31 | BVB    | 0.003334 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.032905 |
| 2024-04-30 | VCB    | 0.287435 | 0.300000 | 0.300000 | 0.300000 | 0.043478 | 0.062450 |
| 2024-04-30 | BID    | 0.163195 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.045776 |
| 2024-04-30 | CTG    | 0.071464 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.047042 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-04-30 | MBB    | 0.054028 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.048199 |
| 2024-04-30 | TCB    | 0.084841 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.042283 |
| 2024-04-30 | ACB    | 0.056078 | 0.300000 | 0.300000 | 0.300000 | 0.043478 | 0.072331 |
| 2024-04-30 | VPB    | 0.077694 | 0.000000 | 0.089093 | 0.000000 | 0.043478 | 0.046905 |
| 2024-04-30 | HDB    | 0.028491 | 0.083667 | 0.006275 | 0.000000 | 0.043478 | 0.053561 |
| 2024-04-30 | VIB    | 0.026572 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.039355 |
| 2024-04-30 | TPB    | 0.019829 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.041649 |
| 2024-04-30 | STB    | 0.032518 | 0.000000 | 0.000000 | 0.100000 | 0.043478 | 0.034951 |
| 2024-04-30 | SHB    | 0.018354 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.035484 |
| 2024-04-30 | LPB    | 0.021573 | 0.120295 | 0.000000 | 0.300000 | 0.043478 | 0.036367 |
| 2024-04-30 | MSB    | 0.013241 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.048634 |
| 2024-04-30 | OCB    | 0.014725 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.047056 |
| 2024-04-30 | NAB    | 0.006653 | 0.098804 | 0.000000 | 0.000000 | 0.043478 | 0.037807 |
| 2024-04-30 | BAB    | 0.005669 | 0.000000 | 0.300000 | 0.000000 | 0.043478 | 0.054532 |
| 2024-04-30 | KLB    | 0.001483 | 0.000000 | 0.004632 | 0.000000 | 0.043478 | 0.032455 |
| 2024-04-30 | PGB    | 0.003821 | 0.097234 | 0.000000 | 0.000000 | 0.043478 | 0.026613 |
| 2024-04-30 | VBB    | 0.002623 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.033159 |
| 2024-04-30 | SGB    | 0.002241 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.040062 |
| 2024-04-30 | ABB    | 0.004181 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.040981 |
| 2024-04-30 | BVB    | 0.003291 | 0.000000 | 0.000000 | 0.000000 | 0.043478 | 0.032345 |
| 2024-05-31 | VCB    | 0.281444 | 0.300000 | 0.300000 | 0.300000 | 0.041667 | 0.057934 |
| 2024-05-31 | BID    | 0.157029 | 0.015931 | 0.000000 | 0.300000 | 0.041667 | 0.042356 |
| 2024-05-31 | CTG    | 0.067202 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.043843 |
| 2024-05-31 | MBB    | 0.048229 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.044511 |
| 2024-05-31 | TCB    | 0.084718 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.038495 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-05-31 | ACB    | 0.053825 | 0.300000 | 0.300000 | 0.300000 | 0.041667 | 0.066942 |
| 2024-05-31 | VPB    | 0.074355 | 0.000000 | 0.016635 | 0.000000 | 0.041667 | 0.044129 |
| 2024-05-31 | HDB    | 0.028445 | 0.230827 | 0.000000 | 0.000000 | 0.041667 | 0.049642 |
| 2024-05-31 | VIB    | 0.023910 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.037466 |
| 2024-05-31 | TPB    | 0.018941 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.038428 |
| 2024-05-31 | STB    | 0.029569 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.032945 |
| 2024-05-31 | SHB    | 0.018769 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.042911 |
| 2024-05-31 | LPB    | 0.026169 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.034371 |
| 2024-05-31 | SSB    | 0.030650 | 0.002775 | 0.083365 | 0.000000 | 0.041667 | 0.043047 |
| 2024-05-31 | MSB    | 0.012610 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.046011 |
| 2024-05-31 | OCB    | 0.013701 | 0.000000 | 0.000000 | 0.100000 | 0.041667 | 0.044917 |
| 2024-05-31 | NAB    | 0.006840 | 0.149295 | 0.000000 | 0.000000 | 0.041667 | 0.035045 |
| 2024-05-31 | BAB    | 0.005637 | 0.000000 | 0.300000 | 0.000000 | 0.041667 | 0.064605 |
| 2024-05-31 | KLB    | 0.001422 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.030762 |
| 2024-05-31 | PGB    | 0.003651 | 0.001171 | 0.000000 | 0.000000 | 0.041667 | 0.024782 |
| 2024-05-31 | VBB    | 0.002966 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.030657 |
| 2024-05-31 | SGB    | 0.002478 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.037204 |
| 2024-05-31 | ABB    | 0.004053 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.038828 |
| 2024-05-31 | BVB    | 0.003384 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.030169 |
| 2024-06-30 | VCB    | 0.270118 | 0.198848 | 0.300000 | 0.300000 | 0.041667 | 0.057436 |
| 2024-06-30 | BID    | 0.150904 | 0.163613 | 0.000000 | 0.300000 | 0.041667 | 0.041983 |
| 2024-06-30 | CTG    | 0.065590 | 0.000000 | 0.000000 | 0.039031 | 0.041667 | 0.043041 |
| 2024-06-30 | MBB    | 0.048372 | 0.000000 | 0.000000 | 0.022493 | 0.041667 | 0.043019 |
| 2024-06-30 | TCB    | 0.088421 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.038217 |
| 2024-06-30 | ACB    | 0.059070 | 0.106564 | 0.300000 | 0.300000 | 0.041667 | 0.065441 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-06-30 | VPB    | 0.076268 | 0.000000 | 0.005910 | 0.000000 | 0.041667 | 0.049741 |
| 2024-06-30 | HDB    | 0.028119 | 0.294089 | 0.000000 | 0.000000 | 0.041667 | 0.049219 |
| 2024-06-30 | VIB    | 0.024730 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.037319 |
| 2024-06-30 | TPB    | 0.018745 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.037945 |
| 2024-06-30 | STB    | 0.029309 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.032587 |
| 2024-06-30 | SHB    | 0.018770 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.042953 |
| 2024-06-30 | LPB    | 0.032435 | 0.064773 | 0.000000 | 0.038476 | 0.041667 | 0.034646 |
| 2024-06-30 | SSB    | 0.030064 | 0.114681 | 0.094090 | 0.000000 | 0.041667 | 0.042968 |
| 2024-06-30 | MSB    | 0.013309 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.045317 |
| 2024-06-30 | OCB    | 0.014852 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.044031 |
| 2024-06-30 | NAB    | 0.006497 | 0.041063 | 0.000000 | 0.000000 | 0.041667 | 0.035241 |
| 2024-06-30 | BAB    | 0.005610 | 0.000000 | 0.300000 | 0.000000 | 0.041667 | 0.065994 |
| 2024-06-30 | KLB    | 0.001517 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.031523 |
| 2024-06-30 | PGB    | 0.003564 | 0.016369 | 0.000000 | 0.000000 | 0.041667 | 0.024645 |
| 2024-06-30 | VBB    | 0.002977 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.030228 |
| 2024-06-30 | SGB    | 0.002326 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.036692 |
| 2024-06-30 | ABB    | 0.004541 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.039839 |
| 2024-06-30 | BVB    | 0.003888 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.029971 |
| 2024-07-31 | VCB    | 0.268964 | 0.300000 | 0.096232 | 0.300000 | 0.041667 | 0.049314 |
| 2024-07-31 | BID    | 0.141208 | 0.000000 | 0.000000 | 0.158870 | 0.041667 | 0.037588 |
| 2024-07-31 | CTG    | 0.064951 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.042018 |
| 2024-07-31 | MBB    | 0.050305 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.040331 |
| 2024-07-31 | TCB    | 0.089551 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.036417 |
| 2024-07-31 | ACB    | 0.058096 | 0.300000 | 0.241881 | 0.300000 | 0.041667 | 0.061764 |
| 2024-07-31 | VPB    | 0.080525 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.045919 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-07-31 | HDB    | 0.028357 | 0.100000 | 0.000000 | 0.000000 | 0.041667 | 0.046705 |
| 2024-07-31 | VIB    | 0.024338 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.034075 |
| 2024-07-31 | TPB    | 0.019134 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.037459 |
| 2024-07-31 | STB    | 0.030889 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.031857 |
| 2024-07-31 | SHB    | 0.019129 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.039976 |
| 2024-07-31 | LPB    | 0.036115 | 0.300000 | 0.000000 | 0.241130 | 0.041667 | 0.032040 |
| 2024-07-31 | SSB    | 0.029636 | 0.000000 | 0.264903 | 0.000000 | 0.041667 | 0.054747 |
| 2024-07-31 | MSB    | 0.013564 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.043420 |
| 2024-07-31 | OCB    | 0.014477 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.040557 |
| 2024-07-31 | NAB    | 0.006645 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.037511 |
| 2024-07-31 | BAB    | 0.005717 | 0.000000 | 0.300000 | 0.000000 | 0.041667 | 0.068722 |
| 2024-07-31 | KLB    | 0.001559 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.028357 |
| 2024-07-31 | PGB    | 0.003590 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.023825 |
| 2024-07-31 | VBB    | 0.002790 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.033718 |
| 2024-07-31 | SGB    | 0.002388 | 0.000000 | 0.096984 | 0.000000 | 0.041667 | 0.050050 |
| 2024-07-31 | ABB    | 0.004358 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.043624 |
| 2024-07-31 | BVB    | 0.003712 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.040005 |
| 2024-08-31 | VCB    | 0.268548 | 0.000000 | 0.065913 | 0.213667 | 0.041667 | 0.051852 |
| 2024-08-31 | BID    | 0.148183 | 0.155839 | 0.000000 | 0.000000 | 0.041667 | 0.036259 |
| 2024-08-31 | CTG    | 0.063949 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.040956 |
| 2024-08-31 | MBB    | 0.052739 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.040640 |
| 2024-08-31 | TCB    | 0.085014 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.035554 |
| 2024-08-31 | ACB    | 0.057157 | 0.297073 | 0.238980 | 0.300000 | 0.041667 | 0.061074 |
| 2024-08-31 | VPB    | 0.078211 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.044654 |
| 2024-08-31 | HDB    | 0.032184 | 0.016297 | 0.000000 | 0.000000 | 0.041667 | 0.045761 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-08-31 | VIB    | 0.023436 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.033231 |
| 2024-08-31 | TPB    | 0.019250 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.036577 |
| 2024-08-31 | STB    | 0.029661 | 0.000000 | 0.000000 | 0.053596 | 0.041667 | 0.031377 |
| 2024-08-31 | SHB    | 0.018617 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.039155 |
| 2024-08-31 | LPB    | 0.035607 | 0.300000 | 0.000000 | 0.300000 | 0.041667 | 0.031073 |
| 2024-08-31 | SSB    | 0.029620 | 0.000000 | 0.265745 | 0.000000 | 0.041667 | 0.053274 |
| 2024-08-31 | MSB    | 0.013245 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.043320 |
| 2024-08-31 | OCB    | 0.014194 | 0.000000 | 0.000000 | 0.132737 | 0.041667 | 0.039763 |
| 2024-08-31 | NAB    | 0.007521 | 0.230790 | 0.000000 | 0.000000 | 0.041667 | 0.037969 |
| 2024-08-31 | BAB    | 0.005364 | 0.000000 | 0.300000 | 0.000000 | 0.041667 | 0.080372 |
| 2024-08-31 | KLB    | 0.001474 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.028000 |
| 2024-08-31 | PGB    | 0.003250 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.023295 |
| 2024-08-31 | VBB    | 0.002661 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.032796 |
| 2024-08-31 | SGB    | 0.002261 | 0.000000 | 0.129363 | 0.000000 | 0.041667 | 0.050915 |
| 2024-08-31 | ABB    | 0.004105 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.042718 |
| 2024-08-31 | BVB    | 0.003749 | 0.000000 | 0.000000 | 0.000000 | 0.041667 | 0.039414 |
| 2024-09-30 | VCB    | 0.268808 | 0.247946 | 0.164723 | 0.215077 | 0.040000 | 0.054208 |
| 2024-09-30 | BID    | 0.148251 | 0.088752 | 0.000000 | 0.000000 | 0.040000 | 0.034863 |
| 2024-09-30 | CTG    | 0.068195 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040777 |
| 2024-09-30 | MBB    | 0.052286 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038354 |
| 2024-09-30 | TCB    | 0.083150 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033390 |
| 2024-09-30 | ACB    | 0.056223 | 0.000000 | 0.133702 | 0.184923 | 0.040000 | 0.057125 |
| 2024-09-30 | VPB    | 0.075943 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042885 |
| 2024-09-30 | HDB    | 0.033006 | 0.063970 | 0.000000 | 0.000000 | 0.040000 | 0.040581 |
| 2024-09-30 | VIB    | 0.023102 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032402 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-09-30 | TPB    | 0.018478 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034301 |
| 2024-09-30 | STB    | 0.030424 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029445 |
| 2024-09-30 | SHB    | 0.017161 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036690 |
| 2024-09-30 | LPB    | 0.037352 | 0.299332 | 0.000000 | 0.300000 | 0.040000 | 0.029274 |
| 2024-09-30 | SSB    | 0.028855 | 0.000000 | 0.242411 | 0.000000 | 0.040000 | 0.049702 |
| 2024-09-30 | MSB    | 0.013281 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040569 |
| 2024-09-30 | OCB    | 0.013612 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037255 |
| 2024-09-30 | NAB    | 0.007976 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.035173 |
| 2024-09-30 | BAB    | 0.005182 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.083904 |
| 2024-09-30 | KLB    | 0.001567 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.027002 |
| 2024-09-30 | PGB    | 0.003106 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.021920 |
| 2024-09-30 | VAB    | 0.001697 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.038938 |
| 2024-09-30 | VBB    | 0.002516 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.031223 |
| 2024-09-30 | SGB    | 0.002185 | 0.000000 | 0.159164 | 0.000000 | 0.040000 | 0.050229 |
| 2024-09-30 | ABB    | 0.003948 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040968 |
| 2024-09-30 | BVB    | 0.003696 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038820 |
| 2024-10-31 | VCB    | 0.262621 | 0.082972 | 0.147731 | 0.300000 | 0.040000 | 0.053775 |
| 2024-10-31 | BID    | 0.146713 | 0.181170 | 0.000000 | 0.174852 | 0.040000 | 0.034893 |
| 2024-10-31 | CTG    | 0.069998 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040439 |
| 2024-10-31 | MBB    | 0.052642 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038222 |
| 2024-10-31 | TCB    | 0.083914 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033296 |
| 2024-10-31 | ACB    | 0.056836 | 0.000000 | 0.159565 | 0.150890 | 0.040000 | 0.061457 |
| 2024-10-31 | VPB    | 0.078442 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042611 |
| 2024-10-31 | HDB    | 0.032718 | 0.179389 | 0.000000 | 0.000000 | 0.040000 | 0.040411 |
| 2024-10-31 | VIB    | 0.023662 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033099 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-10-31 | TPB    | 0.020753 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034100 |
| 2024-10-31 | STB    | 0.032342 | 0.000000 | 0.000000 | 0.095848 | 0.040000 | 0.029441 |
| 2024-10-31 | SHB    | 0.017422 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036311 |
| 2024-10-31 | LPB    | 0.037003 | 0.300000 | 0.000000 | 0.278410 | 0.040000 | 0.029482 |
| 2024-10-31 | SSB    | 0.024952 | 0.000000 | 0.238152 | 0.000000 | 0.040000 | 0.049408 |
| 2024-10-31 | MSB    | 0.014538 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040359 |
| 2024-10-31 | OCB    | 0.013483 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037212 |
| 2024-10-31 | NAB    | 0.007858 | 0.256468 | 0.000000 | 0.000000 | 0.040000 | 0.034659 |
| 2024-10-31 | BAB    | 0.005130 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.083664 |
| 2024-10-31 | KLB    | 0.001586 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.026570 |
| 2024-10-31 | PGB    | 0.003284 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.021842 |
| 2024-10-31 | VAB    | 0.001783 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038615 |
| 2024-10-31 | VBB    | 0.002545 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.031019 |
| 2024-10-31 | SGB    | 0.002144 | 0.000000 | 0.154552 | 0.000000 | 0.040000 | 0.049889 |
| 2024-10-31 | ABB    | 0.003940 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040696 |
| 2024-10-31 | BVB    | 0.003692 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038529 |
| 2024-11-30 | VCB    | 0.269394 | 0.176749 | 0.145661 | 0.300000 | 0.040000 | 0.054110 |
| 2024-11-30 | BID    | 0.142140 | 0.015066 | 0.000000 | 0.072775 | 0.040000 | 0.034970 |
| 2024-11-30 | CTG    | 0.068212 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040600 |
| 2024-11-30 | MBB    | 0.051749 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038260 |
| 2024-11-30 | TCB    | 0.083547 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033329 |
| 2024-11-30 | ACB    | 0.056536 | 0.172242 | 0.169369 | 0.162632 | 0.040000 | 0.061526 |
| 2024-11-30 | VPB    | 0.080492 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042826 |
| 2024-11-30 | HDB    | 0.031460 | 0.193681 | 0.000000 | 0.000000 | 0.040000 | 0.040618 |
| 2024-11-30 | VIB    | 0.023425 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033100 |



| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-11-30 | TPB    | 0.021048 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035267 |
| 2024-11-30 | STB    | 0.034419 | 0.000000 | 0.000000 | 0.035397 | 0.040000 | 0.029425 |
| 2024-11-30 | SHB    | 0.016934 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036250 |
| 2024-11-30 | LPB    | 0.038472 | 0.300000 | 0.000000 | 0.300000 | 0.040000 | 0.029810 |
| 2024-11-30 | SSB    | 0.024422 | 0.000000 | 0.227986 | 0.129196 | 0.040000 | 0.047375 |
| 2024-11-30 | MSB    | 0.013877 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039330 |
| 2024-11-30 | OCB    | 0.012866 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037492 |
| 2024-11-30 | NAB    | 0.007596 | 0.142262 | 0.000000 | 0.000000 | 0.040000 | 0.034797 |
| 2024-11-30 | BAB    | 0.005173 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.083292 |
| 2024-11-30 | KLB    | 0.001432 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.026586 |
| 2024-11-30 | PGB    | 0.003032 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.021842 |
| 2024-11-30 | VAB    | 0.001722 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038098 |
| 2024-11-30 | VBB    | 0.002518 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.030991 |
| 2024-11-30 | SGB    | 0.002096 | 0.000000 | 0.156984 | 0.000000 | 0.040000 | 0.050056 |
| 2024-11-30 | ABB    | 0.003779 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040896 |
| 2024-11-30 | BVB    | 0.003658 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039153 |
| 2024-12-31 | VCB    | 0.274011 | 0.300000 | 0.134575 | 0.300000 | 0.040000 | 0.053946 |
| 2024-12-31 | BID    | 0.141995 | 0.000000 | 0.000000 | 0.066425 | 0.040000 | 0.034769 |
| 2024-12-31 | CTG    | 0.069690 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040388 |
| 2024-12-31 | MBB    | 0.050901 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038087 |
| 2024-12-31 | TCB    | 0.084207 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033308 |
| 2024-12-31 | ACB    | 0.057228 | 0.103517 | 0.173917 | 0.119741 | 0.040000 | 0.061580 |
| 2024-12-31 | VPB    | 0.076929 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042845 |
| 2024-12-31 | HDB    | 0.030264 | 0.278888 | 0.000000 | 0.000000 | 0.040000 | 0.040186 |
| 2024-12-31 | VIB    | 0.023967 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033069 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2024-12-31 | TPB    | 0.020108 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035503 |
| 2024-12-31 | STB    | 0.033226 | 0.000000 | 0.000000 | 0.081165 | 0.040000 | 0.029291 |
| 2024-12-31 | SHB    | 0.016785 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036333 |
| 2024-12-31 | LPB    | 0.038436 | 0.300000 | 0.000000 | 0.300000 | 0.040000 | 0.029881 |
| 2024-12-31 | SSB    | 0.025598 | 0.000000 | 0.241182 | 0.132669 | 0.040000 | 0.048572 |
| 2024-12-31 | MSB    | 0.013183 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039008 |
| 2024-12-31 | OCB    | 0.012320 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038134 |
| 2024-12-31 | NAB    | 0.007628 | 0.017595 | 0.000000 | 0.000000 | 0.040000 | 0.034583 |
| 2024-12-31 | BAB    | 0.005101 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.082941 |
| 2024-12-31 | KLB    | 0.001450 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.027323 |
| 2024-12-31 | PGB    | 0.003035 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.021680 |
| 2024-12-31 | VAB    | 0.001720 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038042 |
| 2024-12-31 | VBB    | 0.002925 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.030901 |
| 2024-12-31 | SGB    | 0.002037 | 0.000000 | 0.150325 | 0.000000 | 0.040000 | 0.049831 |
| 2024-12-31 | ABB    | 0.003654 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040761 |
| 2024-12-31 | BVB    | 0.003605 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039038 |
| 2025-01-31 | VCB    | 0.263005 | 0.300000 | 0.124959 | 0.032437 | 0.040000 | 0.052915 |
| 2025-01-31 | BID    | 0.135537 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034239 |
| 2025-01-31 | CTG    | 0.072372 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039935 |
| 2025-01-31 | MBB    | 0.051952 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037261 |
| 2025-01-31 | TCB    | 0.086365 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032690 |
| 2025-01-31 | ACB    | 0.057561 | 0.300000 | 0.150359 | 0.000000 | 0.040000 | 0.060360 |
| 2025-01-31 | VPB    | 0.075711 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041660 |
| 2025-01-31 | HDB    | 0.035868 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042021 |
| 2025-01-31 | VIB    | 0.024403 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033338 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-01-31 | TPB    | 0.020295 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034600 |
| 2025-01-31 | STB    | 0.036156 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.028827 |
| 2025-01-31 | SHB    | 0.016398 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035819 |
| 2025-01-31 | LPB    | 0.042713 | 0.300000 | 0.000000 | 0.300000 | 0.040000 | 0.029338 |
| 2025-01-31 | SSB    | 0.024768 | 0.000000 | 0.260570 | 0.179310 | 0.040000 | 0.048335 |
| 2025-01-31 | MSB    | 0.013122 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039401 |
| 2025-01-31 | OCB    | 0.012829 | 0.000000 | 0.000000 | 0.188253 | 0.040000 | 0.037441 |
| 2025-01-31 | NAB    | 0.007583 | 0.100000 | 0.000000 | 0.000000 | 0.040000 | 0.034361 |
| 2025-01-31 | BAB    | 0.005139 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.086509 |
| 2025-01-31 | KLB    | 0.001352 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.026823 |
| 2025-01-31 | PGB    | 0.002869 | 0.000000 | 0.004336 | 0.000000 | 0.040000 | 0.024902 |
| 2025-01-31 | VAB    | 0.001746 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037496 |
| 2025-01-31 | VBB    | 0.002872 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.030004 |
| 2025-01-31 | SGB    | 0.002083 | 0.000000 | 0.159777 | 0.000000 | 0.040000 | 0.049562 |
| 2025-01-31 | ABB    | 0.003637 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042021 |
| 2025-01-31 | BVB    | 0.003666 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040144 |
| 2025-02-28 | VCB    | 0.262105 | 0.300000 | 0.122524 | 0.278878 | 0.040000 | 0.052565 |
| 2025-02-28 | BID    | 0.141699 | 0.000000 | 0.000000 | 0.119912 | 0.040000 | 0.034131 |
| 2025-02-28 | CTG    | 0.071390 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.039625 |
| 2025-02-28 | MBB    | 0.052330 | 0.000000 | 0.000000 | 0.018560 | 0.040000 | 0.037039 |
| 2025-02-28 | TCB    | 0.085092 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032525 |
| 2025-02-28 | ACB    | 0.055500 | 0.156307 | 0.140167 | 0.000000 | 0.040000 | 0.060275 |
| 2025-02-28 | VPB    | 0.073177 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041731 |
| 2025-02-28 | HDB    | 0.031283 | 0.186809 | 0.000000 | 0.111966 | 0.040000 | 0.038840 |
| 2025-02-28 | VIB    | 0.025107 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033864 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-02-28 | TPB    | 0.019677 | 0.000000 | 0.000000 | 0.079175 | 0.040000 | 0.037367 |
| 2025-02-28 | STB    | 0.035584 | 0.000000 | 0.000000 | 0.091510 | 0.040000 | 0.028611 |
| 2025-02-28 | SHB    | 0.016178 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035778 |
| 2025-02-28 | LPB    | 0.048095 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.028910 |
| 2025-02-28 | SSB    | 0.027213 | 0.000000 | 0.295831 | 0.000000 | 0.040000 | 0.052323 |
| 2025-02-28 | MSB    | 0.012322 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040492 |
| 2025-02-28 | OCB    | 0.012378 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036964 |
| 2025-02-28 | NAB    | 0.007891 | 0.056883 | 0.000000 | 0.000000 | 0.040000 | 0.034393 |
| 2025-02-28 | BAB    | 0.005044 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.084975 |
| 2025-02-28 | KLB    | 0.001327 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.026857 |
| 2025-02-28 | PGB    | 0.002778 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.025125 |
| 2025-02-28 | VAB    | 0.001658 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037194 |
| 2025-02-28 | VBB    | 0.002728 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029939 |
| 2025-02-28 | SGB    | 0.002141 | 0.000000 | 0.141478 | 0.000000 | 0.040000 | 0.048957 |
| 2025-02-28 | ABB    | 0.003427 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041751 |
| 2025-02-28 | BVB    | 0.003876 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039769 |
| 2025-03-31 | VCB    | 0.256873 | 0.100000 | 0.115961 | 0.283763 | 0.040000 | 0.055151 |
| 2025-03-31 | BID    | 0.141092 | 0.000000 | 0.000000 | 0.158483 | 0.040000 | 0.037908 |
| 2025-03-31 | CTG    | 0.075829 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039925 |
| 2025-03-31 | MBB    | 0.052259 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038469 |
| 2025-03-31 | TCB    | 0.087283 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032443 |
| 2025-03-31 | ACB    | 0.055356 | 0.300000 | 0.179171 | 0.000000 | 0.040000 | 0.059653 |
| 2025-03-31 | VPB    | 0.072668 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041497 |
| 2025-03-31 | HDB    | 0.030948 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037239 |
| 2025-03-31 | VIB    | 0.024537 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033489 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-03-31 | TPB    | 0.019440 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037164 |
| 2025-03-31 | STB    | 0.035778 | 0.000000 | 0.000000 | 0.257754 | 0.040000 | 0.028890 |
| 2025-03-31 | SHB    | 0.016945 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035530 |
| 2025-03-31 | LPB    | 0.047163 | 0.300000 | 0.000000 | 0.300000 | 0.040000 | 0.028546 |
| 2025-03-31 | SSB    | 0.027385 | 0.000000 | 0.268444 | 0.000000 | 0.040000 | 0.049902 |
| 2025-03-31 | MSB    | 0.012204 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040388 |
| 2025-03-31 | OCB    | 0.012088 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036727 |
| 2025-03-31 | NAB    | 0.008036 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.034039 |
| 2025-03-31 | BAB    | 0.005205 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.084285 |
| 2025-03-31 | KLB    | 0.001405 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.027078 |
| 2025-03-31 | PGB    | 0.002791 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.025046 |
| 2025-03-31 | VAB    | 0.001843 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036919 |
| 2025-03-31 | VBB    | 0.002802 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029930 |
| 2025-03-31 | SGB    | 0.002068 | 0.000000 | 0.136424 | 0.000000 | 0.040000 | 0.048136 |
| 2025-03-31 | ABB    | 0.003569 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041863 |
| 2025-03-31 | BVB    | 0.004436 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039785 |
| 2025-04-30 | VCB    | 0.263224 | 0.220671 | 0.134200 | 0.242212 | 0.040000 | 0.056256 |
| 2025-04-30 | BID    | 0.133396 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038846 |
| 2025-04-30 | CTG    | 0.075692 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040702 |
| 2025-04-30 | MBB    | 0.054730 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038616 |
| 2025-04-30 | TCB    | 0.091910 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032554 |
| 2025-04-30 | ACB    | 0.055314 | 0.300000 | 0.159560 | 0.000000 | 0.040000 | 0.060044 |
| 2025-04-30 | VPB    | 0.071473 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041784 |
| 2025-04-30 | HDB    | 0.029655 | 0.068048 | 0.000000 | 0.000000 | 0.040000 | 0.037909 |
| 2025-04-30 | VIB    | 0.023455 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033639 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-04-30 | TPB    | 0.016635 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037300 |
| 2025-04-30 | STB    | 0.035798 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.029222 |
| 2025-04-30 | SHB    | 0.021085 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035268 |
| 2025-04-30 | LPB    | 0.043351 | 0.300000 | 0.000000 | 0.300000 | 0.040000 | 0.029234 |
| 2025-04-30 | SSB    | 0.027576 | 0.000000 | 0.278663 | 0.100955 | 0.040000 | 0.051726 |
| 2025-04-30 | MSB    | 0.013110 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040496 |
| 2025-04-30 | OCB    | 0.011968 | 0.000000 | 0.000000 | 0.056833 | 0.040000 | 0.036953 |
| 2025-04-30 | NAB    | 0.007553 | 0.111280 | 0.000000 | 0.000000 | 0.040000 | 0.034144 |
| 2025-04-30 | BAB    | 0.005201 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.080482 |
| 2025-04-30 | KLB    | 0.001324 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.026860 |
| 2025-04-30 | PGB    | 0.002843 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.025329 |
| 2025-04-30 | VAB    | 0.001858 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035606 |
| 2025-04-30 | VBB    | 0.002949 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029894 |
| 2025-04-30 | SGB    | 0.002081 | 0.000000 | 0.127577 | 0.000000 | 0.040000 | 0.048344 |
| 2025-04-30 | ABB    | 0.003659 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041214 |
| 2025-04-30 | BVB    | 0.004161 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037575 |
| 2025-05-31 | VCB    | 0.255298 | 0.300000 | 0.126505 | 0.000000 | 0.040000 | 0.056516 |
| 2025-05-31 | BID    | 0.128853 | 0.000000 | 0.000000 | 0.279713 | 0.040000 | 0.038521 |
| 2025-05-31 | CTG    | 0.073572 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040770 |
| 2025-05-31 | MBB    | 0.057943 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038638 |
| 2025-05-31 | TCB    | 0.095199 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032438 |
| 2025-05-31 | ACB    | 0.055423 | 0.300000 | 0.158165 | 0.000000 | 0.040000 | 0.060374 |
| 2025-05-31 | VPB    | 0.067453 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041784 |
| 2025-05-31 | HDB    | 0.031181 | 0.091731 | 0.000000 | 0.000000 | 0.040000 | 0.037728 |
| 2025-05-31 | VIB    | 0.023088 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033529 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-05-31 | TPB    | 0.017126 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035891 |
| 2025-05-31 | STB    | 0.039588 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.029270 |
| 2025-05-31 | SHB    | 0.023124 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032403 |
| 2025-05-31 | LPB    | 0.045965 | 0.300000 | 0.000000 | 0.298693 | 0.040000 | 0.028960 |
| 2025-05-31 | SSB    | 0.028114 | 0.000000 | 0.288013 | 0.000000 | 0.040000 | 0.052898 |
| 2025-05-31 | MSB    | 0.013028 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040032 |
| 2025-05-31 | OCB    | 0.012091 | 0.000000 | 0.000000 | 0.121594 | 0.040000 | 0.037041 |
| 2025-05-31 | NAB    | 0.008131 | 0.008269 | 0.000000 | 0.000000 | 0.040000 | 0.033814 |
| 2025-05-31 | BAB    | 0.005300 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.080393 |
| 2025-05-31 | KLB    | 0.001495 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.031245 |
| 2025-05-31 | PGB    | 0.002855 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.025437 |
| 2025-05-31 | VAB    | 0.002111 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035639 |
| 2025-05-31 | VBB    | 0.003098 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029770 |
| 2025-05-31 | SGB    | 0.002204 | 0.000000 | 0.127317 | 0.000000 | 0.040000 | 0.048319 |
| 2025-05-31 | ABB    | 0.003758 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041141 |
| 2025-05-31 | BVB    | 0.004001 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037450 |
| 2025-06-30 | VCB    | 0.241100 | 0.057003 | 0.089438 | 0.233078 | 0.040000 | 0.051950 |
| 2025-06-30 | BID    | 0.128818 | 0.000000 | 0.000000 | 0.130218 | 0.040000 | 0.038528 |
| 2025-06-30 | CTG    | 0.072701 | 0.138049 | 0.000000 | 0.000000 | 0.040000 | 0.041168 |
| 2025-06-30 | MBB    | 0.057487 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038579 |
| 2025-06-30 | TCB    | 0.105768 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032272 |
| 2025-06-30 | ACB    | 0.055856 | 0.000000 | 0.133803 | 0.000000 | 0.040000 | 0.056765 |
| 2025-06-30 | VPB    | 0.072126 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039066 |
| 2025-06-30 | HDB    | 0.030264 | 0.211354 | 0.000000 | 0.000000 | 0.040000 | 0.037914 |
| 2025-06-30 | VIB    | 0.022861 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032643 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-06-30 | TPB    | 0.017183 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036912 |
| 2025-06-30 | STB    | 0.039542 | 0.000000 | 0.000000 | 0.271928 | 0.040000 | 0.029290 |
| 2025-06-30 | SHB    | 0.023865 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035982 |
| 2025-06-30 | LPB    | 0.046262 | 0.300000 | 0.000000 | 0.199158 | 0.040000 | 0.029778 |
| 2025-06-30 | SSB    | 0.026978 | 0.000000 | 0.281121 | 0.000000 | 0.040000 | 0.051132 |
| 2025-06-30 | MSB    | 0.012890 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039324 |
| 2025-06-30 | OCB    | 0.011831 | 0.000000 | 0.000000 | 0.165617 | 0.040000 | 0.036046 |
| 2025-06-30 | NAB    | 0.007901 | 0.293593 | 0.000000 | 0.000000 | 0.040000 | 0.034201 |
| 2025-06-30 | BAB    | 0.005268 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.084631 |
| 2025-06-30 | KLB    | 0.002142 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.030623 |
| 2025-06-30 | PGB    | 0.002858 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.024742 |
| 2025-06-30 | VAB    | 0.002613 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035713 |
| 2025-06-30 | VBB    | 0.003528 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.030012 |
| 2025-06-30 | SGB    | 0.002099 | 0.000000 | 0.195638 | 0.000000 | 0.040000 | 0.050766 |
| 2025-06-30 | ABB    | 0.003899 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.043896 |
| 2025-06-30 | BVB    | 0.004159 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038066 |
| 2025-07-31 | VCB    | 0.234626 | 0.078296 | 0.081971 | 0.000000 | 0.040000 | 0.051640 |
| 2025-07-31 | BID    | 0.125075 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038459 |
| 2025-07-31 | CTG    | 0.076580 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040809 |
| 2025-07-31 | MBB    | 0.058651 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038787 |
| 2025-07-31 | TCB    | 0.114387 | 0.010777 | 0.000000 | 0.000000 | 0.040000 | 0.032734 |
| 2025-07-31 | ACB    | 0.054289 | 0.300000 | 0.113315 | 0.000000 | 0.040000 | 0.056409 |
| 2025-07-31 | VPB    | 0.071571 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040135 |
| 2025-07-31 | HDB    | 0.029278 | 0.007236 | 0.000000 | 0.000000 | 0.040000 | 0.037637 |
| 2025-07-31 | VIB    | 0.022499 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034025 |



| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-07-31 | TPB    | 0.016793 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036497 |
| 2025-07-31 | STB    | 0.043685 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.031236 |
| 2025-07-31 | SHB    | 0.022696 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036173 |
| 2025-07-31 | LPB    | 0.045180 | 0.300000 | 0.000000 | 0.111878 | 0.040000 | 0.030183 |
| 2025-07-31 | SSB    | 0.025622 | 0.000000 | 0.288162 | 0.286412 | 0.040000 | 0.052048 |
| 2025-07-31 | MSB    | 0.012901 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040001 |
| 2025-07-31 | OCB    | 0.012578 | 0.000000 | 0.000883 | 0.000000 | 0.040000 | 0.038243 |
| 2025-07-31 | NAB    | 0.007607 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.035404 |
| 2025-07-31 | BAB    | 0.005334 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.082635 |
| 2025-07-31 | KLB    | 0.001959 | 0.000000 | 0.000000 | 0.087649 | 0.040000 | 0.025405 |
| 2025-07-31 | PGB    | 0.002637 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.025601 |
| 2025-07-31 | VAB    | 0.002693 | 0.003692 | 0.000000 | 0.300000 | 0.040000 | 0.032617 |
| 2025-07-31 | VBB    | 0.003497 | 0.000000 | 0.000000 | 0.214062 | 0.040000 | 0.028988 |
| 2025-07-31 | SGB    | 0.002006 | 0.000000 | 0.215669 | 0.000000 | 0.040000 | 0.052733 |
| 2025-07-31 | ABB    | 0.003944 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.043298 |
| 2025-07-31 | BVB    | 0.003912 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038303 |
| 2025-08-31 | VCB    | 0.218100 | 0.135595 | 0.087518 | 0.000000 | 0.040000 | 0.050685 |
| 2025-08-31 | BID    | 0.114771 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037504 |
| 2025-08-31 | CTG    | 0.073824 | 0.070569 | 0.000000 | 0.000000 | 0.040000 | 0.039483 |
| 2025-08-31 | MBB    | 0.072464 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039455 |
| 2025-08-31 | TCB    | 0.100086 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.031313 |
| 2025-08-31 | ACB    | 0.051594 | 0.057543 | 0.118684 | 0.000000 | 0.040000 | 0.055561 |
| 2025-08-31 | VPB    | 0.087833 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039153 |
| 2025-08-31 | HDB    | 0.031271 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037151 |
| 2025-08-31 | VIB    | 0.027338 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036428 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-08-31 | TPB    | 0.017249 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037683 |
| 2025-08-31 | STB    | 0.041123 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029709 |
| 2025-08-31 | SHB    | 0.030095 | 0.000000 | 0.000000 | 0.149902 | 0.040000 | 0.035714 |
| 2025-08-31 | LPB    | 0.042164 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.030591 |
| 2025-08-31 | SSB    | 0.024227 | 0.000000 | 0.272315 | 0.108702 | 0.040000 | 0.053331 |
| 2025-08-31 | MSB    | 0.013058 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042496 |
| 2025-08-31 | OCB    | 0.014165 | 0.000000 | 0.000964 | 0.000000 | 0.040000 | 0.038783 |
| 2025-08-31 | NAB    | 0.009426 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.036180 |
| 2025-08-31 | BAB    | 0.005700 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.082136 |
| 2025-08-31 | KLB    | 0.003656 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.024691 |
| 2025-08-31 | PGB    | 0.002673 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.026554 |
| 2025-08-31 | VAB    | 0.003961 | 0.136293 | 0.000000 | 0.300000 | 0.040000 | 0.032551 |
| 2025-08-31 | VBB    | 0.003991 | 0.000000 | 0.000000 | 0.054734 | 0.040000 | 0.029743 |
| 2025-08-31 | SGB    | 0.001848 | 0.000000 | 0.220520 | 0.086662 | 0.040000 | 0.051395 |
| 2025-08-31 | ABB    | 0.005411 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.043626 |
| 2025-08-31 | BVB    | 0.003974 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038085 |
| 2025-09-30 | VCB    | 0.208596 | 0.141576 | 0.108528 | 0.000000 | 0.040000 | 0.053076 |
| 2025-09-30 | BID    | 0.109067 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040416 |
| 2025-09-30 | CTG    | 0.069264 | 0.063901 | 0.000000 | 0.000000 | 0.040000 | 0.041161 |
| 2025-09-30 | MBB    | 0.081930 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041637 |
| 2025-09-30 | TCB    | 0.097831 | 0.000000 | 0.000000 | 0.162487 | 0.040000 | 0.033148 |
| 2025-09-30 | ACB    | 0.052341 | 0.008688 | 0.105533 | 0.000000 | 0.040000 | 0.057494 |
| 2025-09-30 | VPB    | 0.100008 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033307 |
| 2025-09-30 | HDB    | 0.033181 | 0.053700 | 0.000000 | 0.000000 | 0.040000 | 0.036709 |
| 2025-09-30 | VIB    | 0.026738 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039283 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-09-30 | TPB    | 0.019280 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037824 |
| 2025-09-30 | STB    | 0.038419 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.031933 |
| 2025-09-30 | SHB    | 0.031203 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033112 |
| 2025-09-30 | LPB    | 0.046557 | 0.300000 | 0.000000 | 0.039029 | 0.040000 | 0.032944 |
| 2025-09-30 | SSB    | 0.022316 | 0.000000 | 0.253360 | 0.000000 | 0.040000 | 0.054817 |
| 2025-09-30 | MSB    | 0.014057 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.043474 |
| 2025-09-30 | OCB    | 0.013324 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038918 |
| 2025-09-30 | NAB    | 0.008408 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.037552 |
| 2025-09-30 | BAB    | 0.004990 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.070948 |
| 2025-09-30 | KLB    | 0.002975 | 0.000000 | 0.000000 | 0.198485 | 0.040000 | 0.024130 |
| 2025-09-30 | PGB    | 0.002382 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.028898 |
| 2025-09-30 | VAB    | 0.003845 | 0.132136 | 0.000000 | 0.300000 | 0.040000 | 0.033974 |
| 2025-09-30 | VBB    | 0.003494 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.030875 |
| 2025-09-30 | SGB    | 0.001668 | 0.000000 | 0.232579 | 0.000000 | 0.040000 | 0.053880 |
| 2025-09-30 | ABB    | 0.004370 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.031115 |
| 2025-09-30 | BVB    | 0.003758 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039376 |
| 2025-10-31 | VCB    | 0.200932 | 0.000000 | 0.174164 | 0.000000 | 0.040000 | 0.054919 |
| 2025-10-31 | BID    | 0.108522 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041401 |
| 2025-10-31 | CTG    | 0.073095 | 0.074087 | 0.000000 | 0.000000 | 0.040000 | 0.042257 |
| 2025-10-31 | MBB    | 0.082453 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037330 |
| 2025-10-31 | TCB    | 0.102321 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033761 |
| 2025-10-31 | ACB    | 0.051175 | 0.208478 | 0.000000 | 0.000000 | 0.040000 | 0.052878 |
| 2025-10-31 | VPB    | 0.094728 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.031159 |
| 2025-10-31 | HDB    | 0.032776 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035756 |
| 2025-10-31 | VIB    | 0.025096 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039856 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-10-31 | TPB    | 0.018528 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034448 |
| 2025-10-31 | STB    | 0.041909 | 0.000000 | 0.000000 | 0.025275 | 0.040000 | 0.033102 |
| 2025-10-31 | SHB    | 0.029904 | 0.000000 | 0.000000 | 0.142927 | 0.040000 | 0.032872 |
| 2025-10-31 | LPB    | 0.054598 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.031860 |
| 2025-10-31 | SSB    | 0.021453 | 0.000000 | 0.258903 | 0.000000 | 0.040000 | 0.055223 |
| 2025-10-31 | MSB    | 0.013358 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039590 |
| 2025-10-31 | OCB    | 0.013162 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039965 |
| 2025-10-31 | NAB    | 0.008211 | 0.300000 | 0.000000 | 0.000000 | 0.040000 | 0.039273 |
| 2025-10-31 | BAB    | 0.004950 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.073383 |
| 2025-10-31 | KLB    | 0.003571 | 0.000000 | 0.000000 | 0.281035 | 0.040000 | 0.025394 |
| 2025-10-31 | PGB    | 0.002346 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.030233 |
| 2025-10-31 | VAB    | 0.003700 | 0.117435 | 0.000000 | 0.261173 | 0.040000 | 0.034624 |
| 2025-10-31 | VBB    | 0.003509 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032324 |
| 2025-10-31 | SGB    | 0.001628 | 0.000000 | 0.266933 | 0.000000 | 0.040000 | 0.055580 |
| 2025-10-31 | ABB    | 0.004545 | 0.000000 | 0.000000 | 0.289590 | 0.040000 | 0.032512 |
| 2025-10-31 | BVB    | 0.003530 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040299 |
| 2025-11-30 | VCB    | 0.192849 | 0.000000 | 0.210785 | 0.000000 | 0.040000 | 0.055668 |
| 2025-11-30 | BID    | 0.101691 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042872 |
| 2025-11-30 | CTG    | 0.101902 | 0.298112 | 0.000000 | 0.000000 | 0.040000 | 0.046853 |
| 2025-11-30 | MBB    | 0.073615 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038754 |
| 2025-11-30 | TCB    | 0.094343 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035104 |
| 2025-11-30 | ACB    | 0.050724 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.052328 |
| 2025-11-30 | VPB    | 0.086642 | 0.000000 | 0.000000 | 0.099812 | 0.040000 | 0.030931 |
| 2025-11-30 | HDB    | 0.047837 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035293 |
| 2025-11-30 | VIB    | 0.023187 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038815 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-11-30 | TPB    | 0.017886 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033662 |
| 2025-11-30 | STB    | 0.040518 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034606 |
| 2025-11-30 | SHB    | 0.028839 | 0.000395 | 0.000000 | 0.240479 | 0.040000 | 0.034401 |
| 2025-11-30 | LPB    | 0.055423 | 0.300000 | 0.056895 | 0.000000 | 0.040000 | 0.034058 |
| 2025-11-30 | SSB    | 0.019500 | 0.000000 | 0.217299 | 0.000000 | 0.040000 | 0.052141 |
| 2025-11-30 | MSB    | 0.014559 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039088 |
| 2025-11-30 | OCB    | 0.013149 | 0.000000 | 0.000000 | 0.091431 | 0.040000 | 0.040393 |
| 2025-11-30 | NAB    | 0.007886 | 0.288984 | 0.000000 | 0.000000 | 0.040000 | 0.039519 |
| 2025-11-30 | BAB    | 0.004798 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.070758 |
| 2025-11-30 | KLB    | 0.003765 | 0.000000 | 0.026435 | 0.246365 | 0.040000 | 0.024442 |
| 2025-11-30 | PGB    | 0.002790 | 0.000000 | 0.016447 | 0.000000 | 0.040000 | 0.029830 |
| 2025-11-30 | VAB    | 0.003588 | 0.021532 | 0.000000 | 0.214572 | 0.040000 | 0.033380 |
| 2025-11-30 | VBB    | 0.004721 | 0.090977 | 0.000000 | 0.000000 | 0.040000 | 0.033105 |
| 2025-11-30 | SGB    | 0.001650 | 0.000000 | 0.172138 | 0.000000 | 0.040000 | 0.052591 |
| 2025-11-30 | ABB    | 0.004762 | 0.000000 | 0.000000 | 0.107342 | 0.040000 | 0.032358 |
| 2025-11-30 | BVB    | 0.003375 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.039049 |
| 2025-12-31 | VCB    | 0.189498 | 0.000000 | 0.176834 | 0.000000 | 0.040000 | 0.053238 |
| 2025-12-31 | BID    | 0.102922 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.040832 |
| 2025-12-31 | CTG    | 0.103969 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044961 |
| 2025-12-31 | MBB    | 0.073995 | 0.000000 | 0.000000 | 0.180838 | 0.040000 | 0.037745 |
| 2025-12-31 | TCB    | 0.092534 | 0.170506 | 0.000000 | 0.080874 | 0.040000 | 0.040337 |
| 2025-12-31 | ACB    | 0.049216 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.050392 |
| 2025-12-31 | VPB    | 0.090092 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029958 |
| 2025-12-31 | HDB    | 0.048807 | 0.017732 | 0.000000 | 0.000000 | 0.040000 | 0.036257 |
| 2025-12-31 | VIB    | 0.023590 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038225 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2025-12-31 | TPB    | 0.018961 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034347 |
| 2025-12-31 | STB    | 0.036200 | 0.143214 | 0.051154 | 0.000000 | 0.040000 | 0.038504 |
| 2025-12-31 | SHB    | 0.030060 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033687 |
| 2025-12-31 | LPB    | 0.054211 | 0.300000 | 0.073663 | 0.067354 | 0.040000 | 0.034080 |
| 2025-12-31 | SSB    | 0.019446 | 0.000000 | 0.185126 | 0.000000 | 0.040000 | 0.049068 |
| 2025-12-31 | MSB    | 0.016210 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036997 |
| 2025-12-31 | OCB    | 0.012942 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041647 |
| 2025-12-31 | NAB    | 0.008080 | 0.257511 | 0.000000 | 0.000000 | 0.040000 | 0.039441 |
| 2025-12-31 | BAB    | 0.004598 | 0.000000 | 0.300000 | 0.117064 | 0.040000 | 0.073699 |
| 2025-12-31 | KLB    | 0.003519 | 0.000000 | 0.032932 | 0.300000 | 0.040000 | 0.023684 |
| 2025-12-31 | PGB    | 0.002760 | 0.038384 | 0.034501 | 0.039620 | 0.040000 | 0.028706 |
| 2025-12-31 | VAB    | 0.003500 | 0.006197 | 0.000000 | 0.010263 | 0.040000 | 0.033687 |
| 2025-12-31 | VBB    | 0.004468 | 0.066456 | 0.000000 | 0.048374 | 0.040000 | 0.031914 |
| 2025-12-31 | SGB    | 0.001672 | 0.000000 | 0.145791 | 0.000000 | 0.040000 | 0.050702 |
| 2025-12-31 | ABB    | 0.005382 | 0.000000 | 0.000000 | 0.155613 | 0.040000 | 0.033264 |
| 2025-12-31 | BVB    | 0.003367 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044630 |
| 2026-01-31 | VCB    | 0.185156 | 0.000000 | 0.154779 | 0.000000 | 0.040000 | 0.053947 |
| 2026-01-31 | BID    | 0.105259 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.043454 |
| 2026-01-31 | CTG    | 0.107008 | 0.000000 | 0.000000 | 0.054987 | 0.040000 | 0.045281 |
| 2026-01-31 | MBB    | 0.078537 | 0.000000 | 0.000000 | 0.033939 | 0.040000 | 0.037280 |
| 2026-01-31 | TCB    | 0.093342 | 0.167819 | 0.000000 | 0.136527 | 0.040000 | 0.039405 |
| 2026-01-31 | ACB    | 0.047509 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.049047 |
| 2026-01-31 | VPB    | 0.086071 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029582 |
| 2026-01-31 | HDB    | 0.057289 | 0.067623 | 0.000000 | 0.000000 | 0.040000 | 0.036249 |
| 2026-01-31 | VIB    | 0.022078 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037735 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2026-01-31 | TPB    | 0.018281 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034007 |
| 2026-01-31 | STB    | 0.042138 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.038704 |
| 2026-01-31 | SHB    | 0.028523 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033268 |
| 2026-01-31 | LPB    | 0.045474 | 0.300000 | 0.056810 | 0.052926 | 0.040000 | 0.033280 |
| 2026-01-31 | SSB    | 0.019132 | 0.000000 | 0.204459 | 0.000000 | 0.040000 | 0.048467 |
| 2026-01-31 | MSB    | 0.014910 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036188 |
| 2026-01-31 | OCB    | 0.012264 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.042800 |
| 2026-01-31 | NAB    | 0.007881 | 0.300000 | 0.000000 | 0.016235 | 0.040000 | 0.039268 |
| 2026-01-31 | BAB    | 0.004415 | 0.000000 | 0.300000 | 0.194698 | 0.040000 | 0.070862 |
| 2026-01-31 | KLB    | 0.003769 | 0.000000 | 0.045887 | 0.300000 | 0.040000 | 0.026473 |
| 2026-01-31 | PGB    | 0.002628 | 0.058273 | 0.028130 | 0.026802 | 0.040000 | 0.029561 |
| 2026-01-31 | VAB    | 0.003272 | 0.048768 | 0.000000 | 0.000000 | 0.040000 | 0.034257 |
| 2026-01-31 | VBB    | 0.004316 | 0.030685 | 0.000000 | 0.023565 | 0.040000 | 0.031577 |
| 2026-01-31 | SGB    | 0.001619 | 0.000000 | 0.209935 | 0.000000 | 0.040000 | 0.052525 |
| 2026-01-31 | ABB    | 0.005868 | 0.026833 | 0.000000 | 0.160322 | 0.040000 | 0.032593 |
| 2026-01-31 | BVB    | 0.003260 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044191 |
| 2026-02-28 | VCB    | 0.205519 | 0.000000 | 0.148265 | 0.000000 | 0.040000 | 0.053706 |
| 2026-02-28 | BID    | 0.136900 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.043634 |
| 2026-02-28 | CTG    | 0.105003 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044906 |
| 2026-02-28 | MBB    | 0.076439 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036944 |
| 2026-02-28 | TCB    | 0.086925 | 0.182342 | 0.000000 | 0.000000 | 0.040000 | 0.039255 |
| 2026-02-28 | ACB    | 0.043190 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.048741 |
| 2026-02-28 | VPB    | 0.076148 | 0.000000 | 0.000000 | 0.169108 | 0.040000 | 0.029430 |
| 2026-02-28 | HDB    | 0.049419 | 0.186305 | 0.000000 | 0.108614 | 0.040000 | 0.034425 |
| 2026-02-28 | VIB    | 0.020047 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037757 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2026-02-28 | TPB    | 0.016647 | 0.000000 | 0.000000 | 0.114602 | 0.040000 | 0.033975 |
| 2026-02-28 | STB    | 0.041436 | 0.000000 | 0.007043 | 0.000000 | 0.040000 | 0.037217 |
| 2026-02-28 | SHB    | 0.025277 | 0.000000 | 0.000000 | 0.071684 | 0.040000 | 0.033003 |
| 2026-02-28 | LPB    | 0.041022 | 0.300000 | 0.044972 | 0.122961 | 0.040000 | 0.031320 |
| 2026-02-28 | SSB    | 0.017023 | 0.000000 | 0.213821 | 0.000000 | 0.040000 | 0.049832 |
| 2026-02-28 | MSB    | 0.013498 | 0.000000 | 0.000000 | 0.196805 | 0.040000 | 0.035791 |
| 2026-02-28 | OCB    | 0.010963 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.045472 |
| 2026-02-28 | NAB    | 0.007057 | 0.240903 | 0.000000 | 0.000000 | 0.040000 | 0.039138 |
| 2026-02-28 | BAB    | 0.004515 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.071130 |
| 2026-02-28 | KLB    | 0.003087 | 0.033220 | 0.046682 | 0.000593 | 0.040000 | 0.028447 |
| 2026-02-28 | PGB    | 0.002949 | 0.000000 | 0.023988 | 0.000000 | 0.040000 | 0.029335 |
| 2026-02-28 | VAB    | 0.003218 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036185 |
| 2026-02-28 | VBB    | 0.003983 | 0.012059 | 0.000000 | 0.000000 | 0.040000 | 0.031586 |
| 2026-02-28 | SGB    | 0.001586 | 0.000000 | 0.215229 | 0.000000 | 0.040000 | 0.052266 |
| 2026-02-28 | ABB    | 0.005019 | 0.045171 | 0.000000 | 0.215633 | 0.040000 | 0.032007 |
| 2026-02-28 | BVB    | 0.003130 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044499 |
| 2026-03-31 | VCB    | 0.193651 | 0.000000 | 0.069068 | 0.000000 | 0.040000 | 0.048334 |
| 2026-03-31 | BID    | 0.124527 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035332 |
| 2026-03-31 | CTG    | 0.106090 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044941 |
| 2026-03-31 | MBB    | 0.081979 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.037167 |
| 2026-03-31 | TCB    | 0.089833 | 0.178748 | 0.000000 | 0.000000 | 0.040000 | 0.039677 |
| 2026-03-31 | ACB    | 0.045032 | 0.000000 | 0.002997 | 0.000000 | 0.040000 | 0.053368 |
| 2026-03-31 | VPB    | 0.080577 | 0.000000 | 0.000000 | 0.162391 | 0.040000 | 0.029116 |
| 2026-03-31 | HDB    | 0.049958 | 0.138152 | 0.000000 | 0.275245 | 0.040000 | 0.034485 |
| 2026-03-31 | VIB    | 0.019935 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041200 |



| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2026-03-31 | TPB    | 0.018227 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035148 |
| 2026-03-31 | STB    | 0.044096 | 0.027699 | 0.060705 | 0.000000 | 0.040000 | 0.038660 |
| 2026-03-31 | SHB    | 0.025790 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.032714 |
| 2026-03-31 | LPB    | 0.043396 | 0.300000 | 0.065898 | 0.000000 | 0.040000 | 0.031036 |
| 2026-03-31 | SSB    | 0.017170 | 0.000000 | 0.221723 | 0.000000 | 0.040000 | 0.049337 |
| 2026-03-31 | MSB    | 0.013593 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035564 |
| 2026-03-31 | OCB    | 0.011079 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044918 |
| 2026-03-31 | NAB    | 0.007150 | 0.262583 | 0.045402 | 0.000000 | 0.040000 | 0.043600 |
| 2026-03-31 | BAB    | 0.004594 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.069478 |
| 2026-03-31 | KLB    | 0.003170 | 0.000000 | 0.008083 | 0.300000 | 0.040000 | 0.027508 |
| 2026-03-31 | PGB    | 0.003115 | 0.000000 | 0.025683 | 0.000000 | 0.040000 | 0.028926 |
| 2026-03-31 | VAB    | 0.003148 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035549 |
| 2026-03-31 | VBB    | 0.004038 | 0.067016 | 0.068944 | 0.000000 | 0.040000 | 0.036464 |
| 2026-03-31 | SGB    | 0.001624 | 0.000000 | 0.131497 | 0.000000 | 0.040000 | 0.051450 |
| 2026-03-31 | ABB    | 0.005138 | 0.025803 | 0.000000 | 0.262364 | 0.040000 | 0.031512 |
| 2026-03-31 | BVB    | 0.003089 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044513 |
| 2026-04-30 | VCB    | 0.191340 | 0.000000 | 0.034094 | 0.078654 | 0.040000 | 0.045718 |
| 2026-04-30 | BID    | 0.113053 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.033365 |
| 2026-04-30 | CTG    | 0.105919 | 0.000000 | 0.000000 | 0.092571 | 0.040000 | 0.046248 |
| 2026-04-30 | MBB    | 0.083973 | 0.000000 | 0.000000 | 0.185321 | 0.040000 | 0.038349 |
| 2026-04-30 | TCB    | 0.083984 | 0.192158 | 0.000000 | 0.000000 | 0.040000 | 0.040518 |
| 2026-04-30 | ACB    | 0.047678 | 0.000000 | 0.042229 | 0.000000 | 0.040000 | 0.053642 |
| 2026-04-30 | VPB    | 0.082023 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.029283 |
| 2026-04-30 | HDB    | 0.050207 | 0.070429 | 0.000000 | 0.043454 | 0.040000 | 0.034504 |
| 2026-04-30 | VIB    | 0.021560 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041216 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2026-04-30 | TPB    | 0.017822 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034538 |
| 2026-04-30 | STB    | 0.046217 | 0.100852 | 0.072188 | 0.300000 | 0.040000 | 0.040231 |
| 2026-04-30 | SHB    | 0.027759 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033028 |
| 2026-04-30 | LPB    | 0.046625 | 0.300000 | 0.055763 | 0.000000 | 0.040000 | 0.030759 |
| 2026-04-30 | SSB    | 0.019006 | 0.000000 | 0.231462 | 0.000000 | 0.040000 | 0.048643 |
| 2026-04-30 | MSB    | 0.014203 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035803 |
| 2026-04-30 | OCB    | 0.011703 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.047657 |
| 2026-04-30 | NAB    | 0.007350 | 0.258381 | 0.051370 | 0.000000 | 0.040000 | 0.044595 |
| 2026-04-30 | BAB    | 0.004691 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.067703 |
| 2026-04-30 | KLB    | 0.002994 | 0.000000 | 0.023740 | 0.000000 | 0.040000 | 0.027510 |
| 2026-04-30 | PGB    | 0.003224 | 0.000000 | 0.012073 | 0.000000 | 0.040000 | 0.028279 |
| 2026-04-30 | VAB    | 0.003250 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.036028 |
| 2026-04-30 | VBB    | 0.004499 | 0.000000 | 0.027886 | 0.000000 | 0.040000 | 0.037161 |
| 2026-04-30 | SGB    | 0.001749 | 0.000000 | 0.149194 | 0.000000 | 0.040000 | 0.050563 |
| 2026-04-30 | ABB    | 0.006038 | 0.078180 | 0.000000 | 0.000000 | 0.040000 | 0.031555 |
| 2026-04-30 | BVB    | 0.003132 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.043103 |
| 2026-05-31 | VCB    | 0.191279 | 0.000000 | 0.000000 | 0.017306 | 0.040000 | 0.044190 |
| 2026-05-31 | BID    | 0.111754 | 0.000000 | 0.000000 | 0.147149 | 0.040000 | 0.031357 |
| 2026-05-31 | CTG    | 0.103916 | 0.000000 | 0.000000 | 0.300000 | 0.040000 | 0.044625 |
| 2026-05-31 | MBB    | 0.080326 | 0.120225 | 0.000000 | 0.235545 | 0.040000 | 0.037771 |
| 2026-05-31 | TCB    | 0.089926 | 0.014423 | 0.000000 | 0.000000 | 0.040000 | 0.037320 |
| 2026-05-31 | ACB    | 0.046210 | 0.000000 | 0.051955 | 0.000000 | 0.040000 | 0.053378 |
| 2026-05-31 | VPB    | 0.079088 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.030974 |
| 2026-05-31 | HDB    | 0.050968 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.034366 |
| 2026-05-31 | VIB    | 0.021240 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.041616 |

| Date       | Ticker | w_mkt    | w_TAN    | w_MV     | w_BL     | w_EW     | w_RP     |
|------------|--------|----------|----------|----------|----------|----------|----------|
| 2026-05-31 | TPB    | 0.017256 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033777 |
| 2026-05-31 | STB    | 0.048713 | 0.131561 | 0.077103 | 0.300000 | 0.040000 | 0.040531 |
| 2026-05-31 | SHB    | 0.025589 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.033494 |
| 2026-05-31 | LPB    | 0.051117 | 0.300000 | 0.045324 | 0.000000 | 0.040000 | 0.030975 |
| 2026-05-31 | SSB    | 0.018188 | 0.000000 | 0.265335 | 0.000000 | 0.040000 | 0.049096 |
| 2026-05-31 | MSB    | 0.014930 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035901 |
| 2026-05-31 | OCB    | 0.011469 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.047612 |
| 2026-05-31 | NAB    | 0.007717 | 0.259867 | 0.016185 | 0.000000 | 0.040000 | 0.043596 |
| 2026-05-31 | BAB    | 0.004638 | 0.000000 | 0.300000 | 0.000000 | 0.040000 | 0.065296 |
| 2026-05-31 | KLB    | 0.003265 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.026757 |
| 2026-05-31 | PGB    | 0.003053 | 0.000000 | 0.028042 | 0.000000 | 0.040000 | 0.031467 |
| 2026-05-31 | VAB    | 0.003203 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.035856 |
| 2026-05-31 | VBB    | 0.005524 | 0.000000 | 0.009129 | 0.000000 | 0.040000 | 0.040016 |
| 2026-05-31 | SGB    | 0.001685 | 0.000000 | 0.195197 | 0.000000 | 0.040000 | 0.053941 |
| 2026-05-31 | ABB    | 0.005904 | 0.173923 | 0.011730 | 0.000000 | 0.040000 | 0.031777 |
| 2026-05-31 | BVB    | 0.003042 | 0.000000 | 0.000000 | 0.000000 | 0.040000 | 0.044311 |

## 10. Phan tich Information Coefficient (IC) cua tin hieu

IC (Spearman rank correlation) đo lường mức độ tin hieu idiosyncratic momentum dự báo dung chieu loi nhuan thực tế tháng kế tiếp. IC > 0 nghĩa là tin hieu xếp hạng có phieu dung chieu voi ket qua thực tế. Hit rate: tỷ lệ bước rebalancing mà cluster Long thực sự vượt cluster Short.

| Metric       | Value   |
|--------------|---------|
| Mean IC      | +0.0667 |
| IC > 0 (%)   | 50.0%   |
| Hit Rate (%) | 50.9%   |
| IC Std       | 0.3618  |
| N steps      | 106     |

### IC per rebalancing step (full series):

| Date       | IC        | Hit | N_stocks |
|------------|-----------|-----|----------|
| 2017-08-31 | -0.357143 | 1   | 7        |
| 2017-09-30 | -0.071429 | 0   | 7        |
| 2017-10-31 | -0.107143 | 0   | 7        |
| 2017-11-30 | -0.321429 | 0   | 7        |
| 2017-12-31 | -0.464286 | 0   | 7        |
| 2018-01-31 | 0.071429  | 0   | 7        |
| 2018-02-28 | 0.535714  | 1   | 7        |
| 2018-03-31 | 0.178571  | 1   | 7        |
| 2018-04-30 | 0.000000  | 1   | 7        |
| 2018-05-31 | -0.357143 | 1   | 7        |
| 2018-06-30 | -0.250000 | 0   | 7        |
| 2018-07-31 | -0.178571 | 0   | 7        |
| 2018-08-31 | 0.392857  | 1   | 7        |

| Date       | IC        | Hit | N_stocks |
|------------|-----------|-----|----------|
| 2018-09-30 | -0.821429 | 0   | 7        |
| 2018-10-31 | 0.357143  | 1   | 7        |
| 2018-11-30 | 0.750000  | 1   | 7        |
| 2018-12-31 | 0.500000  | 1   | 7        |
| 2019-01-31 | 0.607143  | 1   | 7        |
| 2019-02-28 | 0.142857  | 1   | 7        |
| 2019-03-31 | 0.000000  | 1   | 7        |
| 2019-04-30 | -0.018019 | 0   | 7        |
| 2019-05-31 | 0.535714  | 1   | 7        |
| 2019-06-30 | 0.892857  | 1   | 7        |
| 2019-07-31 | 0.642857  | 1   | 7        |
| 2019-08-31 | 0.321429  | 0   | 7        |
| 2019-09-30 | 0.821429  | 1   | 7        |
| 2019-10-31 | 0.678571  | 1   | 7        |
| 2019-11-30 | 0.857143  | 1   | 7        |
| 2019-12-31 | -0.035714 | 0   | 7        |
| 2020-01-31 | -0.642857 | 0   | 7        |
| 2020-02-29 | -0.964286 | 0   | 7        |
| 2020-03-31 | 0.214286  | 0   | 8        |
| 2020-04-30 | -0.071429 | 0   | 8        |
| 2020-05-31 | -0.142857 | 0   | 8        |
| 2020-06-30 | -0.738095 | 0   | 8        |
| 2020-07-31 | -0.095238 | 0   | 8        |
| 2020-08-31 | -0.150000 | 0   | 9        |
| 2020-09-30 | 0.466667  | 1   | 9        |

| Date       | IC        | Hit | N_stocks |
|------------|-----------|-----|----------|
| 2020-10-31 | 0.406061  | 0   | 10       |
| 2020-11-30 | -0.260606 | 0   | 10       |
| 2020-12-31 | -0.072727 | 0   | 11       |
| 2021-01-31 | 0.763636  | 1   | 11       |
| 2021-02-28 | 0.482517  | 1   | 12       |
| 2021-03-31 | 0.252747  | 1   | 13       |
| 2021-04-30 | 0.615385  | 1   | 13       |
| 2021-05-31 | 0.489011  | 1   | 13       |
| 2021-06-30 | -0.358242 | 0   | 14       |
| 2021-07-31 | 0.037363  | 1   | 14       |
| 2021-08-31 | -0.142857 | 0   | 15       |
| 2021-09-30 | -0.342857 | 0   | 15       |
| 2021-10-31 | -0.135714 | 0   | 15       |
| 2021-11-30 | 0.050000  | 0   | 15       |
| 2021-12-31 | -0.150000 | 1   | 15       |
| 2022-01-31 | -0.010714 | 0   | 15       |
| 2022-02-28 | 0.428571  | 0   | 15       |
| 2022-03-31 | 0.210714  | 0   | 15       |
| 2022-04-30 | -0.182143 | 0   | 15       |
| 2022-05-31 | -0.203571 | 1   | 15       |
| 2022-06-30 | -0.221429 | 0   | 15       |
| 2022-07-31 | -0.150000 | 1   | 15       |
| 2022-08-31 | 0.428571  | 1   | 15       |
| 2022-09-30 | 0.564706  | 1   | 16       |
| 2022-10-31 | 0.229412  | 1   | 16       |

| Date       | IC        | Hit | N_stocks |
|------------|-----------|-----|----------|
| 2022-11-30 | -0.079412 | 1   | 16       |
| 2022-12-31 | 0.258824  | 1   | 16       |
| 2023-01-31 | -0.208824 | 0   | 16       |
| 2023-02-28 | 0.085294  | 0   | 16       |
| 2023-03-31 | 0.282353  | 1   | 16       |
| 2023-04-30 | -0.573529 | 0   | 16       |
| 2023-05-31 | -0.044118 | 1   | 16       |
| 2023-06-30 | -0.282353 | 0   | 16       |
| 2023-07-31 | 0.250000  | 1   | 16       |
| 2023-08-31 | 0.023529  | 0   | 16       |
| 2023-09-30 | -0.247549 | 1   | 17       |
| 2023-10-31 | -0.058824 | 0   | 17       |
| 2023-11-30 | 0.061275  | 0   | 17       |
| 2023-12-31 | -0.091747 | 1   | 19       |
| 2024-01-31 | 0.277193  | 1   | 19       |
| 2024-02-29 | 0.113495  | 0   | 22       |
| 2024-03-31 | 0.250556  | 1   | 23       |
| 2024-04-30 | -0.170949 | 1   | 23       |
| 2024-05-31 | -0.030435 | 0   | 24       |
| 2024-06-30 | -0.049608 | 1   | 24       |
| 2024-07-31 | 0.374783  | 1   | 24       |
| 2024-08-31 | 0.320870  | 0   | 24       |
| 2024-09-30 | -0.008462 | 0   | 25       |
| 2024-10-31 | 0.166154  | 1   | 25       |
| 2024-11-30 | 0.135385  | 0   | 25       |

| Date       | IC        | Hit | N_stocks |
|------------|-----------|-----|----------|
| 2024-12-31 | 0.373846  | 1   | 25       |
| 2025-01-31 | -0.030390 | 1   | 25       |
| 2025-02-28 | -0.227692 | 0   | 25       |
| 2025-03-31 | -0.395845 | 0   | 25       |
| 2025-04-30 | -0.026154 | 1   | 25       |
| 2025-05-31 | 0.258462  | 0   | 25       |
| 2025-06-30 | 0.160769  | 1   | 25       |
| 2025-07-31 | -0.259231 | 0   | 25       |
| 2025-08-31 | -0.230769 | 0   | 25       |
| 2025-09-30 | 0.390769  | 1   | 25       |
| 2025-10-31 | 0.214615  | 1   | 25       |
| 2025-11-30 | 0.056549  | 1   | 25       |
| 2025-12-31 | -0.056923 | 1   | 25       |
| 2026-01-31 | -0.414695 | 0   | 25       |
| 2026-02-28 | 0.448932  | 1   | 25       |
| 2026-03-31 | -0.039231 | 0   | 25       |
| 2026-04-30 | 0.136923  | 1   | 25       |
| 2026-05-31 | 0.049231  | 0   | 25       |



## 11. Phan tich Tinh vung theo Giai doan (Sub-period)

Chia giai doan OOS thanh 2 nua bang nhau de kiem dinh tinh vung. Neu BL thuong EW o ca hai nua, ket qua khong phai may man. Period 1: truoc COVID / tang truong; Period 2: hau COVID / bien dong.

| Period   | Period_Start | Period_End | N_months | Portfolio | Ann_Return | Ann_Vol  | Sharpe   | Sortino  | MDD       | Calmar   |
|----------|--------------|------------|----------|-----------|------------|----------|----------|----------|-----------|----------|
| Full     | 2017-08-31   | 2026-05-31 | 106      | BL        | 0.301188   | 0.281283 | 0.939644 | 1.6281   | -0.306594 | 0.982370 |
| Full     | 2017-08-31   | 2026-05-31 | 106      | EW        | 0.266157   | 0.273821 | 0.837317 | 1.4530   | -0.365473 | 0.728253 |
| Period_1 | 2017-08-31   | 2021-12-31 | 53       | BL        | 0.414678   | 0.351477 | 1.0696   | 1.9065   | -0.306594 | 1.3525   |
| Period_1 | 2017-08-31   | 2021-12-31 | 53       | EW        | 0.433110   | 0.320610 | 1.2301   | 2.2541   | -0.303570 | 1.4267   |
| Period_2 | 2022-01-31   | 2026-05-31 | 53       | BL        | 0.196101   | 0.187020 | 0.861236 | 1.3701   | -0.222165 | 0.882679 |
| Period_2 | 2022-01-31   | 2026-05-31 | 53       | EW        | 0.117209   | 0.214168 | 0.383702 | 0.593285 | -0.365473 | 0.320704 |

## 12. Kiem dinh so cum K toi uu (Elbow + Silhouette)

Xac dinh k toi uu cho K-means bang hai phuong phap: (1) Elbow method - inertia giam nhanh roi cham lai tai 'knee'; (2) Silhouette score (Rousseeuw 1987) - do muc do tach biet giua cac cum. Ket qua OOS cho thay k=4 toi uu ve Sharpe, nhât quan voi analysis nay.

| k | Mean_Inertia | Mean_Silhouette | Std_Silhouette |
|---|--------------|-----------------|----------------|
| 2 | 19.4828      | 0.420100        | 0.089200       |
| 3 | 11.0918      | 0.405700        | 0.053100       |
| 4 | 7.3568       | 0.389100        | 0.048000       |
| 5 | 5.0307       | 0.382500        | 0.060200       |
| 6 | 3.5828       | 0.371600        | 0.066600       |

Phụ lục – Lãi suất Phi rủi ro VN10Y theo Năm

Lãi suất trái phiếu chính phủ Việt Nam kỳ hạn 10 năm (VN10Y) được dùng làm lãi suất phi rủi ro. Lãi suất tháng = VN10Y\_năm / 12.

| Năm  | VN10Y (% năm) | RF tháng (%) |
|------|---------------|--------------|
| 2014 | 6.50          | 0.5417       |
| 2015 | 6.50          | 0.5417       |
| 2016 | 6.20          | 0.5167       |
| 2017 | 5.80          | 0.4833       |
| 2018 | 4.58          | 0.3817       |
| 2019 | 4.58          | 0.3817       |
| 2020 | 3.00          | 0.2500       |
| 2021 | 2.53          | 0.2108       |
| 2022 | 4.00          | 0.3333       |
| 2023 | 3.50          | 0.2917       |
| 2024 | 3.06          | 0.2550       |
| 2025 | 3.10          | 0.2583       |
| 2026 | 4.35          | 0.3625       |